Project PyroSat

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CubeSat Mission Parameters	Value
Mission Name	PyroSat
Mass (kg)	3.775 $\pm$ 0.245 kg
Cube Size	3U
Rail length (mm)	340.5
Readiness Date for integration to Dispenser	April 2029
Desired Orbital Lifetime after Deployment (years)	0.5
Designated Operational Lifetime (years)	0.5
Orbital Parameters	Desired/Preferred Range
Altitude (km)	375-450 km
Inclination (degrees)	33.4-53

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Proposal Abstract

PyroSat is a 3U CubeSat mission that measures soil moisture and land surface temperature, two precursors of wildfire risk that existing models underuse. Most wildfire satellites detect and track fires after ignition, and the missions that observe pre-fire conditions fly on much larger platforms. PyroSat combines Global Navigation Satellite System reflectometry (GNSS-R) and long-wave thermal infrared imaging on one small satellite. The GNSS-R receiver records navigation signals reflected from the ground and uses them to estimate soil moisture, while a thermal camera measures land surface temperature. Both measurements feed a monthly wildfire-probability model. Built around a modified SquidSat bus developed by the Sun Devil Satellite Laboratory (SDSL) at Arizona State University, PyroSat will operate from the International Space Station orbit, with observations processed onboard and downlinked to the SDSL ground station over S-band. The mission has two goals: to show that accurate wildfire risk prediction is possible with a low-cost satellite, and to give ASU students hands-on experience designing, building, and operating a spacecraft. PyroSat is a proof of concept, not an operational warning system. It will assess how well a small satellite can measure pre-fire conditions and whether combining the two measurements improves wildfire-probability estimates over models that use either one alone.

Mission Overview

Problem Statement

Satellites have supported wildfire management for decades, in most cases after a fire has already begun. Landsat, a joint National Aeronautics and Space Administration (NASA) and U.S. Geological Survey (USGS) program, provides a 30 m burned-area product with records back to 1984 [81]. FireSat, a constellation led by the nonprofit Earth Fire Alliance and built by Muon Space with Google Research, detects fires as small as 5 x 5 m. Its first three operational satellites launched in July 2026 and are planned to deliver data to early adopters at least twice a day, with a global revisit of one hour or less targeted for 2029 [82]. Greece's Hellenic Fire System uses four 8U CubeSats with thermal infrared imagers to detect and track active wildfires [83]. ECOSTRESS measures land surface temperature at about 70 m from the International Space Station (ISS) and supports plant water stress mapping [84], and the Copernicus Sentinel satellites image burned land, smoke aerosols, and fire-emitted gases [85]. PyroSat's contribution is cost and form factor: a 3U CubeSat that measures the pre-ignition precursors, soil moisture and surface temperature, that neither the larger missions nor the active-fire systems target together.

Fire risk management frameworks still largely rely on combining topographical models, historical climate data, vegetation, and fuel potential to determine which areas are at risk of wildfires, while underutilizing direct measurements of the conditions that precede fires.

The underlying science is well-established. Wildfire fuel is split into two categories: dead fuel, which exchanges moisture directly with the soil beneath it; and live fuel, which draws water through its roots, becomes drought-stressed, and as a result, more flammable as soil moisture drops. One Sierra Nevada study found a direct, quantified relationship: a 1% increase in soil moisture corresponded to roughly a 0.6% increase in dead fuel moisture [22]. Additionally, soil moisture anomalies have been found to emerge five to six months prior to large wildfires, making it a useful early signal for prediction [23]. Surface temperature complements this on an even faster timescale. When vegetation is water-stressed, it reduces transpiration, causing surface warming that can be measured and precedes other signs of drought stress that typically lead to fires [25, 53].

A 2025 collaborative study between USGS, NASA, and the University of Southern California researchers tested this idea directly by integrating Cyclone Global Navigation Satellite System (CYGNSS)-derived soil moisture into the USGS Wildland Fire Potential Index. Tested against 1,591 known large fires from 2018-2022, the soil-moisture-enhanced index produced small, directionally consistent shifts: slightly lower danger values at the moment of ignition and higher danger values as fires grew over time. The authors described these findings as preliminary evidence of model stability rather than a confirmed improvement in accuracy, with further

validation identified as a necessary next step [24]. These results establish a technical baseline for PyroSat's core hypothesis and justify further development.

Access to wildfire early warning is also uneven: well-resourced nations operate sophisticated early-warning systems, but researchers building low-cost alternatives note that replicating such alternatives in developing countries remains difficult because of financial constraints [89]. Even where better satellite data exists, local experts caution that it is not sufficient on its own.

PyroSat is not intended to close this gap directly. As a single 3U CubeSat with a short mission life, it serves as a low-cost proof of concept for testing whether direct measurements of fuel-moisture precursors can meaningfully strengthen existing wildfire-risk tools.

To make this test concrete, PyroSat's measurements feed a logistic model that combines soil moisture, thermal stress, and weather and vegetation conditions into a monthly probability of wildfire, given in Eq. (1).

$$\eta = b_0 + b_1 z_1 + b_2 z_5 + b_3 R_T + b_4(-z_1)R_T + \gamma \cdot X, \quad P = \frac{1}{1 + e^{-\eta}}$$

Eq. (1). Wildfire Probability Estimation Model

Here, P is the probability of a wildfire in a given cell during a given month, and η is the linear predictor (the log-odds of fire), which the logistic function P = 1/(1 + e^(−η)) converts into a probability between 0 and 1, where e is Euler's number. A cell is one square of a fixed latitude and longitude grid, and PyroSat's soil moisture and thermal measurements are averaged to each cell which is described in Spatial Co-registration and Boson overlap. PyroSat's cell size has not yet been decided; the baseline analysis behind Eq. (1) used cells of 9 x 9 km [29]. The soil moisture inputs z_1 and z_5 are GNSS-R soil moisture anomalies 1 month and 5 months before the forecast month. Each is the cell's monthly mean soil moisture minus its average for that calendar month over the record, divided by the standard deviation for that month, so a negative value means drier than normal. R_T is a thermal stress score from 0 to 1, the average of two thermal stress indices computed from PyroSat's land surface temperature and gridMET air temperature as discussed in Thermal Data Processing. The term $b_4(-z_1)R_T$ is an interaction that is large and positive only when the soil is drier than normal and thermal stress is high. The remaining inputs, grouped as the vector X, are vapor pressure deficit, wind speed, and a vegetation index (NDVI) used as a fuel proxy, each standardized by its training mean and standard deviation. Vapor pressure deficit and wind speed are obtained from gridMET weather data, while NDVI is taken from NASA's MODIS monthly vegetation index product (MOD13A3). The intercept b_0, the coefficients b_1 to b_4 on the soil moisture, thermal stress and interaction terms, and the coefficient vector γ, which multiplies X, are fitted beforehand by maximum likelihood on past fire records and are not inputs at run time [29].

Team Structure Overview

PyroSat is a student-led mission organized by the student organization Sun Devil Satellite Laboratory (SDSL) and the Arizona State University Interplanetary Lab (IPL). SDSL, in collaboration with the IPL, has previously built and launched the Phoenix CubeSat in 2019, LightCube in 2023, DORA in 2024, and Coconut CubeSat in 2026. Students will participate in the full development, assembly, and testing process prior to launch. The student team will also oversee all collaborating parties on the satellite, including affiliate laboratories and industry contacts. One goal of PyroSat is to create learning opportunities for students to get involved with satellites and space technologies that will give them the skills and knowledge they need, along with their university education, to be prepared to enter the space industry. The specific leadership structure for PyroSat is outlined in the figure below.

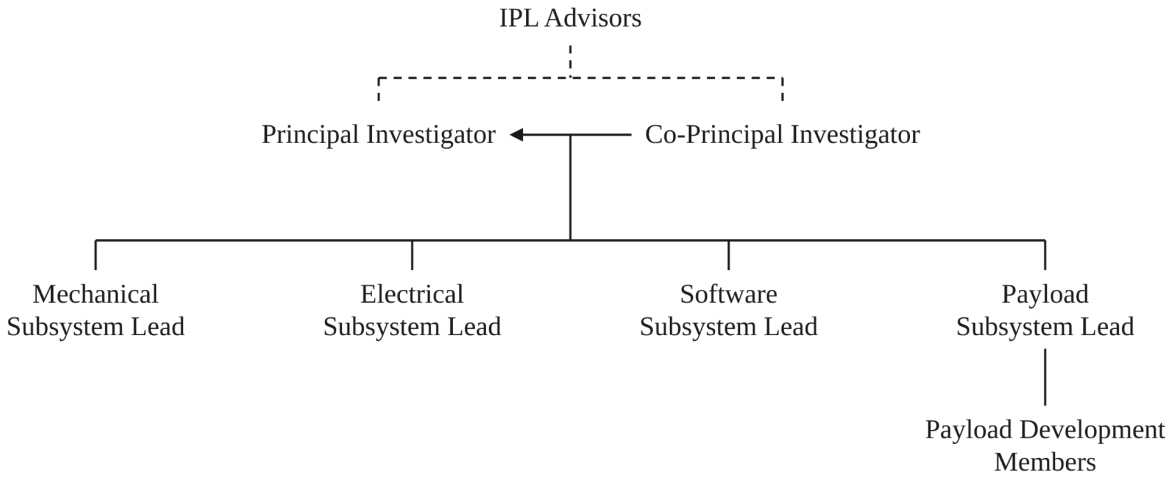


Figure 1. PyroSat leadership structure

Concept of Operations

After PyroSat is deployed, it will wait a specified time determined by the launch provider prior to antenna and solar panel deployment. Immediately after deployment, the attitude determination and control system (ADCS) will begin pointing the satellite toward nadir (straight down toward Earth), which the GNSS-R and thermal payloads need to initiate data collection. PyroSat's position will then be calculated using on-board Global Positioning System (GPS) trilateration and public Two-Line Element set (TLEs) data, and the times when it passes over the Arizona State University ground station will be calculated. After an initial commissioning period, during which the team verifies satellite health, confirms nadir pointing and ground-station communication, and checks the operation of both the GNSS-R receiver and the thermal camera, PyroSat will record synchronized GNSS-R delay-Doppler maps from the GNSS-transmitting satellite and PyroSat's own GNSS-R receiver, as well as thermal radiometric

data over land, and store them with time, position, and attitude data [88]. The duration of the commissioning period has not yet been decided and will be set during mission operations planning. When the satellite is over the ASU ground station, it will enter a dedicated communication mode, listening for commands and transmitting all stored science data to the ground. On the ground, the data will be processed by a terrain-refinement algorithm that improves the geolocation of each GNSS-R observation using a digital elevation model (DEM), such as the Shuttle Radar Topography Mission (SRTM) DEM [5]. The refined observations will then be used as inputs to the wildfire-probability model, Eq. (1).

Payload

Payload Overview

GNSS Reflectometry & LWIR Thermal Sensor

PyroSat's payload will collect two primary remote-sensing measurements required to investigate the mission's wildfire-risk index: surface soil moisture and land surface thermal measurements. The payload will integrate a software-defined GNSS-R instrument with a radiometric long-wave infrared (LWIR) imager within the 2U payload volume of the SquidSat bus, a space of about 9.84 by 9.84 by 18.6 cm [1]. In the stowed launch configuration of the PyroSat CAD model (Figure 10), the payload section fills about 9.0 by 9.6 by 21.4 cm. The bus changes that make room are described in Bus Architecture and in Structures & Mechanical Integration. These instruments will operate as complementary sensing systems, with GNSS-R providing information on surface reflectivity and soil moisture, and the LWIR imager providing land-surface temperature measurements. The payload's onboard processing focuses on three tasks: generating GNSS-R delay-Doppler maps (DDMs) and estimating where each one was recorded, matching each observation to the corresponding thermal image, and storing the resulting science data so it can be refined with higher accuracy on the ground.

LWIR Thermal Sensor

For PyroSat's thermal LWIR instrument, the FLIR Boson+ 640 Radiometric camera will be used, selected due to the Boson camera core family's heritage in CubeSat missions such as BeaverCube [41]. The Boson+ utilizes a 640 x 512 resolution with a 12 μm pixel pitch. The proposed configuration is the 18mm radiometric configuration (22640A024-6IARX), which provides a horizontal field of view (HFOV) of approximately 24° [27]. At an anticipated ISS orbit of about 400 km, this corresponds to a nadir footprint of approximately 170 km and a horizontal ground sampling distance of approximately 270 m/pixel. The camera core measures approximately 21 x 21 x 11 mm and weighs 7.5 g without the lens. FLIR advertises the camera's configuration-dependent power consumption, with a power draw of approximately 1.0 - 1.3 W

during continuous measurement [3]. FLIR lists the complete camera with this lens at approximately 41 g, with an overall length of approximately 38 mm, and a maximum diameter of 27.6 mm [27]. FLIR also identifies the camera's A024 configuration as radiometry-capable, required for PyroSat's quantitative thermal measurements. Because the Boson's specified radiometric accuracy is about ±5 °C, PyroSat will use an onboard blackbody reference to calibrate the camera to about 1 K, the provisional accuracy target set from sensitivity tests of the thermal stress score.

The values above are manufacturer ratings for ground use, not spaceflight specifications, because the Boson+ needs mechanical, thermal, and environmental adaptation for launch and orbit. The feasibility of adapting a Boson core camera for aerospace environments has previously been demonstrated by NASA Marshall Space Flight Center through its *Ruggedized Infrared Camera* technology. NASA developed a ruggedized assembly around a commercial FLIR Boson 640 camera with an 8.7mm lens, incorporating an aerospace-grade aluminum housing, a mechanical lens-retention system, a conductive thermal-management approach, and a ruggedized electrical interface to address the environmental conditions of spaceflight, such as vibrations, shock, thermal, and vacuum requirements. This assembly had a reported mass of approximately 163 g and occupied less than 131 cm^3, including camera, lens, housing, and supporting electronics [26].

To protect the FLIR Boson camera and prevent altering the LWIR data, a similar assembly will be created around the camera, clamping the lens and camera to a shared aluminum housing and integrating electronic connections. Radiometric accuracy depends on the temperature of this housing, and while the Boson+ operates from -40° to 80° C [3], its radiometric performance degrades as the housing temperature drifts. The thermal design therefore aims to hold the housing within a temperature range of about 1 K during each data capture, and Figure 2 shows how it is arranged. It follows three rules: isolate, use one heat path, and heat to a setpoint.

The housing is isolated from the bus. It is mounted on PEEK standoffs, which have a low thermal conductivity of about 0.29 W/(m·K) at 23 °C, and the assembly is wrapped in multi-layer insulation (MLI) blankets of polyimide film to keep heat from the bus electronics and the S-band radio from reaching the camera [42], [43]. The Boson, lens, mount, and housing are also bolted to a block of aluminum 6061 that acts as a heat reservoir, evening out temperature gradients and slowing temperature changes over an orbit. A copper thermal strap connects the aluminum block to a radiating side panel of the bus, which is the assembly's only passive means of rejecting heat [42], [44], [45]. A thermistor on the aluminum block feeds a thermostat that switches a polyimide thermofoil heater on whenever the temperature falls below the setpoint, such as during long eclipses [46], [47]. The strap is sized so that, with the heater off, the assembly would settle slightly below the setpoint. The heater therefore only ever needs to add heat, and the design never has to remove it actively. This approach builds on the thermal isolation used on Phoenix, SDSL's previous CubeSat, and on ASTERIA, a 6U CubeSat that used

passive cooling and trim heaters to keep its focal plane temperature stable [48], [49]. The stability target, heat budget, and radiating-panel sizing are covered in Thermal Management.

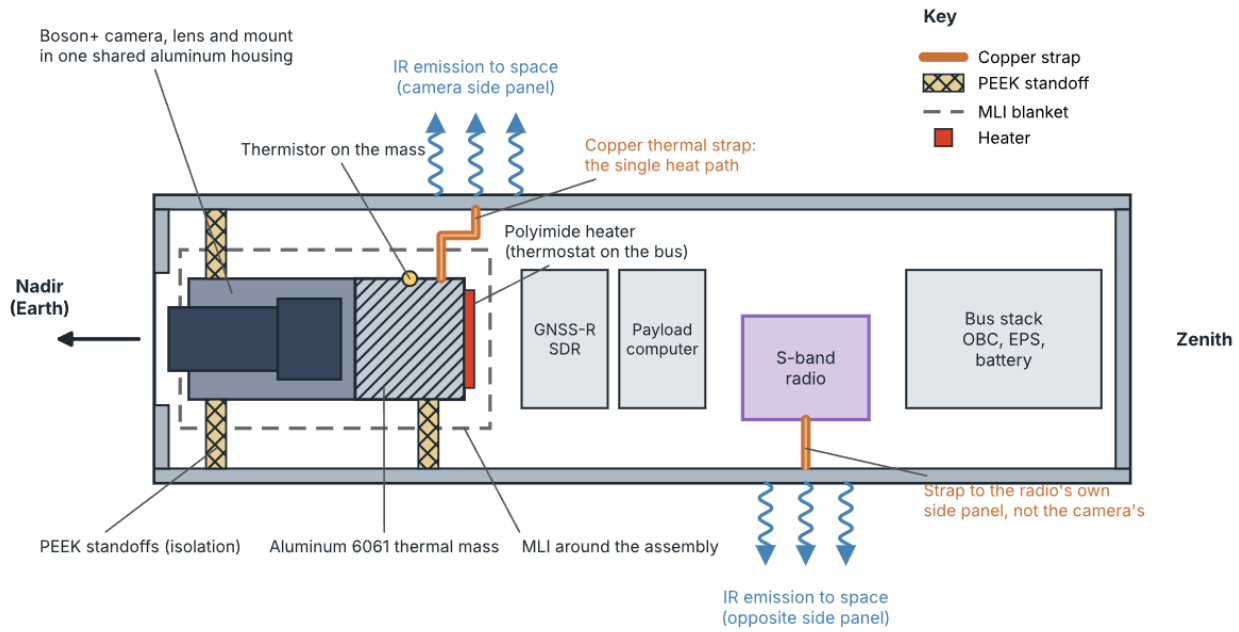


Figure 2. Thermal arrangement of the Boson+ camera assembly (side view, deployed wings not shown). The camera housing and aluminum thermal mass are isolated from the frame by PEEK standoffs and MLI, with a single copper strap to a radiating side panel and a thermostat-controlled heater on the mass. The S-band radio has its own strap to the opposite side panel.

To limit outgassing and keep the imager lens from clouding, the FLIR Boson camera, lens, and housing will undergo a bakeout before launch to remove contaminants [50]. After deployment and once the satellite is stable and the camera is off, the camera heater will raise the camera and lens housing to a warm decontamination temperature for a set period of time so that remaining volatiles outgas to space instead of condensing on the lens. .

Calibration & Validation

The Boson requires extensive on-ground and in-orbit calibration to reduce its radiometric error to PyroSat's ±1 K target. CIRiS [30] flew a similar uncooled microbolometer with an on-board calibration system: a scene-select mirror alternating between Earth, deep space and two carbon-nanotube blackbodies. PyroSat will carry a reduced version of this system with a single internal blackbody. Testing will verify whether it reaches ±1 K.

GNSS-R Antennas & RF Front End

Nadir Antenna Array

The FMPL-2 paper lists an antenna design based on its requirements of both an LHCP GPS L1 1575 MHz and the 1400-1417 MHz L-Band radiometer as shown in Figure 3 [6]. Since PyroSat does not use a radiometer layer, the radiometer plane is removed, reducing the antenna board from five layers to four. Additionally, PyroSat will utilize two separate 3-patch antenna arrays instead of FMPL-2's 6-patch design to fit within the 3U bus constraint. The geometry details are listed in the PyroSat CAD model as represented in Figure 11. The resulting PCB stackup will require electromagnetic simulation and optimization to determine whether the remaining ground plane spacing, substrate properties, and feed structure provide the required L1 antenna performance.

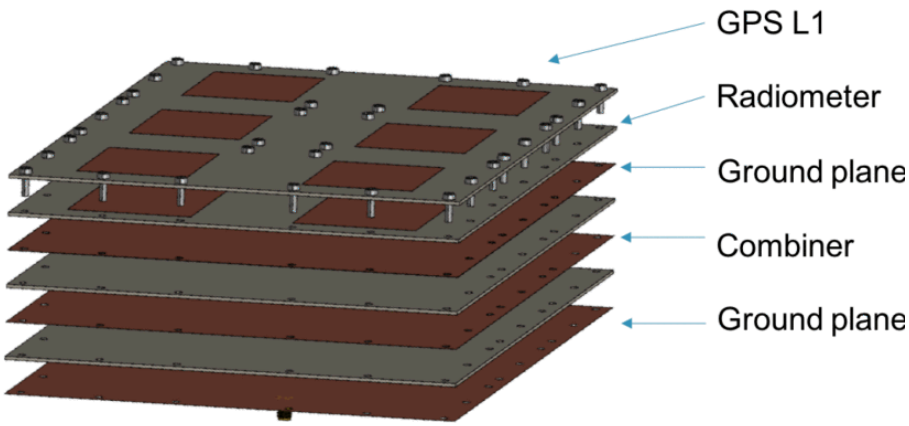


Figure 3. FMPL-2 Nadir antenna design including its patches and combiner in the internal layers, which map a GNSS L1 and radiometer patches.

Each PyroSat subarray will contain three microstrip patch elements designed to receive reflected GNSS L1 signals centered at 1575.42 MHz. The geometry of each plane, dielectric properties of the substrate, feed location, and spacing from the ground plane determine its resonant frequency, impedance, bandwidth, and polarization response. The nadir-facing antennas will be designed as left-hand circular polarization (LHCP), corresponding to the dominant polarization expected from reflected GNSS signals.

A conductive ground plane is positioned beneath the patch elements and provides the RF reference required for the microstrip antenna structure. Together with the dielectric substrate, the spacing between each patch and the ground plane influences the antenna's resonant frequency, impedance, bandwidth, and radiation characteristics. The ground plane also helps suppress radiation toward the spacecraft and directs the antenna response toward the nadir-facing observation region.

The RF signal produced by each of the three patch elements is routed through a feed network to a 3:1 combining network. The combiner links the three element signals into a single

RF output for the subarray. The relative electrical path lengths and impedances between the elements and combiner must be controlled so that signals arriving from the desired observation direction combine constructively. Consequently, the feed-network geometry affects the subarray's gain, impedance matching, and overall efficiency. Each three-element subarray will then produce one conditioned RF output representing the combined response of its three L1 patch elements. PyroSat will use two equivalent three-element subarrays, producing two independent RF outputs. These signals will be routed through their respective RF front-end chains and simultaneously digitized by the two receive channels of the AD9361. The resulting complex I/Q streams will then be phase- and amplitude-aligned and digitally combined within the software-defined radio (SDR) processing architecture.

Lastly, the initial dimensions of the proposed antenna elements will be based on the values demonstrated by FMPL-2. Each FMPL-2 GNSS L1 patch measured approximately 51.96 x 52.84 mm, with a 100 x 100 mm ground plane used during individual-element development and characterization. These dimensions provide an initial design reference for PyroSat's three-element subarrays. However, the final PyroSat patch dimensions, element spacing, substrate properties, and ground-plane geometry will require re-optimization following the result of the removal of the radiometer layer and the modification of the antenna array to two separate 3-patch arrays, as these changes will alter the electromagnetic properties of the antenna.

Zenith GNSS Antenna

For the zenith RHCP GPS L1 antenna patch for receiving a direct GNSS signal, PyroSat's CAD baseline for the zenith-facing antenna is the SkyFox Labs nanoPATCH-L1 (50 x 50 x 12.5 mm, 29 g), which fits the 88 mm body-panel band. The larger piPATCH-L1E1/FM flight model, described below, was the earlier candidate and is kept as a reference until the nanoPATCH-L1 datasheet is confirmed. The piPatch-L1E1/FM is an active CubeSat GNSS antenna designed for mounting on the Z+/Z- face of a nadir- or zenith-oriented CubeSat, making it compatible with PyroSat's intended spacecraft orientation [32].

The piPATCH-L1E1/FM flight model occupies a 98 x 98 x 13 mm envelope and weighs approximately 50 g, allowing it to fit within a standard CubeSat Z-face. The piPatch-L1E1/FM incorporates its own low-noise amplifier (LNA) and L1/E1 Saw Filtering. The integrated LNA amplifies the weak direct GNSS signals near the antenna before losses are introduced, while the SAW filter suppresses signals outside the desired L1/E1 frequency range. The antenna requires a 2.7 to 5.7 V supply, with an expected operating point of 3.3 V at approximately 20 mA, corresponding to approximately 0.066W.

Within PyroSat, the zenith antenna will receive the direct GNSS signals used to identify and track available transmitting satellites and to determine their navigation information. The direct GNSS RF signal can subsequently be routed to the OEM719 receiver while the two nadir-facing subarrays independently receive the much weaker reflected signals used for DDM generation.

GNSS-R RF Front End & Receiver

The figure 4 seen below is derived from the FMPL-2 RF front end. The proposed architecture retains separate direct RHCP and LHCP signal paths while eliminating the 1413 MHz L-Band radiometry channel. This elimination drastically simplifies the RF front end compared to the one described in the FMPL-2 paper, since there is no mixing or splitting of the L-Band signal [6]. The proposed replacement for the L-Band's production microwave brightness temperature will be the Boson's land-surface temperature sensing for the mission's wildfire assessment.

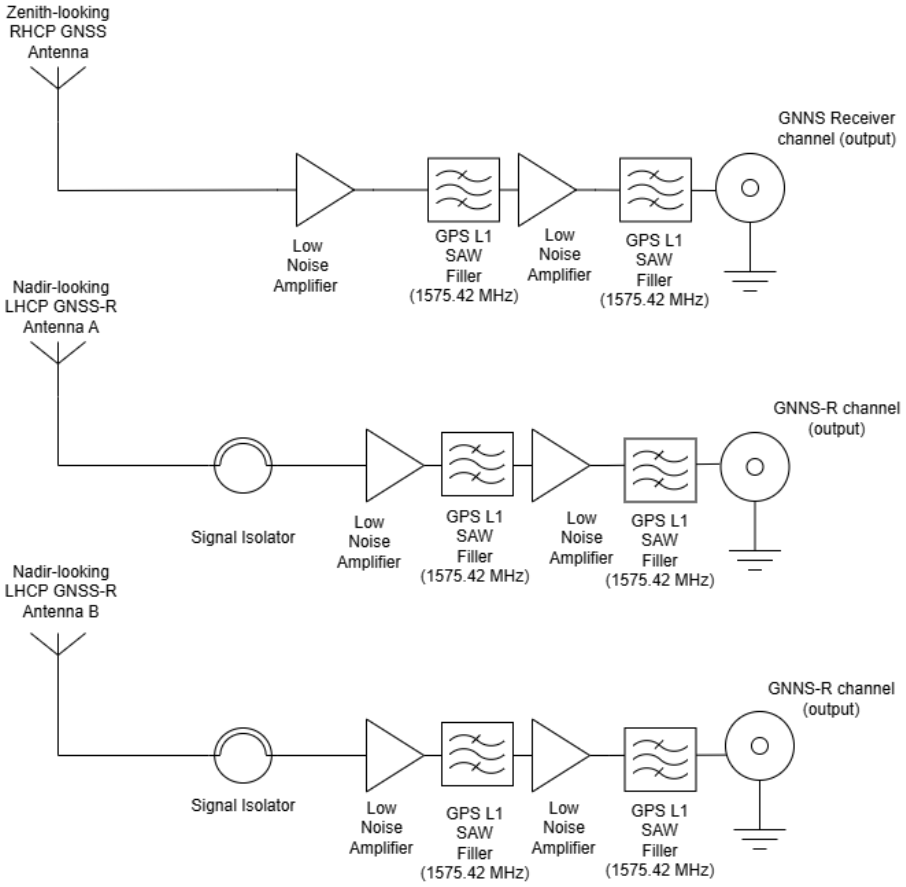


Figure 4. RF-FE receiving chain adapted from FMPL-2 [6]

The preliminary GNSS-R front end consists of direct and reflected GNSS receiver channels operating near the L1 frequency of 1575.42 MHz. The zenith-facing RHCP antenna receives direct GNSS signals from transmitting GPS satellites above, while the two nadir-facing LHCP three-patch subarrays receive signals reflected from the Earth's surface. Each channel uses low-noise amplification and L1 filtering to increase the desired signal level while suppressing interference prior to signal processing. Final amplifier gain, noise figure, filter bandwidth, insertion loss, and board requirements will be established through future experimental analysis.

Once the signals are processed through the RF-FE chain, they will be sent to their respective data pipelines. The zenith-looking antenna's purpose is to produce a conditioned analog L1 RF signal that is sent directly for signal processing to the GNSS receiver (OEM719) to be used in developing the DDM. Furthermore, the nadir-looking antenna's purpose is to detect the GNSS-R reflected signals, which are sent directly to the AD9361 for digitization into complex samples $I[n] + jQ[n]$. These samples are then transferred to Zynq-7045 for GNSS-R signal processing. Using the identity of the transmitting GNSS satellite, the processor generates a local replica of the corresponding PRN code and correlates it against the reflected I/Q samples over a bin of candidate delays and Doppler frequencies. The resulting correlation powers are used to construct the Delay-Doppler map (DDM).

Onboard Computing & Data

Computations & Controls

This section describes the computing hardware that runs the payload and the spacecraft, how processing is divided between the Zynq FPGA, the ARM processor, and the bus computer, and how data moves between them. Real-time signal processing runs on the FPGA, payload control and device management run on the ARM processor, and spacecraft-level commanding and housekeeping stay with the RP2350 onboard computer.

The proposed onboard computing architecture is based around a Xilinx Zynq-7045 system on chip (SoC), which combines a programmable FPGA with a dual-core ARM processor. This combination enables the payload to perform both GNSS-R signal processing in the FPGA and software-based instrument control on the ARM processor. FMPL-2 used a similar Zynq-7045 architecture, with FPGA resources performing GNSS-R signal processing while the ARM processor supported higher-level payload operations [6].

The Zynq FPGA will support real-time GNSS-R processing. Reflected GNSS signals received by the nadir-facing antenna array pass through the RF front end and the AD9361 RF transceiver, which down-converts and digitizes them into I/Q samples. These samples are then passed to the FPGA, which performs the cross-correlation and signal processing that generate Delay-Doppler Maps (DDMs), reducing the computational burden on the general-purpose ARM processor in the same way as the FMPL-2 processing architecture [6].

The ARM processor architecture will be responsible for higher-level payload control and data management. This includes commanding and acquiring radiometric frames from the Boson+ 640, receiving spacecraft navigation data and timestamps, associating measurements with spacecraft altitude, storing data, and executing preliminary GNSS-R geolocation algorithms. Once an FPGA generates a DDM, the ARM processor can combine this DDM with the known location of PyroSat and the transmitting GNSS satellite to estimate the specular reflection point and determine whether it plausibly overlaps the Boson's thermal observation frame.

The payload will interface with the primary RP2350 onboard computer (OBC) as opposed to replacing it, and the RP2350 will handle spacecraft-level command, state management, payload mode control, and housekeeping. The NanoCom SDR MK3 will serve as the primary science-processing and storage platform. GNSS-R I/Q samples will be processed by the NanoCom's Zynq-7045 into Delay-Doppler Maps (DDMs), while Boson+ 640 thermal frames and associated observation metadata will be collected and stored locally using the NanoCom's integrated 70 GB pSLC and 233 GB MLC eMMC storage [35]. Command traffic between the RP2350 and the NanoCom is described in Command & Data Handling, and the downlink path is described in S-Band Science Downlink.

Data Management & Storage

This section covers the science data from the moment that it is produced until deletion. Bus housekeeping data is described in Command & Data Handling. Science data remains on the payload and never travels over the CAN bus.

The Boson+ 640 produces 640 x 512 pixel frames at 16 bits per pixel, or about 655 kB per raw frame [3]. The capture strategy is to image only while over target land, about 3 frames per observation, which is roughly 2 MB per observation. The GNSS-R receiver's PRN data produces delay-Doppler maps and the metadata needed to locate each specular point. For sizing, PyroSat assumes an FMPL-2-like delay-Doppler map of 500 cells (100 delay bins x 5 Doppler bins) at 4 bytes per cell, about 2 kB before metadata, produced at up to 4 maps/second, the rate reported for each CYGNSS observatory. At this rate, the GNSS-R receiver would produce approximately 8 kB of delay-Doppler map data per second, or approximately 4.8 MB over the 10 minutes of Observation mode in each orbit. These figures are sizing estimates and will be replaced by measurements of the receiver's actual output..

At the targeted 1 Mbps, the S-band link carries about 52 MB over a 7-minute pass described in S-Band Science Downlink, which is the hard ceiling per pass. Capturing continuously would produce far more data than this. At the assumed rate of 8 kB per second, the GNSS-R receiver alone would generate about 44 MB per orbit, or about 690 MB per day over roughly 15.6 orbits, while the ground station provides only 2 to 5 passes per day. Capturing only over land, about 27% of the ground track, reduces the volume, but the thermal frame rate has not yet been set, so the land-only total cannot yet be quantified. It is expected to remain larger than the downlink can return. The payload therefore selects which observations to keep instead of recording everything. Each thermal observation is about 2 MB (three frames of about 655 kB), so one 7-minute pass at 1 Mbps can carry approximately 26 observations if nothing else is transmitted. The actual number will be lower because GNSS-R data shares the same downlink. If the downlink becomes the limiting factor, the payload computer can compress the Boson+ frames losslessly before transmission to fit more observations into each pass, which keeps the raw counts that the ground calibration and geolocation depend on. The achievable ratio will be measured on real Boson+ data, and the 2 MB per observation figure above conservatively

assumes no compression. The 52 MB per-pass limit assumes a 7-minute contact at 1 Mbps. The actual limit will depend on the pass durations in the ground-station contact plan. The baseline is 2 usable passes per day with the ASU ground station alone because PyroSat will roughly get one pass every 10 orbits, at about 7 minutes each, which returns about 105 MB per day.

“Science data is stored on the NanoCom SDR MK3’s integrated eMMC, which has 70 GB of pSLC and 233 GB of MLC storage, about 300 GB in total [35], and is managed locally by the Zynq. At the budgeted 1 to 2 GB for 1,500 to 3,000 frames (about 0.67 MB per frame), the 233 GB MLC area alone holds about 350,000 frames, so storage capacity does not constrain the number of observations that can wait for a missed pass. Each observation is written as a self-describing file with a header holding the GPS time, position, attitude quaternion, gain and exposure settings, and a cyclic redundancy check (CRC). Writes are append-only, journaled and wear-leveled. The payload keeps each file until the ground station acknowledges receipt, after which the file is marked deletable and removed by ground command, and frames are transmitted in blocks so that an interrupted pass resumes at the last good block instead of restarting. If storage reaches a defined fill threshold, the payload stops recording new observations and reports the condition in telemetry until space is freed. If the eMMC reports errors or the file system fails to mount, the payload reports the fault over the CAN bus, stops writing, and downlinks from a small RAM buffer until the OBC commands a restart, as described in Payload Control & Fault Handling.

Science Processing

GNSS-R Signal Processing & DDM generation

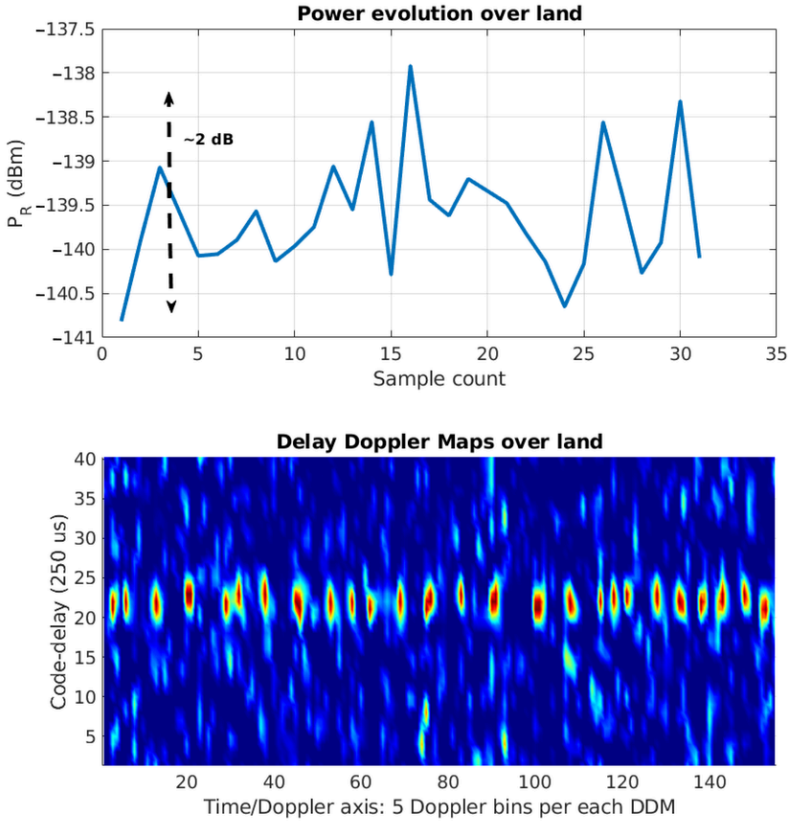


Figure 5. FMPL-2 measured reflected-power evolution and sequence of Delay-Doppler Maps over land [80].

Figure 5 provides an overview of the GNSS-R data product that is expected from PyroSat given its basis in FMPL-2's processing architecture. The lower plot shows a sequence of GPS L1 Delay-Doppler Map collected by FMPL-2 overland by its GNSS-R patch antennas, with each DDM consisting of five doppler bins across the horizontal axis and code delay represented vertically. The concentrated high-power response visible across successive observations represents the detected reflected GNSS signal. The upper plot shows the corresponding variation in received reflected power along the observation track, demonstrating that the strength of the land reflection changes between measurements. FMPL-2 therefore provides flight-demonstrated evidence that an AD9361/Zynq-based CubeSat GNSS-R architecture can resolve measurable GPS L1 reflections over land [80].

PyroSat will use a software-defined GNSS reflectometry architecture based on the Flexible Microwave Payload-2 (FMPL-2) [6]. Reflected GNSS L1 signals are received by the two nadir-facing three-patch LHCP subarrays, passed through their own RF front-end chains to

the two receive channels of the AD9361 RF transceiver, which downconverts and digitizes the received signals into (I/Q) samples. The digitized samples are then transferred to the Zynq-7045 SOC for signal processing and DDM generation. FMPL-2 generates DDMs using a modified Parallel Code Phase Search (PCPS) algorithm. Because each GNSS satellite transmits a known pseudorandom (PRN) code, the payload processor can generate a local replica of the transmitting satellite's PRN code and correlate it with the received reflected signal. Correlation is performed over a range of code delays and Doppler to determine the distribution of reflected signal intensity over the map [6].

The received complex signal can be represented as,

$$x[n] = I[n] + jQ[n] \tag{2}$$

where $I[n]$ & $Q[n]$ are the in-phase and quadrature components, j is the imaginary unit, and n is the sample index. These two components are required to preserve the information required for GNSS signal geolocation and DDM generation.

The I/Q samples are then transferred to the Zynq-7045 for correlation with locally generated replicas of the GNSS signals. Consequently, the reflected signal from one particular satellite can be identified by correlating the received signal with a local replica of that satellite's PRN code. Rather than searching only for the presence of the PRN code, GNSS-R processing evaluates correlations across multiple possible signal delays and Doppler frequencies. The resulting correlation power forms a two-dimensional Delay-Doppler Map (DDM). FMPL-2 defines the DDM as the distribution of the received signals in the delay and Doppler domains.

Mathematically, it is also defined as,

$$\left\langle \left| Y_{xy}(\tau,\nu) \right|^2 \right\rangle = \left\langle \left| \frac{1}{T_c} \int_0^{T_c} x(t) y^*(t-\tau) e^{-j2\pi\nu t}\, dt \right|^2 \right\rangle \tag{3}$$

Where x(t) is the received reflected GNSS signal, y(t) is a copy of the observed transmitting satellite's known PRN code generated onboard and used as a reference to identify the reflected GNSS signal, τ is the candidate code delay, ν is the candidate Doppler frequency, and T_c is the coherent integration time. The superscript * denotes complex conjugation, while the angle brackets ⟨⟩ represent incoherent averaging of the resulting correlation power [6].

The two search variables in equation (3), τ and ν, represent two distinct properties of the reflected signal. The delay τ represents the difference in signal arrival time resulting from the additional distance traveled by the reflected GNSS signal from the transmitter to the Earth's surface, and finally to PyroSat. The Doppler frequency ν is the frequency shift caused by the rate of change of the reflected signal path length as the GNSS satellite and PyroSat move relative to the reflecting specular point on Earth. By searching across combinations of delay and Doppler

frequency, as this equation does, the processor determines where the reflected GNSS signal produces the strongest correlation with the known PRN code.

Rather than evaluating equation (3) for every possible delay-Doppler combination, the FMPL-2 paper details the implementation of a modified Parallel Code Phase Search (PCPS) algorithm to reduce the computational burden of this correlation process. As shown in *Figure 6, the circular cross-correlation is accelerated using the programmable logic of the Zynq-7045*. The received I/Q samples are transformed into the frequency-domain representation of the corresponding PRN code, then transformed back into the time domain using an inverse FFT (IFFT). This operation produces the correlation across the searched code delays for a given Doppler frequency and is repeated across the required Doppler search bins [6].

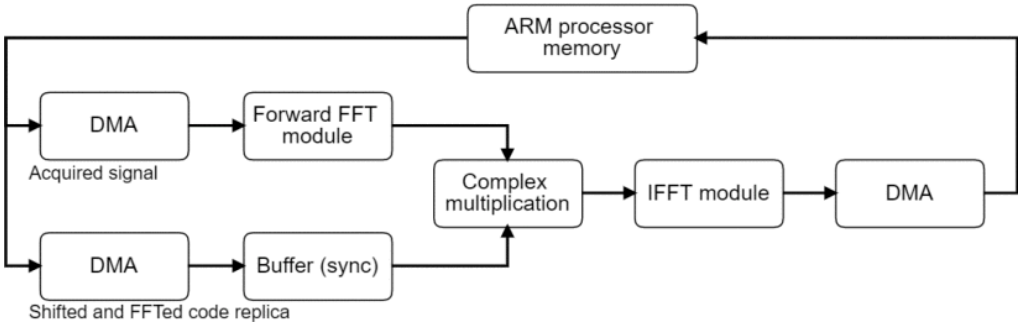


Figure 6. Circular Correlation implemented in FPGA from FMPL-2 paper [6]

The resulting correlation values are transferred to the ARM processor, where the squared magnitude in equation (3) is calculated to obtain the correlation power, and successive results are accumulated via incoherent integration. Repeating this process across the selected delay and Doppler bins yields the two-dimensional Delay-Doppler Map (DDM), in which each cell represents the measured correlation power for a particular delay-Doppler combination [6]. PyroSat will adapt this FMPL-2 processing architecture to use the AD9361 transceiver and the Zynq-7045 SOC. The resulting DDM will be stored with its observation time, satellite PVT, altitude, GNSS transmitter information, and instrument metadata. This data will then support preliminary GNSS-R geolocation and calibration; once transmitted back to the ground, further refinement and processing will be done to estimate surface reflectivity and retrieve soil moisture.

For its initial implementation, PyroSat will use an FMPL-2-based GPS L1 DDM configuration consisting of 100 code-delay bins and five Doppler bins, producing 500 correlation-power cells per DDM. FMPL-2 used a default delay spacing of 0.25 GPS C/A chips and a Doppler spacing of 500 Hz [6]. These parameters provide PyroSat with an initial processing baseline and may later be adjusted according to measured signal performance, FPGA processing requirements, and the desired delay-Doppler resolution.

The DDM grid is constructed by repeatedly executing the PCPS correlation process. For each candidate Doppler bin, the incoming reflected I/Q samples are compensated for the corresponding Doppler frequency and correlated with the locally generated PRN replica. The

FPGA's FFT-based correlation produces correlated values across the searched code-delay bins. These values are then transferred to the ARM processor, where their squared magnitudes are calculated and accumulated via incoherent integration, as shown in equation (3). Repeating this process for each of the five Doppler bins produces the complete 100 x 5 DDM, each cell representing the measured correlation power associated with a candidate delay-Doppler combination.

The measured correlation-power distribution within each DDM will serve as the primary data product collected from the GNSS-R antennas. Therefore, it is the basis for subsequent geolocation calibration and surface-reflectivity retrieval calculations.

Geolocation of GNSS-R signal

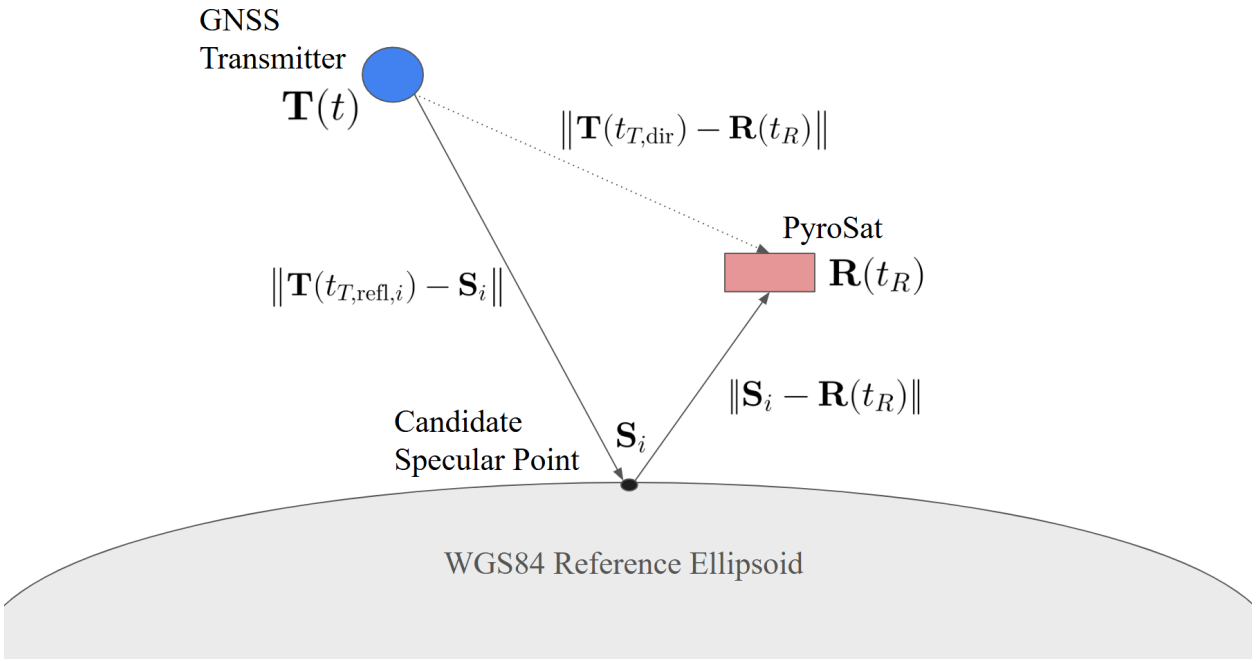


Figure 7. GNSS-R observation geometry showing the direct signal path and candidate reflected signal path through S_i on the WGS84 reference ellipsoid.

One of the primary challenges associated with PyroSat's GNSS-R payload is determining the geographic region represented by each measured DDM. Unlike the Boson+ thermal imager, the GNSS-R instrument does not produce a conventional two-dimensional geographic image. Instead, the location of a GNSS-R observation is determined from the bistatic geometry between the transmitting GNSS satellite, Earth's reflecting surface, and PyroSat. PyroSat will therefore calculate a preliminary onboard geolocation for each suitable GNSS-R observation and store this solution with corresponding DDM for subsequent ground refinement.

PyroSat's design target is to locate each GNSS-R observation to within about 1 km on the ground. This matches the roughly 1 km area from which a coherent land reflection is received

and the roughly 1 km geolocation accuracy for which the CYGNSS land processing was designed [5]. At an altitude of 400 km, a 1 km ground error corresponds to an angle of arctan(1 km / 400 km) ≈ 0.143 degrees, so the spacecraft's attitude knowledge for the GNSS-R payload must be at least this good. The ADCS section uses this value as the GNSS-R pointing-knowledge requirement.

The geolocation calculator must account for the finite propagation time of the GNSS signal. A reflected signal received by PyroSat at reception time t_R was transmitted by the GNSS satellite at an earlier time $t_{T,refl}$, reached the reflecting surface at an intermediate time t_S, and subsequently arrived at PyroSat at t_R. The corresponding direct GNSS signal received at the same reception frame follows a shorter transmitter-to-receiver path and was therefore transmitted at a later time $t_{T,dir}$. Consequently, $t_{T,refl} < t_{T,dir} < t_R$.

PyroSat's position, velocity, and time (PVT) at t_R will be determined by the onboard navigation system. The transmitting GNSS satellite's position at the appropriate transmit frame will be reconstructed from its broadcast ephemeris received by the OEM719. Accounting for these distinct time frames prevents the transmitter from being incorrectly treated as stationary while the GNSS signals propagate.

For an initial geolocation solution, Earth's surface will be represented using the WGS84 reference ellipsoid. A WGS84 reference ellipsoid is a mathematical approximation of the Earth's shape used for global positioning and navigation. Rather than representing Earth as a perfect sphere, it represents it as its more accurate form, an oblate ellipsoid, providing a geometric surface on which an initial predicted specular point can be approximated. The predicted specular point is the surface location at which the transmitter-to-surface and surface-to-receiver geometry satisfies the law of specular reflection $\theta_t = \theta_r$.

Let $T(t_{T,refl})$ represent the position of the transmitting GNSS satellite at the reflected signal's transmit frame, $R(t_R)$ represent the position of PyroSat at the reception frame, and S represent a candidate location on the WGS84 ellipsoid. The unit vectors from the candidate surface location toward the transmitter and receiver are then,

$$\mathbf{u}_T(\mathbf{S}) = \frac{\mathbf{T}(t_{T,\mathrm{refl}}) - \mathbf{S}}{\left\|\mathbf{T}(t_{T,\mathrm{refl}}) - \mathbf{S}\right\|} \tag{4}$$

$$\mathbf{u}_R(\mathbf{S}) = \frac{\mathbf{R}(t_R) - \mathbf{S}}{\left\|\mathbf{R}(t_R) - \mathbf{S}\right\|} \tag{5}$$

The local surface normal at the ideal specular point will be parallel to the bisector of these two directions:

$$\mathbf{n}(\mathbf{S}) \parallel \mathbf{u}_T(\mathbf{S}) + \mathbf{u}_R(\mathbf{S}) \quad (6)$$

Equivalently, the specular point can be formulated as,

$$\mathbf{n}(\mathbf{S}) \times [\mathbf{u}_T(\mathbf{S}) + \mathbf{u}_R(\mathbf{S})] = \mathbf{0} \quad (7)$$

An initial estimate is obtained from the transmitter and receiver geometry relative to the WGS84 ellipsoid. A numerical solver will iteratively adjust the candidate latitude and longitude until the reflection condition in Eq. (7) is satisfied within a predefined tolerance. Because the transmitter position depends on the signal transmit frame, the corresponding time and transmitter state can be updated iteratively as the candidate solution is refined. The resulting solution, S_0, represents PyroSat's geometry-derived estimate of the instantaneous specular reflection location. The specific numerical solver, convergence tolerance, and time iteration procedure will be selected and validated during algorithm development.

This estimate provides the center of the subsequent DDM-assisted refinement. Nearby candidate surface locations S_i are evaluated around S_0. Each S_i represents a fixed geographic candidate on the WGS84 ellipsoid during its evaluation. For each candidate, PyroSat predicts the excess delay and Doppler coordinates at which a reflection from that location should appear within the measured DDM.

For a signal received by PyroSat at t_R, the reflected propagation associated with candidate S_i is,

$$L_{\text{refl},i} = \left\|\mathbf{T}(t_{T,\text{refl},i}) - \mathbf{S}_i\right\| + \left\|\mathbf{S}_i - \mathbf{R}(t_R)\right\| \quad (8)$$

The corresponding direct signal received at the same reception frame follows the path

$$L_{\text{dir}} = \left\|\mathbf{T}(t_{T,\text{dir}}) - \mathbf{R}(t_R)\right\| \quad (9)$$

The two transmitter positions in Eqs. (8) and (9) are evaluated at different frames of time because the reflected and direct signals travel different distances before arriving at PyroSat. Their approximate transmit frames satisfy the light-time relationships

$$t_{T,\text{refl},i} = t_R - \frac{L_{\text{refl},i}}{c} \quad (10)$$

$$t_{T,\text{dir}} = t_R - \frac{L_{\text{dir}}}{c} \quad (11)$$

where c is the speed of light. Because the path lengths themselves depend on the transmitter position at their respective timeframes, the transit times can be determined iteratively using the GNSS broadcast ephemeris.

The difference between the reflected and direct propagation paths represents the excess distance traveled by the reflected signal. The predicted excess delay for candidate S_i is therefore

$$\Delta\tau_i = \frac{L_{\mathrm{refl},i} - L_{\mathrm{dir}}}{c} \quad (12)$$

This excess delay is the quantity relevant to the delay dimension of the DDM. In other words, PyroSat is not simply measuring how long the reflected signal took to travel from the GNSS satellite to the receiver; it is measuring the additional propagation delay of the reflected signal relative to the corresponding direct GNSS reference signal.

The Doppler coordinate is similarly determined from the rate at which the propagation geometry changes. As the GNSS transmitter and PyroSat move, the direct and reflected path lengths vary with time. If the DDM Doppler coordinate is refined relative to the direct GNSS reference signal, the predicted excess Doppler for candidate S_i can be represented as,

$$f_{D,i} = -\frac{1}{\lambda}\frac{d}{dt_R}\left[L_{\mathrm{refl},i} - L_{\mathrm{dir}}\right] \quad (13)$$

where λ is the wavelength of the GNSS signal at the carrier frequency of 1575.42 MHz which in the equation below is represented by L1 for the fact that it is the L1 band frequency picked up by the nadir patch antenna array.

$$\lambda = \frac{c}{f_{L1}} \approx 0.1903 \text{ m} \quad (14)$$

Each candidate location S_i therefore predicts a corresponding location in delay-Doppler space, $S_i \rightarrow (\Delta\tau_i, f_{D,i})$. These predicted coordinates can then be compared with the measured correlation-power distribution $P(\tau, \nu)$ contained within the DDM. Candidates whose predicted delay and Doppler coordinates coincide more closely with the region of measured reflected-signal power are favored, while candidates inconsistent with the measured response are rejected. Evaluating nearby candidates around S_0 therefore provides a DDM-assisted refinement of the initial geometric specular-point solution.

The DDM-refined geolocation will then be compared with the georeferenced Boson+ field of view to determine whether the GNSS-R and thermal measurements overlap at the observation. Observations whose refined GNSS-R locations remain within the thermal frame will be marked as simultaneous co-located observations and given the highest science priority. If DDM-assisted refinement places the reflection outside the corresponding Boson+ frame, the DDM will not be immediately discarded. Because the DDM has already been generated and represents a small science product, it can instead be retained with a non-overlap flag and its

associated navigation and instrumentation metadata. Following downlink, these observations may still support GNSS-R calibration and soil-moisture retrieval or be spatially paired with suitable thermal observations acquired at another time. Final terrain-aware geolocation and spatial co-registration will be performed on the ground using the DEM-based refinement described below.

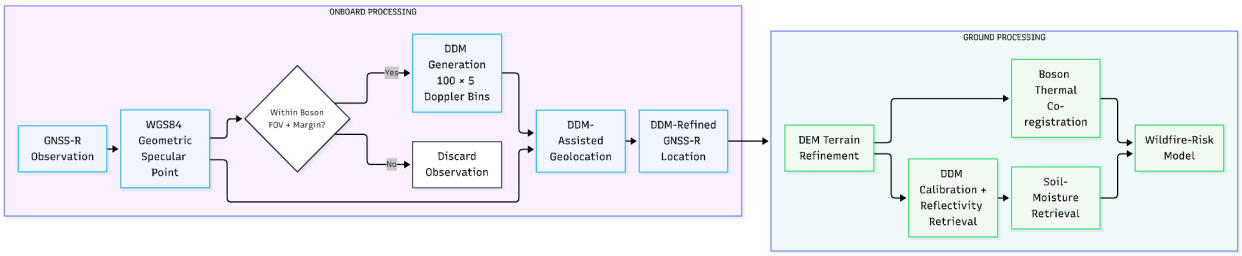


Figure 8. Geolocation flowchart describing both onboard and ground processing.

The DDM-refined location calculated onboard PyroSat will remain a preliminary geolocation because the onboard solution represents the Earth's surface using the WGS84 reference ellipsoid and does not account for local terrain elevation. This preliminary location is sufficient for determining whether a GNSS-R observation plausibly overlaps the corresponding Boson+ thermal frame. Following downlink, a Digital Elevation Model (DEM) will be incorporated to refine the observation location using the actual terrain surface. PyroSat's ground-based terrain refinement will be based on the approach demonstrated for CYGNSS land observations by Gleason et al. [5]. The DDM-refined onboard solution provides the center of a bounded geographic search region. Candidate latitude and longitude locations surrounding this preliminary solution are first represented on the WGS84 reference ellipsoid and are then displaced according to the corresponding terrain elevation obtained from the DEM.
Gleason et al. represent this terrain-adjusted candidate location using Equation 5 of that paper, shown here as Eq. (15):

$$S = S_{\mathrm{W84}}(\phi, \lambda) + \frac{S_{\mathrm{W84}}(\phi, \lambda)}{\|S_{\mathrm{W84}}(\phi, \lambda)\|}\Delta H \quad , \tag{15}$$

Where $S_{W84}(\phi, \lambda)$ represents the candidate position on the WGS84 ellipsoid at latitude ϕ and longitude λ, and ΔH represents the DEM-derived terrain elevation at that corresponding location. Eq. (15) therefore transforms each candidate from an idealized WGS84 surface position into a three-dimensional candidate located on the estimated physical terrain surface. For each terrain-adjusted candidate, the local terrain geometry can then be evaluated to determine which surface locations remain physically consistent with the bistatic GNSS reflection geometry. This allows the ground-processing algorithm to refine the preliminary onboard solution using the actual terrain rather than assuming that the reflection occurred directly on the WGS84 ellipsoid.

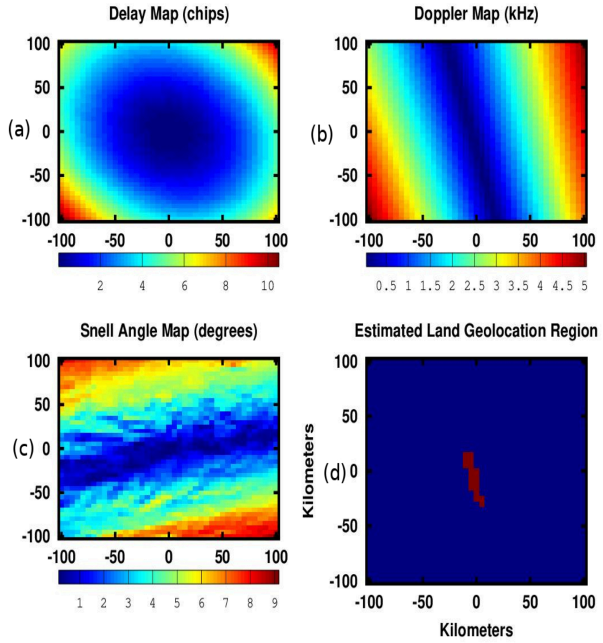


Figure 9. Conceptual terrain-based refinement of a GNSS-R candidate reflection [5]

The resulting terrain-aware solution provides a refined estimate of the GNSS-R reflection location. The final spatial characterization must additionally account for the fact that received reflection originates from a finite surface region rather than a single point. For coherent GNSS-R, this region can be approximated using the first Fresnel zone surrounding the refined specular location, with the effective footprint additionally influenced by the motion of the specular point during measurement interval. The resulting GNSS-R measurement will then be spatially compared with the independently georeferenced Boson+ thermal image. Boson pixels contained within the corresponding GNSS-R footprint will be aggregated to produce a thermal measurement representative of approximately the same geographic area. Following this terrain-aware geolocation and spatial co-registration, the GNSS-R measurement can undergo surface reflectivity and soil-moisture retrieval, with the spatially matched soil-moisture and thermal measurements subsequently serving as inputs to PyroSat's wildfire index.

Instrument Calibration & Soil Moisture Retrieval

Following the final geolocation of the GNSS-R observation, the measured DDM must be calibrated before it can be used for soil-moisture retrieval. The DDM contains only the calibrated reflected power distributed across delay and Doppler, $P(\tau, \upsilon)$, not soil moisture directly. PyroSat will then use the measured reflected signal, receiver calibration information, observation geometry (e.g., a digital elevation model), and antenna characteristics to further estimate the surface reflectivity for each observation. It is important to note that this calibration will be done on the ground. The transmitted observation will be this measured DDM.

Once the observation is transmitted to the ground, a calibration methodology based on the approach presented in Gleason et al. [5] for CYGNSS land observations will be investigated. For a predominantly coherent reflection, the relationship between detected power and surface reflectivity can be represented using this coherent model modeled from the Friis transmission equation:

$$P^{\mathrm{coh}} = \frac{YG_r\lambda^2\Gamma}{(4\pi)^2(R_R + R_T)^2} \tag{16}$$

Where p^{coh} is the calibrated coherent reflected power, and Y is the GNSS transmitter effective isotropic power (EIRP), G_r is the receive-antenna gain in the direction of the reflection, λ is the GNSS signal wavelength, R_T is the distance from the GNSS transmitter to the reflection point, R_R is the distance from the reflection point to PyroSat, and Γ is the surface reflectivity [5].

Because PyroSat's objective is to determine surface reflectivity from the measured reflected power, this relationship can be rearranged to solve for Γ:

$$\Gamma = \frac{P^{\mathrm{coh}}(4\pi)^2(R_R + R_T)^2}{YG_r\lambda^2} \tag{17}$$

The required quantities can be obtained from the calibrated GNSS-R observation and its metadata. The coherent reflected power p^{coh} is determined from the calibrated DDM, while R_T and R_R are calculated using the final reflection point candidate calculated from the refinement process and the known positions of the GNSS transmitter and PyroSat. The GNSS wavelength λ is known and is about 19 cm, derived from the GNSS-R antenna's frequency of 1575.42 MHz. The transmitter (EIRP) Y is obtained from the GNSS transmitter model, and G_r is determined from the characterized receive-antenna pattern and observation geometry. Together, these quantities allow an estimate of surface reflectivity to be produced for each suitable observation.

It is important to note that Eq. (16) represents the coherent component of the reflected GNSS signal. GNSS reflections from land may contain contributions from both coherent and incoherent scattering. Coherent scattering occurs when signals reflected from the surface maintain most of their phase coherence, combining strongly near the specular reflection direction, and is generally observed on smooth surfaces. As surface roughness increases, scattering can become increasingly diffuse or incoherent, causing reflected signals to combine with different phases and cover a broader range of directions. Consequently, these scattering and surface characteristics must be considered when interpreting GNSS-R data for soil moisture retrieval.

Following the calculation of surface reflectivity, PyroSat will initially investigate an empirical soil-moisture retrieval based on the relationship between transient changes in GNSS-R

reflectivity and temporal changes in surface soil moisture. Chew and Small demonstrated this relationship using CYGNSS reflectivity and SMAP soil moisture measurements. Rather than assuming that a single absolute reflectivity value corresponds to a unique soil-moisture value, repeated observations across a geographic region can be used to determine how reflectivity changes over time [31]. For a given geographic region, the change in reflectivity can be expressed relative to reference reflectivity as,

$$\Delta\Gamma(t) = \Gamma(t) - \Gamma_{\text{ref}} \tag{18}$$

Where $\Gamma(t)$ is the calibrated reflectivity measured by PyroSat at time *t,* and Γ_{ref} represents a reference reflectivity for that region. The corresponding change in soil moisture can then be related empirically to the change in reflectivity:

$$\Delta SM(t) = a\,\Delta\Gamma(t) \tag{19}$$

Where $\Delta SM(t)$ is the change in surface soil moisture and *a* is an empirically determined sensitivity relating changes in GNSS-R reflectivity to changes in soil moisture. The coefficient *a* will not be assumed prior to the mission but can instead be determined by comparing repeated PyroSat reflectivity measurements with spatially coincident reference soil-moisture measurements, such as those from SMAP and other satellites. Once a reference soil-moisture value is established, the retrieved soil moisture can conceptually be expressed as:

$$SM(t) = SM_{\text{ref}} + \Delta SM(t) \tag{20}$$

Where SM_{ref} is the reference soil moisture corresponding to Γ_{ref}. Thus, the coherent reflection model is first used to convert the calibrated GNSS-R measurement into surface reflectivity, while the empirical retrieval relates changes in that reflectivity to changes in soil moisture.

Because GNSS-R reflectivity is also influenced by surface roughness, vegetation, incidence angle, and scattering regime, the performance of this empirical retrieval will be evaluated across different land-surface conditions. If these effects introduce significant errors, additional surface and vegetation can be incorporated into subsequent versions of the retrieval algorithm. The resulting PyroSat soil-moisture estimates will be validated against SMAP and available environmental measurements.

Thermal Data Processing

The Boson+ outputs each frame as a 640x512 array of 16-bit digital counts, about 655 kB per frame, which the radiometric configuration can convert to scene temperature. Onboard processing will be kept to a minimum. The payload computer will be in control of the camera, running its flat-field correction at the scheduled times away from the science captures (the camera briefly holds the image during a correction), and attaching each frame's capture time,

position, velocity, attitude, and internal camera temperature. Frames will be stored uncropped so that all radiometric correction and geolocation can be done on the ground, using better terrain, atmosphere, and calibration data than are available onboard.

On the ground, raw counts are first converted to brightness temperature using the on-ground calibration and the in-orbit blackbody measurements described above, with corrections for internal temperature drift, pixel non-uniformity, and inaccurate pixel data. Brightness temperature is not yet land surface temperature (LST). From 400 km, the signal has been altered by the atmosphere, which weakens the surface emission and adds its own, and by surface emissivity, which sets how efficiently the ground radiates. Because the Boson+ is a single broadband imager (about 7.5 to 13.5 μm), split-window methods are not available. PyroSat will correct for the atmosphere using reanalysis profiles run through a radiative transfer model, integrated over the Boson+'s full spectral response (about 7.5 to 13.5 μm) because it is a single broadband imager. Emissivity will be taken from a land product with time-varying emissivity, converted to an effective broadband value over the same spectral response. Static products such as the ASTER Global Emissivity Dataset, which averages observations from roughly 2000 to 2008, are used only as a fallback [51]. Because a single thermal band cannot reliably separate cloud from cold ground, cloud screening on the ground will combine departure from the expected clear-sky temperature in reanalysis, an independent cloud mask from other satellites, spatial texture tests, and consistency with earlier clear-sky observations. Pixels that fail are removed before LST is used, and the screening is deliberately conservative.

Finally, the corrected LST pixels that are free of clouds are averaged over each soil moisture grid cell, and each average is stored with its local solar time. These cell values feed the thermal stress score R_T in equation (1). R_T combines two indices, the Crop Water Stress Index (CWSI) and the Normalized Difference Temperature Index (NDTI), which pair PyroSat's LST with air temperature and modeled hot and cool surface limits from outside weather and energy balance data [29]. Because both indices start from the same LST, any bias in the temperature measurement moves both together, which makes LST accuracy the most important requirement for the thermal payload. The risk analysis found that LST error near 1 K keeps R_T useful, while error near 5 K makes it nearly uninformative [29].

Spatial Co-registration / Boson overlap

The risk model scores each ground cell from two inputs that come from different instruments: soil moisture from GNSS-R and land-surface temperature from the Boson+640. Each input is only useful if both describe the same patch of ground, therefore this section defines how the two data streams are located, compared, and paired.

The Boson looks straight down and sees a frame of about 24° by 19.5°, which is roughly 170 km by 137 km on the ground from a 400 km orbit, with a pixel of about 270 m. A GNSS-R specular point is where a GPS signal bounces toward the satellite, and it is almost never under the satellite. The PyroSat team ran a Monte Carlo Simulation of 1,000,000 random GPS

geometries per altitude on the WGS84 reference ellipsoid, at 400, 500 and 600 km, with PyroSat on a 51.6° inclined orbit. aThe simulation code was written with AI assistance, and the team checked its outputs. The median specular point sits 58° to 61° off the nadir and 830 to 1,140 km from the sub-satellite point. The table below shows how often a reflection lands inside the Boson frame at the same instant.

Reflections Considered	**Share inside the Boson Frame**
All reflections, any angle	About 1.65% to 1.7%
Reflections within 30° of nadir	About 16%

Because of this, simultaneous overlap is a minority case, and the science product cannot depend on it. Co-registration happens on the ground segment, not in the sky. These figures assume ideal conditions, and the true figures depend on the final orbit, the GNSS satellite geometry over the mission and the attitude solution.

Both datasets are first geolocated independently. For each Boson frame, the four corners and every pixel are projected to the ground from the timestamped orbit and attitude solution. For each GNSS-R delay-Doppler map, the specular point is solved on the WGS84 ellipsoid from the GPS and spacecraft positions at the same timestamp [5]. Both are then binned onto one fixed latitude and longitude grid, whose cell size is set by the model (the model scores each cell on a monthly basis). A cell is scored when it holds at least one valid GNSS-R reflection and at least one valid thermal observation in the same month, and the two are paired by cell and date rather than by instant. The full thermal frame is stored and binned without cropping, so any cell the GNSS-R track later visits can be paired.

A 1 km GNSS-R footprint (the design target, which is about 4 Boson pixels of 270 m, or 1080 m) covers roughly 4 by 4 Boson pixels, so thermal pixels are averaged to the GNSS-R cell with a median and a spread, and the spread is kept as a quality flag. Cells with cloud-contaminated or sensor-saturated pixels are dropped before averaging.

A specular point 1,000 km from the ground track is reached by the Boson roughly 140 seconds before or after the reflection, at a ground-track speed of about 7 km/s. Soil moisture and land-surface temperature change little over minutes compared with the monthly scoring window, so a same-pass or same-day pairing window is acceptable. The pairing window is a ground-segment parameter and can be tightened after launch.

If the downlink budget forces a choice, the payload computer can keep only reflections whose specular point falls within the Boson frame, or within a margin of it. This raises the paired fraction at the cost of GNSS-R volume. It is held in reserve as a software setting, not a baseline requirement.

Co-registration quality is set by the attitude knowledge requirement in the ADCS section. At the 60 arcsecond design accuracy of the baseline star tracker the attitude contribution alone is

about 115 m at 400 km, under half a pixel; the 0.04 degree (144 arcsecond) requirement corresponds to about 280 m, about one pixel.

Payload control & fault handling

PyroSat inherits the SquidSat approach of a bus that can power, command and reset the payload without trusting it. The payload is treated as an unreliable guest: the bus is always able to cut its power, and the payload is always able to report that it is unwell.

The payload is fed through the SquidSat electrical power system load switches. Each rail is limited to 2.5 A, with a combined bus limit of 8 A [1]. Each payload element (Boson and payload computer, GNSS-R receiver, S-band radio) sits behind its own switch so a fault in one can be isolated without losing the others. The bus commands switches over its own Controller Area Network (CAN) node, and a switch also opens on over-current or latch-up detection without waiting for a command.

The payload runs in four states: Off, Standby (computer alive, sensors off), Science, and Downlink. The S-band radio draws about 10.8 W typical and up to 15 W peak [17], which would collapse the roughly 7 W orbit-average load budget if it overlapped with GNSS-R or camera operation. The payload computer enforces the exclusion in software, and the bus enforces it again by refusing to close both switches at once as shown in the communications risk table. A state change is requested by CAN command and acknowledged before the next step begins. The payload instruments and the S-band antenna are on the nadir face, and the solar panels are on the zenith face. In Sun pointing, the bus points the zenith face at the Sun, so the nadir face points away from Earth and neither payload nor the S-band antenna can see the ground. The bus therefore holds the payload in Standby or Off for as long as the satellite is in Sun pointing, and returns to nadir, or to the tilt needed for a ground pass, before Science or Downlink resumes. Sun pointing is a bus attitude mode, not a payload state.

The payload follows the SquidSat state machine, which has three states: Initialization, Nominal and Safe [1]. After release from the ISS and a set delay, the on-board computer wakes, checks the health of the other bus boards in Initialization, and begins beaconing a heartbeat. The ground commands the transition to Nominal once it has verified that all statuses are healthy. Payload operations take place only in Nominal, and only when scheduled or commanded from the ground. An operation starts only when the battery is above its threshold, the scheduled time has arrived and no significant error is active. While it runs, the EPS relaxes the limits on the payload's power supply, and the on-board computer restores them when the payload reports over CAN that the operation has finished. Within Nominal, the payload runs in the four states described above. Errors are ranked from 0 (most severe) to 2, and an error ranked 0 or 1 puts the spacecraft in Safe. In Safe, any running operation is stopped, the bus powers the payload off, and the heartbeat slows to conserve power. The spacecraft stays in Safe until the ground resolves the problem. The ground then brings the payload back up in a fixed sequence, checking telemetry at

each step: computer, then Boson, then GNSS-R, then S-band only when a pass is scheduled. The three independent inhibits and the 45-minute RF silence after deployment keep the payload off until the dispenser requirements are satisfied [2, 2.3.7, 2.4.5].

Two layers apply. The payload computer runs a hardware watchdog that resets if the main loop stalls. The bus also expects a heartbeat of CAN at a fixed interval, and if three consecutive heartbeats are missed it power-cycles the payload rail. A payload can be power-cycled a limited number of times per orbit before the bus keeps it off and reports the lockout to the ground, which prevents a reboot loop from draining the battery.

The table below shows the faults the design can handle and its response. The spacecraft acts autonomously only to protect itself. After any major fault, restarting or resuming the affected function is done by ground command during a contact, so the spacecraft never assumes a recovery or enters a restart loop.

Fault	**Detection**	**Response**
Latch-up or over-current	Load switch current limit and current sensor	Switch opens, rail restarts after a hold-off, repeated events lock the element out
Payload computer hang	Hardware watchdog and CAN heartbeat	Reset, then bus power cycle
Storage failure	File system error or CRC mismatch	Stop writing, report fault, hold RAM buffer until a ground downlink command
Boson over-temperature or image loss	Housekeeping telemetry	Boson off, mode drops to Standby
Low battery or brownout	Bus power state	Payload to Off, science descoped
S-band fault	Radio telemetry	Radio off, stored data held until the ground commands a downlink
Over-temperature (S-band radio or any payload element)	Temperature sensors, radio telemetry	Payload off, spacecraft enters safe mode until the ground clears it
Reaction wheel fault or saturation	Wheel speed and current telemetry, momentum estimate	Faulty wheel switched off, magnetorquers desaturate the rest, attitude mode drops to detumble or Sun pointing. Science and S-band downlink are suspended, but command and

		telemetry continue over the UHF link, which does not depend on pointing, and the ground commands the recovery
Star tracker loss or invalid solution	Solution validity flag, residuals against the sun sensors and magnetometer	Science paused to Standby, ADCS switched off, command and telemetry continue over the UHF link until the ground restores the attitude solution
IMU, magnetometer or sun sensor failure	Range checks and cross-checks between sensors	Failed sensor excluded from the attitude filter, estimate continues on the remaining sensors, fault reported to the ground
Magnetorquer over-current or driver fault	Drive current sensor and over-current protection	Driver off, wheels hold attitude, desaturation and detumbling suspended. The driver is re-enabled only by ground command during a contact
ADCS microcontroller hang	Hardware watchdog and CAN heartbeat	Hardware watchdog resets the microcontroller once. If the heartbeat does not return, the bus switches the ADCS off, holds the payload in Standby and reports to the ground. The ADCS is restarted only by ground command during a contact
Loss of attitude control or high body rates	Angular rate above threshold or attitude filter divergence	Payload off, magnetorquer detumble, then Sun pointing for power. Return to nadir and resuming Science happen only by ground command during a contact, after the attitude solution is confirmed valid

In any safe mode, the S-band radio is off and the spacecraft communicates only over the UHF link. The UHF antenna does not need pointing, so the ground can still receive telemetry and send commands while the satellite is tumbling or Sun pointing. The S-band link is brought back only by ground command after the fault is cleared. In a power or thermal emergency, loads are shed in the order set out in the Power section, and the UHF link is the last load shed.

The payload computer and storage are commercial parts in low earth orbit. Radiation upsets are handled by watchdog resets, CRC checking on stored data and a power cycle if a persistent latch-up is detected.

Spacecraft Bus

PyroSat Bus Overview

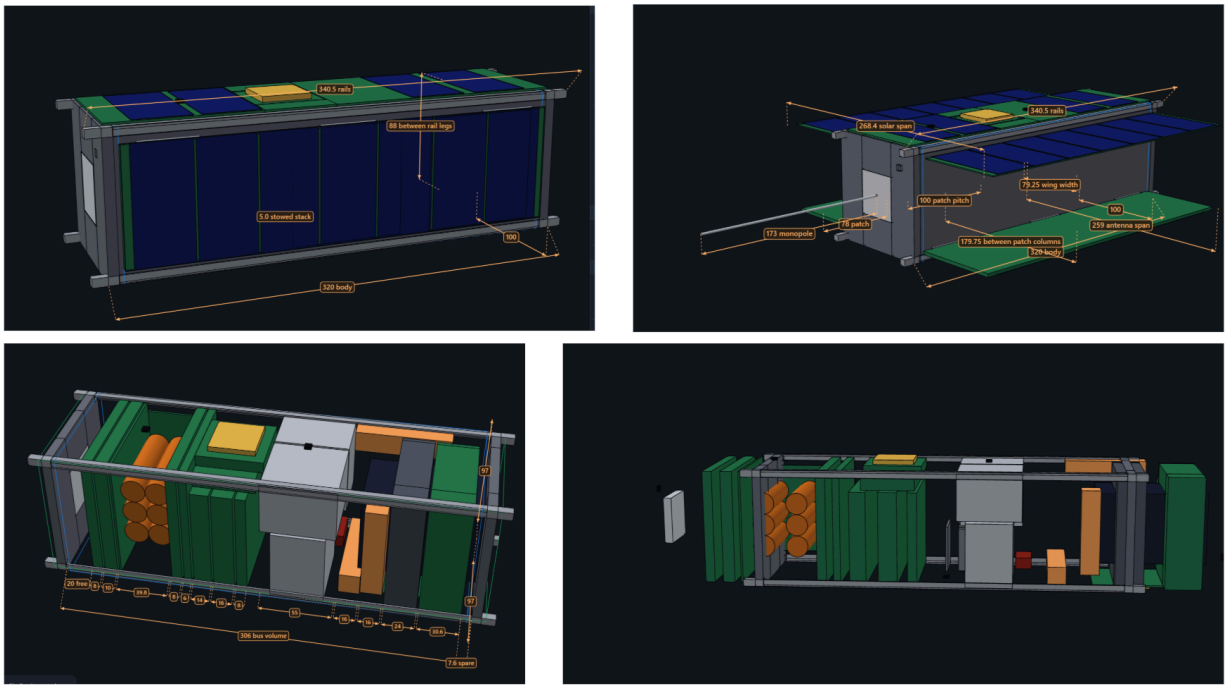


Figure 10. PyroSat 3U concept with the solar wings stacked on top of the antenna wings. The 4 panels show the stowed and deployed configurations.

PyroSat's bus will be a mission-specific derivative formed from SDSL's existing SquidSat Bus. The specific improvements in storage, battery, and electronic configuration will be addressed to match the requirements set by PyroSat's payloads. The bus will be made from 6061 aluminum and anodized using a type-III hard anodizing process. PyroSat adopts SquidSat's hinge for its deployable wings [1] and changes the deployment angle to 90 degrees. Stowed, each wing lies flat against a cross-track side face, with the solar wing stacked on top of the antenna wing. On release, each wing rotates 90 degrees about its hinge and extends outward across the track. The solar wings end up facing zenith in a plane close to the zenith panel, and the antenna wings face nadir in a plane close to the nadir face, for a deployed span of about 268 mm. Deployment combines two mechanisms: a burn wire holds each wing closed through launch and releases it only on command, and a spring in the hinge supplies the torque that opens the wing once the wire is cut. The solar wings release before the antenna wings. The electrical system differs from SquidSat's and is described in Power Systems. PyroSat keeps SquidSat's 433 MHz GMSK radio (19.2 kbps, about 1 W RF output) for telemetry and commands, and as the fallback link whenever pointing is lost. That link cannot carry the science data. A single 655 kB Boson+ frame (5,240 kb) takes about 273 s at 19.2 kbps, so a 7-minute pass returns only about 1 MB, or

roughly 1.5 frames. The mission needs about 1 Mbps, or roughly 50 MB in a 7-minute pass. An additional S-band transceiver is therefore added for science downlink.

Bus Architecture

PyroSat builds on SquidSat, an open-source 3U CubeSat bus from the Sun Devil Satellite Laboratory with a published design and an existing set of boards, firmware and test procedures [1]. SquidSat retains much of the architecture of Coconut, SDSL's earlier 1U CubeSat, which was deployed from the International Space Station in 2026 [86, 87]. SquidSat's first mission is planned to fly multiple payloads [1], so PyroSat can build on its flight results. Starting from this proven foundation lets the team spend its limited time on the science payload, which is the part that is new. SquidSat was designed as a foundation that is easy to modify, and PyroSat extends it with capabilities the original design did not target: tighter attitude knowledge, more power from deployable solar wings to support the S-band downlink and Sun pointing, thermal accommodation for the payload, and deployable antenna wings on the nadir face. Each extension is a mission-specific modification, and this section records what is kept and what changes.

The SquidSat bus provides an RP2350 on-board computer running FreeRTOS with symmetric multiprocessing and a task scheduler called STEVE; an electrical power system and a 433 MHz GMSK deployable-antenna and telemetry board, both built around an STM32F103 running Zephyr; a battery of 72.2 Wh; switchable regulated rails at 3.3, 5 and 12 V, limited to 2.5 A per rail and 8 A in total; a CAN bus joining all boards; and a state machine that governs startup, operation, fault and restart [1]. The bus also defines the payload interface (CAN only, with local payload storage) and the deployment behavior required by the dispenser [1, 2].

The table below summarizes the changes by subsystem. Each is developed in its own section.

Subsystem	Inherited from SquidSat	Mission-specific change
Structure	3U frame, standard rails	Two pairs of deployable wings, mounting for the payload stack, which is about 2.8 cm longer than SquidSat's payload volume (see Structures and Mechanical Integration)
On-board computer	RP2350, FreeRTOS, STEVE	Added tasks for payload mode control and pass scheduling
Power	72.2 Wh battery, 2.5 A rails	Solar wings added to the body panels, maximum power point tracking (MPPT) configured for the new array's voltage and current, load-shed order tuned for the S-band radio, S-band transmit peak checked against the rail and battery limits, with the radio fed from the 12 V rail

Communications	433 MHz GMSK link	Retained for telemetry and commands; high-rate S-band link added in the payload
Attitude	Not sized for 0.04 degree pointing knowledge	Pointing hardware sized to the 0.04 degree knowledge goal (see ADCS)
Thermal	Passive	Unchanged. The payload's thermal control is part of the payload (see Thermal Management)
Payload interface	CAN only	High-speed link added inside the payload; the bus interface is unchanged

Where a change can be placed in the payload instead of the bus, it is. The bus interface, the state machine and the dispenser logic are the parts that were designed and tested for deployment safety, and they stay as they are. The payload carries its own high-speed link, storage and radio, so a payload fault cannot reach the safety-critical bus functions.

Sunlight falls on the solar panels, which charge the battery through the electrical power system. The regulators supply the on-board computer and the payload. The on-board computer coordinates modes over CAN. The payload collects thermal frames and GNSS reflections, stores them, and sends them through the S-band radio when a ground pass is scheduled. Commands and telemetry travel over the UHF link, which is independent of the payload and does not need pointing. Like the S-band link, it works only during a ground contact, so the UHF link is the primary path for uplink and for safety: it carries every command and the health telemetry used to recover the satellite after a fault.

The electrical power system and the ADCS will be built in house by SDSL, with the power system adapted from the SquidSat design and the ADCS developed by the SDSL ADCS team. Their parts budgets are in the Mission Cost section.

SquidSat Interface

PyroSat reuses the SquidSat bus as its host, so the payload must meet the interface that bus already defines. This section lists what the bus offers, what the payload must provide in return, and where the two do not yet match.

The bus offers three regulated rails at 3.3 V, 5 V and 12 V, each limited to 2.5 A, through switchable loads with a combined limit of 8 A [1]. The payload draws from these rails only; it does not connect directly to the battery. The load averaged over a day is about 4.9 W for the whole satellite, while in the worst orbit (science plus downlink) it is about 7.0 W, or about 3.6 W without ADCS (see the Operating Models and Power management table), and the S-band radio and the camera must not run at the same time as GNSS-R (see Payload Control and Fault Handling).

SquidSat connects its payload over CAN only [1]. CAN carries commands, mode changes, heartbeats, housekeeping, and a time reference. This sets a requirement on the payload: it must contain a microcontroller or processor that speaks CAN and answers the bus state machine on its own. The Boson and GNSS-R front end cannot connect to the bus directly. The payload computer is the NanoCom SDR MK3, which the on-board computer reaches over CAN (see Software Architecture).

The payload mounts to the bus stack through the standard connector board and standoffs, within the 3U envelope of 100 by 100 by 340.5 mm [2]. The nadir face carries the thermal-camera opening and the GNSS-R wings, so the payload stack is oriented with its lens end toward the nadir.

Every item that touches the dispenser rails, footswitches or deployment inhibits stays under bus control. The payload does not hold any inhibit and cannot energize itself before the 30-minute wait and 45-minute RF silence have passed [2, 2.4.4, 2.4.5].

The bus changes that PyroSat needs are tabulated in the Bus Architecture section.

Structures & Mechanical Integration

The structure is a 3U CubeSat frame machined from 6061 aluminum with a hard anodize finish on the surfaces that touch the dispenser, which the CubeSat Design Specification requires [2, 2.2.13]. The outer envelope is 100 by 100 by 340.5 mm [2], with L-section rails 6 mm wide and 1.5 mm thick in the current CAD, for a rail length of 340.5 mm and a body length of 320 mm. The 6 mm rail width follows the rail requirement for ISS deployment, which is the launch path this proposal commits to, so the CAD keeps 6 mm rather than the 8.5 mm minimum that the CubeSat Design Specification gives for other dispensers [2, 2.2.5]. The CubeSat Design Specification mass limit is 6.00 kg, but because PyroSat launches from the International Space Station the Nanoracks CubeSat Deployer (NRCSD) limits apply: a 3U may not exceed 4.80 kg, and the center of gravity must stay within 2 cm of the geometric center in X and Y and within 6 cm in Z (NRCSD interface document) [21], [2, 2.2.10, 2.2.11]. The rough mass estimate from the bill of materials is 3534.97 g at the low end and 4017.26 g at the high end, which is 3.53 to 4.02 kg and leaves 0.78 kg of margin at the high end to the 4.80 kg NRCSD limit [21]. The NRCSD document notes that a satellite with attitude control may be held to a stricter mass limit, which the team has not yet raised with the integrator [21].

The body frame has +X along the orbit track, +Y cross-track and +Z toward zenith, so nadir is the -Z face. The nadir face carries the camera opening. The two cross-track side faces (+Y and -Y) carry the deployable wings.

The payload stack mounts in the lower portion of the frame on standoffs, using the standardized mounting methods that SquidSat provides [1]. In the CAD the payload boards sit in one stack along the long axis of the body on standoffs at the SquidSat board pitch, with the GNSS receiver at the top of the stack directly under the zenith patch. The stack ends 7.6 mm

short of the +X end wall and leaves about 20 mm of free length at the -X end. Boards are held with a fixed pitch so their connectors align with the SquidSat connector board. The SquidSat paper gives a payload volume of about 2U (9.84 by 9.84 by 18.6 cm) [1]. In the launch CAD the payload section fills about 9.0 by 9.6 by 21.4 cm, about 2.8 cm longer than that volume. To make room, the bus electronics and battery are kept to about 7 cm at the -X end of the frame, leaving about 21 cm for the payload and ADCS hardware. The Boson and its lens sit at the nadir end of the stack so the lens line of sight needs no fold mirror, and the camera housing is mounted on isolating standoffs so it is thermally separated from the bus as explained in the Thermal Management section.

The nadir face has a circular opening sized to the 18 mm lens (24 degree horizontal field of view, computed from the 12 µm pitch and 640 pixel width in the Boson+ datasheet) with a baffle that keeps stray light and the bus structure out of the field of view [3]. The vertical field of view is about 19.5 degrees, so the opening is checked against both axes. The CAD shows the lens exposed in the nadir opening with no protective element in front of it. If a protective optical element is added to shield the lens, FLIR recommends germanium for Boson-class LWIR cameras because it transmits well in the long-wave band, at the cost of some transmission loss and a radiometric recalibration.

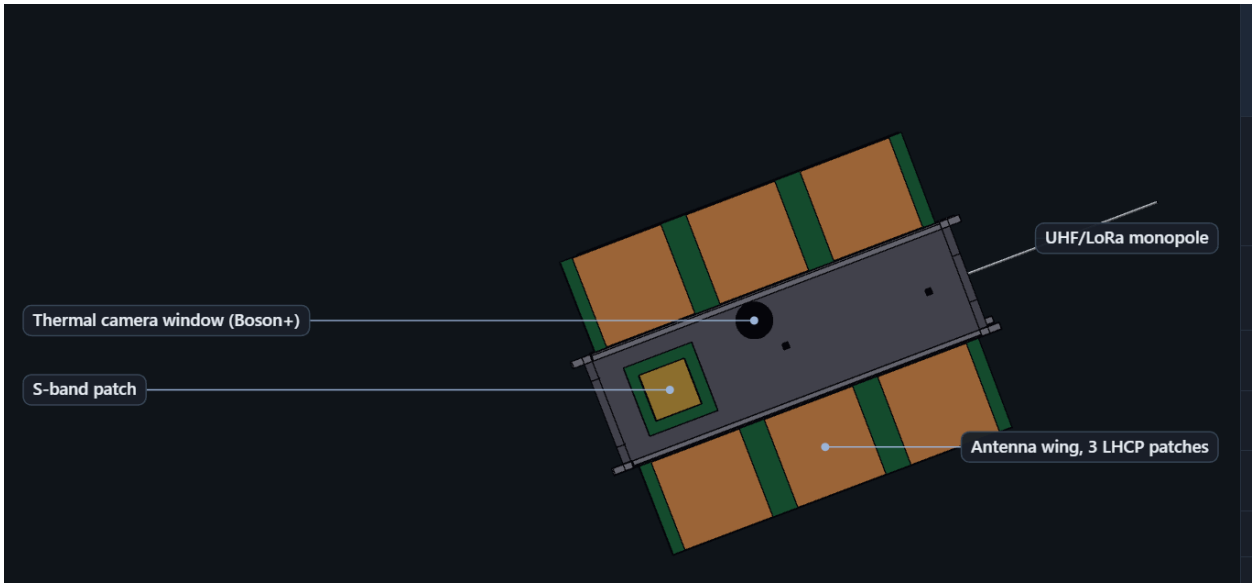


Figure 11. Nadir view of the PyroSat CAD with the antenna wings deployed, showing the Boson+ thermal camera opening, the S-band patch, the antenna wings with three LHCP patches each, and the UHF/LoRa monopole. The solar wings sit behind the antenna wings and are not visible from this side.

Figure 11 shows the nadir face of the CAD with the wings deployed. The GNSS-R antenna is six L1 left-hand circular patches arranged as three along-track on each of two wings, one wing on each cross-track side face. Each side face also carries a hinged solar panel, which sits behind the antenna wing in this view. When stowed, the wings are fitted to the 88 mm clear

band between the rails on each side face (100 mm less two 6 mm rail legs), so no part of a stowed wing or hinge crosses a rail. The key dimensions from the CAD are summarized in the table below.

Mechanical Envelope & Volume Budget

Item	Value
Clear band between rails, each side face	88 mm
Antenna panel width	79.25 mm
Solar panel width	80.5 mm
Patch size	78 mm, on a 100 mm along-track pitch
Hinge barrel radius	0.75 mm
Smallest clearance to a rail (88 mm band, from the panel width and hinge barrel)	3.0 mm
Stowed stack thickness	5.0 mm
Maximum allowed protrusion on side faces	6.5 mm
Deployed span	about 268 mm (plus or minus 134.2 mm)
Spacing between patch columns	about 179.75 mm, 0.94 L1 wavelengths
Interior cavity	97 x 97 x 306 mm
PyroSat payload section (board footprint, CAD)	about 90.2 x 96 x 214 mm, about 1,850 cm^3
SquidSat payload volume [1]	about 98.4 x 98.4 x 186 mm, about 1,800 cm^3

The 5.0 mm stowed stack is 1.5 mm below the 6.5 mm side-face protrusion limit of the CubeSat Design Specification Revision 14.1 [2, 2.2.3]. In the CAD nothing on any face stands more than 5 mm proud of the rail plane, which is the team's own limit. The exact keep-out zones will be checked against the dispenser drawing once the dispenser is chosen.

Wings are held closed by a burn-wire restraint and are released only after the dispenser timing has passed (30-minute wait before any deployment, 45-minute RF silence, three independent inhibits) [2, 2.4.4, 2.4.5, 2.3.8] and the bus state machine commands it. The solar wings release before the antenna wings. The deployed antenna plane sits 6.45 mm from the nadir face plane, per the CAD. Spring-loaded hinges open each wing once the burn wire is cut. Both

are needed: the burn wire restrains the wings through launch and ejection and releases them only on command, but it cannot push, so the spring in each hinge supplies the opening torque.

PyroSat uses SquidSat's existing deployment switch (footswitch) mechanism [1]. The CubeSat Design Specification requires at least one deployment switch that stays actuated while the satellite is in the dispenser, disconnects the power system, and sits at or below the level of the external surface that touches the dispenser, which keeps the satellite powered off until deployment [2, 2.3.2]. The switch placement will be checked against the specification's recommended locations and the dispenser drawing.

The 433 MHz telemetry antenna uses the SquidSat antenna board and its burn-wire release [1]. In the CAD the stowed 173 mm monopole sits in a 40 by 40 by 7 mm pocket in the -X end wall, flush with the end face, so it is on the end of the body and clear of the stowed wings and their hinges.

The bus will go through random vibration, thermal vacuum and deployment tests, as listed in the Verification and Test Plan and Testing tables.

ISS Safety & Compliance

PyroSat will comply with the CubeSat Design Specification, and the ISS integrator's requirements. The satellite is fully off while stowed, held off by deployment switches and a remove-before-flight pin. Three independent inhibits protect against unintended RF emission, and no deployment occurs for at least 30 minutes and no transmission occurs for at least 45 minutes after release. Battery safety follows NASA's ISS requirements for lithium-ion batteries. SquidSat's battery design already implements cell-level protection and independent inhibits [1], and PyroSat inherits it. The safety documentation for PyroSat's configuration, and re-verification of any power-system changes, is part of the development schedule.

Command & Data Handling

The Command & Data Handling system has three parts: the on-board computer that runs the satellite, the CAN bus that links its boards, and a separate high-speed link that moves science data inside the payload. Because high-speed links are vulnerable to radiation-induced errors, the payload link uses error detection and retransmission, and a space-oriented interface such as SpaceWire will be used where the connected hardware supports it. Keeping the third apart from the first two is the main design decision, because it lets a slow, reliable bus manage the satellite while a fast, replaceable link carries the bulk data.

Software Architecture

The flight software is split along the same line as the hardware: a small management layer that runs the satellite, and a separate processing layer that handles the science data. The

split keeps the code that must never fail simple, and it lets the heavy signal processing be developed and updated without touching the satellite's safety logic.

PyroSat keeps the RP2350 used by SquidSat as its on-board computer. The tasks assigned to it are management tasks, not signal processing: deciding which state the satellite is in, monitoring health, running system checks, managing the CAN bus, commanding subsystems and keeping track of what is stored. The chip suits this role. It has two ARM Cortex-M33 cores at 150 MHz, 520 KB of SRAM, a hardware watchdog, direct memory access, and the standard serial interfaces (SPI, I2C, UART, and USB), along with programmable I/O [4]. SquidSat already built its on-board software around this role, so most of the code is inherited [1].

The on-board computer's job reduces to selecting the satellite's state and commanding the right subsystems for it. The inherited states are Initialization, Nominal and Safe [1]. PyroSat adds payload modes under the regular state (standby, science and downlink) as described in Payload Control and Fault Handling, rather than adding new top-level states, so the tested bus logic stays unchanged. The on-board computer selects the payload mode based on the satellite’s health, such as power state, temperature and attitude, and moves the payload to a lower mode whenever health requires it. Returning to a higher mode after a major fault is done by ground command.

Direct memory access lets the RP2350 move blocks of data between memory and peripherals without the cores handling each byte. This is useful for telemetry buffers and for relaying payload file blocks, and it leaves the cores free to run the state machine and answer the watchdog.

The processing the payload does in orbit is deliberately light, and the refinement is done on the ground. The steps are as follows. First, the payload collects a delay-Doppler map and a Boson thermal frame together, recording position and time. Second, it computes the satellite's position and velocity from its GNSS receiver and the transmitting GNSS satellites from its broadcast ephemeris. Third, it predicts the specular point from both states on the WGS84 ellipsoid [5]. Fourth, the ground segment refines that estimate against the measured map and terrain data, which is more complex than is worth doing on board. Fifth, the ground associates the refined specular point with the thermal frame. Because the predicted specular point is not exact, the whole thermal frame is kept instead of being cropped around it; how much margin is needed is a research question that depends on the accuracy of the measured map. The downlink contains the delay-Doppler map, the predicted and refined specular points, position, velocity and time, the GNSS satellite information and the attitude.

The on-board computer schedules payload operations around the orbit: land-mask windows for capture, power constraints for the S-band transmitter, and ground-pass windows for downlink. Schedules are uploaded over the 433 MHz command link and held in a circular buffer on the on-board computer, so a missed pass does not stop the mission.

The baseline freezes flight software at delivery, and on-orbit updates over the 433 MHz link are an option to decide at the design review. Before delivery the team tests the software on a

hardware-in-the-loop bench, with the flight boards on the CAN bus and simulated sensors and ground commands.

Power Systems

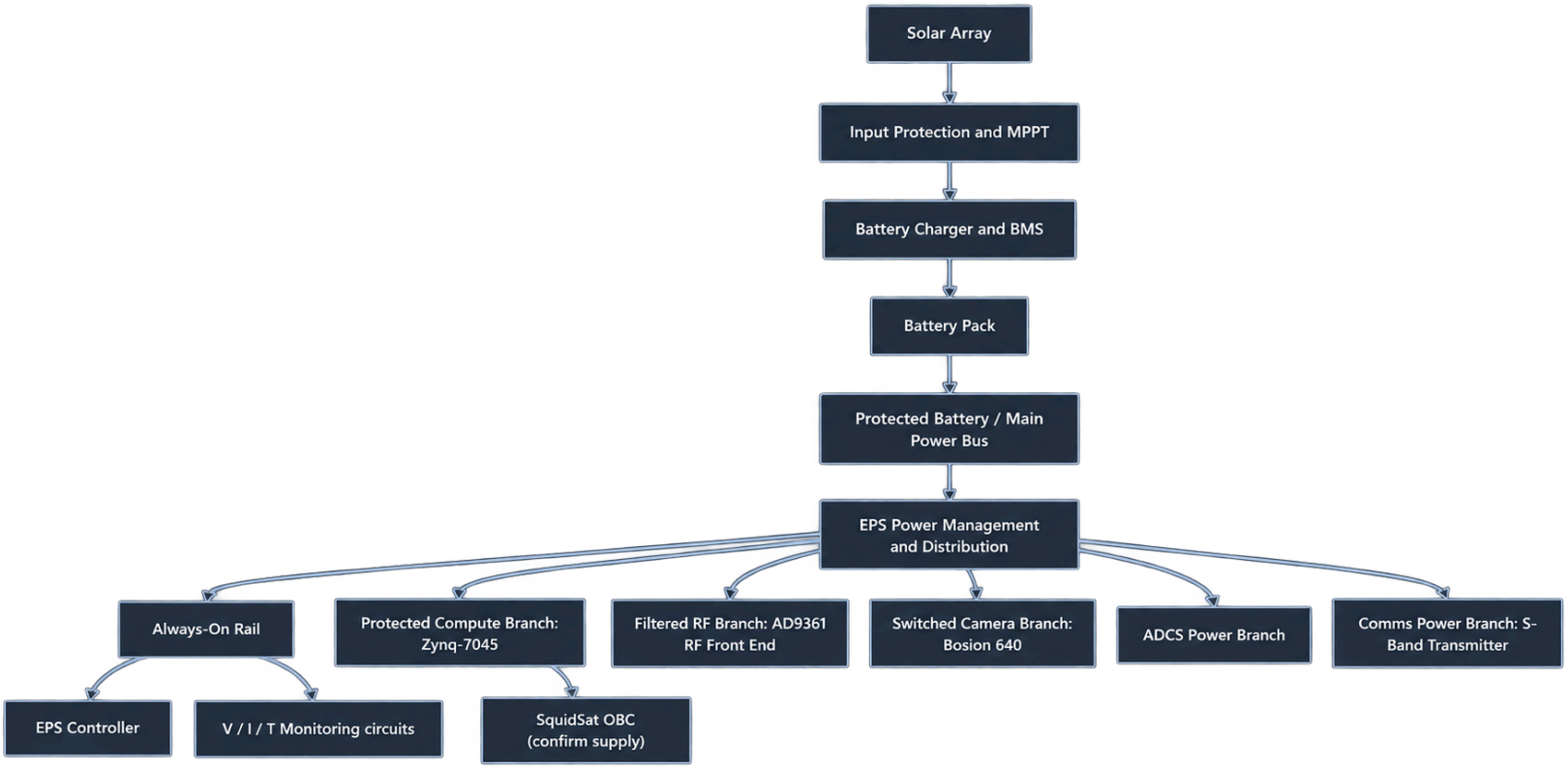
Overview & requirements

The PyroSat electrical power system (EPS) generates power from the solar array, stores energy in a lithium-ion battery pack, regulates it to the rails each subsystem needs, and protects the satellite and its payload from faults. It is sized for the ISS orbit and for the payload's two heaviest loads, GNSS-R processing and thermal imaging, which together with the S-band transmitter set the peak demand. The EPS meets the following requirements.

1. Daily-average generation shall be at least 1.25 times the daily-average load at the worst-case solar beta angle, using Sun pointing for part of the sunlit time as described in the Solar Array section.
2. The EPS shall deliver at least 18 W peak (15 W load + 20% margin) for the duration of the downlink event.
3. The battery shall carry the satellite through the longest eclipse, about 36 minutes, at no more than 30% depth of discharge in any permitted mode.
4. Payload, S-band, and deployment loads shall each be individually switched, current-limited, and protected against latch-up.
5. The satellite shall comply with CubeSat Design Specification and the ISS deployment safety requirements.
6. The EPS shall shed loads autonomously and enter a safe mode at a low state of charge, and shall recover from a fully discharged battery using solar power alone.
7. The EPS shall provide voltage, current and temperature telemetry for every rail, battery, and array.

PyroSat EPS Architecture

Figure 12 shows the PyroSat power architecture overlaid on the SquidSat Bus. The solar array, input protection and MPPT, battery charger and BMS, and protected main bus feed a power management and distribution stage. That Stage supplies an always-on rail(EPS controller and OBC monitoring), a filtered RF branch(RF front-end and AD9361), a protected compute branch(Zynq-7045), a switched camera branch(Boson+ 640), an ADCS branch, and a communications branch (S-Band transmitter). Voltage, current, and temperature monitoring report to the EPS controller, which communicates with the SquidSat OBC over CAN.


Solar Array
Input Protection and MPPT
Battery Charger and BMS
Battery Pack
Protected Battery / Main Power Bus
EPS Power Management and Distribution
Always-On Rail
Protected Compute Branch: Zynq-7045
Filtered RF Branch: AD9361 RF Front End
Switched Camera Branch: Bosion 640
ADCS Power Branch
Comms Power Branch: S-Band Transmitter
EPS Controller
V / I / T Monitoring circuits
SquidSat OBC (confirm supply)

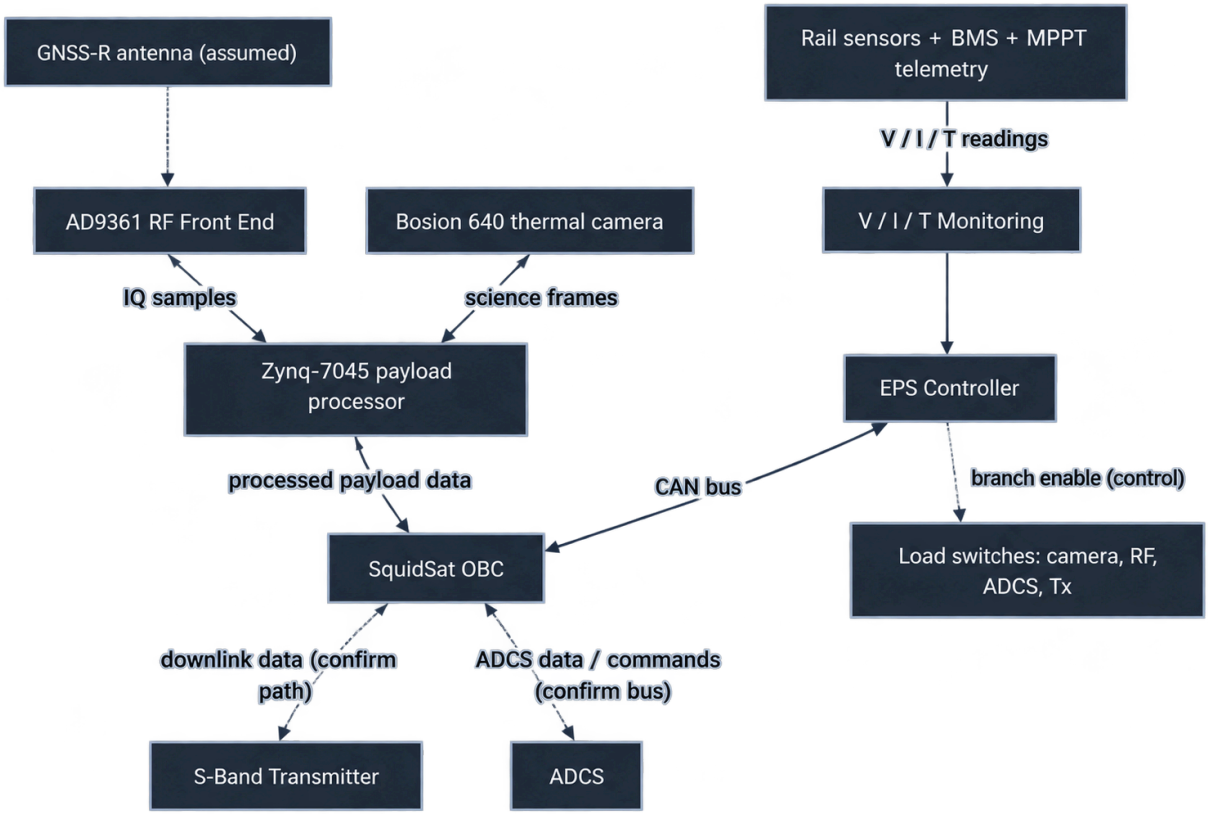

GNSS-R antenna (assumed)
Rail sensors + BMS + MPPT telemetry
V / I / T readings
AD9361 RF Front End
Bosion 640 thermal camera
V / I / T Monitoring
IQ samples
science frames
Zynq-7045 payload processor
EPS Controller
processed payload data
CAN bus
branch enable (control)
SquidSat OBC
Load switches: camera, RF, ADCS, Tx
downlink data (confirm path)
ADCS data / commands (confirm bus)
S-Band Transmitter
ADCS

Figure 12. Proposed EPS architecture for PyroSat

Beyond the retained SquidSat baseline, the PyroSat EPS adds or changes the following.

1. **Array interface:** the documented two input, LTC3119 based maximum power point control is retained unless the generation model shows a need for additional or per panel regulation. Blocking diodes and fusing protect each string.
2. **Distribution:** every payload , S-band, camera, heater, and deployment channel is individually switched with a current limit. A latch-up detector on the payload and storage rails trips on overcurrent and automatically power-cycles the load, which is the mitigation named in the GNSS-R risk assessment.
3. **Independent supervisor:** the documented STM32F103 controller acts as the supervisor, with a watchdog, undervoltage lockout , and the power rules listed under Operating Models and Power management all running independently of the OBC.
4. **Deployment drivers:** redundant burn-wire drivers release the antenna array and the solar panels.
5. **Cold start:** the EPS can restart from solar power alone with a depleted battery

Solar Array

PyroSat proposes 18 AZUR Space 3G30C triple-junction solar cells from DHV Technology, equal to nine DHV-CS-10 two-cell panels. Seven cells go on each of the two deployed side wings and four go on the body zenith panel, inside the fixed 300 x 80.5 mm wing and 316 x 88 mm panel sizes of the launch CAD. Each DHV-CS-10 incorporates two series-connected AZUR Space 3G30C triple-junction solar cells designed for space applications, with a beginning-of-life conversion efficiency of approximately 30% [33]. The solar array will provide electrical power to the spacecraft during illuminated portions of the orbit while excess generated energy will be stored within the spacecraft battery for use during eclipse and periods of high power demand.

The nine panel equivalents provide a combined active solar-cell area of approximately

$$A_{total} = 9(0.006036\, m^2) = 0.054324\, m^2. \quad (21)$$

The maximum beginning-of-life electrical power available from the solar array can first be estimated under ideal illumination conditions. Following the solar-array sizing methodology in *Space Mission Engineering: The New SMAD,* solar-cell power generation depends primarily on the incident solar flux, active solar-cell area, cell efficiency, and solar incidence angle [34]. The ideal maximum power can therefore be represented as

$$P_{max} = S \cdot A_{total} \cdot \eta, \quad (22)$$

Where S is the solar irradiance, A_{total} is the active solar-cell area, and η is the solar-cell conversion efficiency.

Using $S = 1361\ W/m^2$, $A_{total} = 0.054324$, and $\eta = 0.3$, the theoretical maximum beginning-of-life array output is $P_{max} = (1361)(0.054324)(0.3) \simeq 22.2W$

The $22.2W$ value represents an ideal beginning-of-life instantaneous maximum and does not represent the power continuously available to PyroSat. Actual solar-array generation varies with the angle between each panel and the Sun and becomes zero during eclipse. The instantaneous power generated by the array can therefore be represented as,

$$P_{\text{array}}(t) = S \cdot \eta \sum_{i=1}^{8} A_i \max\left[0,\ \hat{n}_i(t) \cdot \hat{s}(t)\right] \tag{23}$$

where A_i is the active area of panel i, $\widehat{n_i}$ is its outward normal vector, and $\widehat{s}$ is the direction toward the Sun. The dot product represents the cosine of the instantaneous solar-incidence angle, while the maximum function prevents a panel facing away from the Sun from contributing power.

For preliminary power budgeting, PyroSat assumes a 93-minute orbital period, rounded from the approximately 92.6-minute period of the assumed 400 km circular orbit. At a beta angle near 0°, PyroSat experiences a near-maximum eclipse-duration condition, with approximately 61% of the orbit illuminated. This corresponds to about 56–57 minutes of sunlight and 35–36 minutes of eclipse per orbit. During eclipse, the solar array produces no electrical power and the satellite must operate using energy stored within its battery.

Table A: Generation against load, per 93-minute orbit

Items	**Orbit-average power (W)**	**Energy (Wh)**	**Energy @ 80% EPS (Wh)**	**Margin vs. load (Wh)**
Nadir pointing, derated, β = 0°	6.0	9.3	7.44	−3.27 to −3.53
Nadir pointing, derated, β = 30°	5.2	8.06	6.45	−4.26 to −4.52
Nadir pointing, derated, β = 60°	3.0	4.65	3.72	−6.99 to −7.25

Nadir pointing, derated, β = 75°	1.6	2.48	1.98	−8.73 to −8.99
Sun pointing all sunlit time, derated (upper bound), β = 0° to 75°	11.5–18.9	17.83–29.3	14.26–23.44	Positive at every β

Table A compares the orbit-average solar-array generation at beta angle β with the satellite power loads calculated in Table D (Operating Models and Power management). These generation estimates are based on the DHV solar-cell specifications and the preliminary PyroSat attitude and orbital model; however, they remain preliminary and will require hardware characterization and further orbital analysis to improve their accuracy. For power-budget comparison, PyroSat currently assumes an overall EPS conversion efficiency of 80% or 20% conversion loss. The current power load is approximately 10.71–10.97 Wh per 93 minute orbit, corresponding to an average load of 6.9–7.1 W. Under these assumptions, continuous nadir pointing does not produce sufficient usable energy to close the orbital energy budget across the evaluated beta angles, making periods of Sun pointing necessary for energy operation. The all-sunlit-time Sun pointing case shown in Table A represents the upper bound rather than an expected case. The required fraction of each illuminated period devoted to Sun pointing will therefore be determined from the final attitude-dependent generation model, operational schedule, and required power margin.

The cell layout keeps the launch CAD dimensions unchanged. DHV still has to confirm that it can build wings and a zenith panel at these sizes and at the stowed stack thickness, and its price for them is not yet quoted. The 80 mm cell width leaves only about 0.25 mm of edge margin on the 80.5 mm wing, so DHV may have to trim the cells slightly. In the CAD, zenith sun sensor 2 has been moved to x = 24 mm so that it no longer sits over a body cell.

Further, Table A demonstrates that continuous nadir pointing does not close PyroSat's orbital energy budget under the current assumptions. At $\beta = 0°$, the most favorable evaluated nadir-pointing case the array provides approximately 7.44 Wh of usable energy after the assumed 80% EPS efficiency (Table A), compared with an orbital load approximately 10.71–10.97 Wh. This produces a deficit of approximately 3.27–3.53 Wh per orbit, or approximately 31–32% required energy. The deficit increases at higher beta angles. However, the modeled Sun-pointing cases produce sufficient energy to close the budget at each evaluated beta angle. PyroSat will therefore use a dedicated Sun-Pointing period during expected operations to recover the energy that cannot be supplied during continuous nadir pointing. The required duration will be refined as the operational, attitude model, and EPS efficiency estimations are matured

Energy Storage

The documented SquidSat pack consists of six Jauch 3350-mAh 18650 Li-ion batteries. Each cell has nominal voltage of 3.6 V and typical capacity of 3350 mAh, giving the 2S3P pack a nominal capacity of 72.4 Wh, consistent with SquidSat's stated 72.2 Wh capacity.

Parameter	Value	Notes
Energy store requirement	At least 30 Wh	Substantial capacity relative to the expected loads
Worst case eclipse energy (standby)	About 2 Wh	About 3% of nominal pack capacity
Worst case eclipse energy, science permitted in eclipse	About 6.9 Wh	About 9% of nominal pack capacity
Battery lifetime	Requires testing	

Each Jauch cell has a PCM that provides hardware level overcharge, overdischarge and overcurrent protection.

Protection function	Threshold
Overcharge detection	4.30 ± 0.05 V
Overdischarge activation	2.50 ± 0.1 V
Overdischarge release	2.9 ± 0.1 V
Overcurrent protection range	7-15 A
Protection layer	Cell level

Charging temperature must also be carefully controlled. The Jauch cells are specified for charging only between 0 °C and +45 °C, with the allowable charging current reduced to 335 mA between 0°C-10°C, 670 mA 10°C-20°C, and 1000 mA 20°C-45°C. [39] SquidSat's battery thermal-control and charge-enable logic must therefore prevent charging below the manufacturer's 0°C limit and maintain the cells within their allowable charging range.

Payload Power Interface (RF Board)

The payload receives power on a dedicated, switchable 5V rail with a current limit and latch-up detector, within the documented limit of 2.5A per rail per payload. The RF low-noise amplifiers are powered from the separate low-noise 5V rail, a linear regulator fed by a filtered 5V line. Switching converters run at fixed, synchronized frequencies chosen so their harmonics

avoid the 1575.42 MHz GNSS L1 band, and they are shielded and filtered. The Boson+ 640 has its own switched rail, with typical and maximum draw to be measured (FLIR lists approximately 1.0 to 1.3 W for continuous measurement). Payload rails are sequenced so the Zynq processing board boots before the RF and camera loads are enabled.

Power System Breakdown

Because instantaneous power consumption may exceed instantaneous solar-array generation particularly during GNSS-R processing and S-band communication, the EPS will rely on the battery to supply short-term duration peak loads. The battery must also provide sufficient energy to operate required systems through the approximately 35-36 minute maximum preliminary eclipse period. EPS sizing will therefore be based primarily on the orbital energy balance,

$$E_{generated/orbit} \geq E_{consumed/orbit} + E_{margin}. \quad (24)$$

Rather than requiring the solar array to continuously exceed the instantaneous power demand. The table gives the loads that matter and what they cost in energy per orbit, using the 93-minute orbit.

Table B: Power loads breakdown per component

Component	**Operating Power**	**Operation**	**Assumed Time / 93-min Orbit**	**Energy / Orbit**
NanoCom SDR MK3	5.46 W [35]	Science	10 min	0.91 Wh
Boson+ 640	1.0–1.3 W [3]	Science	10 min	0.17–0.22 Wh
Nadir RF-FE A	~0.25 W*	Science	10 min	~0.042 Wh
Nadir RF-FE B	~0.25 W*	Science	10 min	~0.042 Wh
OEM719	~0.90 W [90]	Continuous	93 min	~1.40 Wh
piPATCH-L1E1/ FM	0.066 W [32]	Continuous	93 min	~0.10 Wh
RP2350 OBC	~0.10 W [4]	Continuous	93 min	~0.16 Wh
ADCS	3.13 W [17]	Earth / Sun facing	80 min	4.17 Wh
ADCS	3.71 W [17]	GS-pointing	10 min/pass	0.618 Wh

ADCS	5.5 W [17]	Transition	3 min	0.275 Wh
SRS-4 S-band Transceiver	10.8 W [37]	Downlink at 33 dBm	10 min/pass	1.80 Wh
SRS-4 RS-422 interface	0.06 W [37]	Downlink	10 min/pass	~0.01 Wh
SX1278 LoRa Radio	3.25–4.5 W [91]	Transmit	10 min/pass	~0.54–0.75 Wh
SX1278 LoRa Radio	0.065 W [91]	Receive	83 min	~0.090 Wh
SkyFox piPATCH-S-DL/ FM	0 W [38]	Downlink	10 min/pass	0 Wh
Boson thermal-mass heater (HAP6736)	0.317 W [70]	Cold-case thermal control	36 min	0.19 Wh
Battery survival heater (HAP6736)	0.317 W [70]	Thermostat-controlled battery heating	36 min	0.19 Wh
EPS conversion efficiency	80%*	All powered modes	Variable	Applied to mode totals

Based on the preliminary power budget above, PyroSat is expected to consume approximately 10.71–10.97 Wh per 93-minute orbit, corresponding to an orbit-average power load of approximately 6.9–7.1 W. These values represent the electrical energy consumed by the satellite loads before accounting for losses within the electrical power system. For preliminary generation, PyroSat assumes an overall EPS conversion efficiency of 80%, corresponding to a conservative 20% conversion loss. This assumption is used when comparing the satellite load against usable solar-array generation and will be replaced with characterized efficiencies as the EPS design matures. The 10-minute per-orbit communications allocation is a conservative case and does not represent the expected pass frequency. The HAP6736 is operated directly from the 12 V rail throughout the approximately 36-minute maximum eclipse; this however was the worst-case approximation. Actual heater duty cycle will be thermostat-controlled and established through transient thermal analysis and thermal-vacuum testing. Further analysis will also be required as the operation profiles of ADCS, TT&C system, and payload hardware are refined.

The power estimate for the NanoCom SDR Mk3 is derived from its datasheet under the

expected operating condition of 25 °C using the Xilinx Zynq-7045 and one TR600 module [35]. PyroSat is currently expected to utilize only one TR600; however, whether the processor can be maintained near the specified 25 °C operating condition will depend on the final thermal design and must be verified through testing. For high-rate science-data downlink, PyroSat currently proposes the Satlab SRS-4 S-band transceiver, which consumes approximately 10.8 W at 33 dBm RF output, with an additional approximately 60 mW when the RS-422 interface is enabled [37]. Because these represent some of the mission's highest-power payload and communications loads, their operation will be controlled through the satellite's power-management modes rather than treated as continuous loads. The preliminary power budget therefore evaluates energy consumption according to the expected duration of each operating mode rather than assuming simultaneous peak-power operation. Within the 93-minute planning orbit, 80 minutes are currently allocated to steady-state nadir or Sun-pointing operations, 10 minutes to ground-station pointing, and 3 minutes to attitude transitions. The division of orbits between nadir and Sun pointing will depend on the final power-generation and mission-operations schedule. Under continuous nadir pointing, the current array model produces approximately 7.44 Wh of usable energy per orbit in the □ = 0° case after the 80% EPS-efficiency (Table A), compared with a requirement of approximately 10.71–10.97 Wh. This corresponds to an energy deficit of approximately 3.27–3.53 Wh per orbit, or roughly 31–32% of the required orbital energy. The deficit increases at higher beta angles under the current nadir-pointing model, making dedicated Sun-pointing periods necessary for energy-positive operation.

Because the deficit applies to continuous nadir pointing, PyroSat schedules its attitude in whole orbits and balances energy over a day rather than orbit by orbit. At about 15.5 orbits per day, the baseline plan allocates 10.5 orbits to Sun pointing for energy collection and 5 orbits to nadir-pointing data collection, 2 of which also include an S-band downlink to the ASU ground station. Table C gives the resulting daily energy balance, using the load values in Table D and the usable generation in Table A. A Sun-pointing orbit carries only the bus loads (ADCS, the always-on electronics, 433 MHz receive and the thermal-control heaters), about 6.8 Wh. A data orbit adds the 10-minute Observation mode and the slews between attitudes, about 8.3 Wh. A downlink orbit adds the S-band downlink, ground-station pointing and 433 MHz transmit, which is the full 10.71–10.97 Wh in Table D. Together these come to approximately 118 Wh per day, an average of about 4.9 W. Over the same day, the array produces approximately 187, 186, 200 and 256 Wh of usable energy at beta angles of 0°, 30°, 60° and 75°, so the plan is energy-positive at every evaluated beta angle and exceeds the 1.25 times generation-to-load requirement with a ratio of at least 1.57. Energy balance alone needs only about 1 to 5 Sun-pointing orbits per day, and about 5 to 7 with the 1.25 times margin, so the 10.5-orbit allocation leaves room to add science or shorten the Sun-pointing share as the power model matures. At high beta angles the daily total hides a limit on the battery. Five consecutive data orbits (3 data and 2 downlink) draw about 9, 14, 28 and 37 Wh from the battery at $\beta = 0°$, 30°, 60° and 75°, while the science rule in the power rules permits science only above 50% state of charge, about 36 Wh below a full pack. The scheduler therefore interleaves Sun-pointing orbits between data orbits at high beta and does

not run all five data orbits as one block. These values use the conservative 93-minute orbit, the 10-minute pass allocation of Table D and the 80% EPS efficiency, and will be replaced as the flight power model, ground-station contact plan and EPS efficiency are refined.

Table C (daily energy balance)

β	Usable generation (Wh/day)	Load (Wh/day)	Net (Wh/day)	Generation / load	Sun orbits needed to break even
0°	187	118	+68 to +69	1.58	about 0.5
30°	186	118	+67 to +68	1.57	about 2.3
60°	200	118	+81 to +82	1.68	about 4.5
75°	256	118	+138	2.16	about 4.1

Table C: Daily energy balance for 10.5 Sun-pointing orbits, 3 data orbits and 2 downlink orbits per day (15.5 orbits).

For initial communications analysis, PyroSat assumes about 7 minutes of ground-station contact on each of 2 orbits per day, and Table D conservatively budgets 10 minutes of contact in every downlink orbit. This represents a preliminary planning case based on the current assumption of relying primarily on a single ASU ground station. S-band downlink and associated ground-station-pointing operations are therefore budgeted only during this contact period. Actual pass duration, pass frequency, and achievable downlink time will depend on orbital geometry and ground-station visibility and will be refined through mission analysis once the final orbit and ASU ground-station parameters are established.

Operating Models & Power management

Energy producing operations will therefore require active scheduling between nadir and Sun-pointing attitudes. Sun pointing is scheduled in whole orbits, with each orbit allocated either to Earth-facing science operations or to Sun pointing as required by the satellite's energy consumption. The OBC and EPS will use battery state of charge and the scheduled science and communications activities to determine whether science observations can proceed or whether additional Sun-pointing time is required. The final allocation will be established from the flight power model rather than assuming a fixed Sun-pointing duration for every orbit. Table D summarizes PyroSat's resulting load budget.

Table D: Load budget, per 93-minute orbit

Items	Power (W)	Duty cycle on a 93 minute orbit	Energy (Wh)
Observation mode (NanoCom + TR600, Boson+ 640, two RF front ends)	6.96–7.26	10.75% (10 min)	1.16 to 1.21
Always-on (OEM719, nanoPATCH-L1, RP2350 OBC)	1.07	100% (93 min)	1.66
ADCS, Earth or Sun facing	3.13	86.02% (80 min)	4.17
ADCS, ground-station pointing	3.71	10.75% (10 min)	0.62
ADCS, transition (slews)	5.50	3.23% (3 min)	0.28
S-band downlink (SRS-4, RS-422, S-band patch)	10.86	10.75% (10 min)	1.81
433 MHz TT&C, transmit	3.25 to 4.50	10.75% (10 min)	0.54–0.75
433 MHz TT&C, receive	0.065	89.25% (83 min)	0.09
Thermal-control heaters (Boson & Battery)	0.634	38.7% (36 min)	0.38
Total			**10.71–10.97 Wh (6.91–7.08 W average)**

Power rules

1. Science is allowed only above 50% state of charge, and in eclipse only above 70%
2. The satellite enters safe mode below 30% state of charge and returns above 60%
3. Sun-pointing recovery: When the predicted battery state of charge is insufficient to support the scheduled science and communications activities while maintaining the required 30% threshold, the satellite will prioritize solar array Sun-pointing over science collection.

4. Downlink and GNSS-R never run together, which limits peak power and keeps the S-Band transmitter away from the L1 receiver.
5. Loads are shed in order: S-band transmitter, Boson, GNSS-R, payload computer, then ADCS reduced mode, with the battery heater and the UHF link last.
6. The EPS microcontroller enforces these rules independently of the OBC, so a software hang cannot drain the battery.

Communications

S-Band Science Downlink

PyroSat proposes the Satlab SRS-4 S-band Transceiver as its dedicated high-rate science downlink radio. The SRS-4 is a CubeSat-compatible, TRL-9 transceiver with flight heritage. It supports transmission in the 2200–2290 MHz S-band range, adjustable RF output power up to 33 dBm, BPSK, QPSK, and 8PSK modulation, and symbol rates up to 5 MBd. The transceiver also provides CAN and RS-422 interfaces, allowing it to integrate with PyroSat's existing command-and-data architecture [37].

PyroSat will initially target a science downlink data rate of approximately 1 Mbps of science data after coding. For a nominal 7-minute ground-station contact, the theoretical maximum data volume that could be transmitted is

$$D = Rt = (1 \cdot 10^6)(420) = 420\,Mbit \simeq 52.5\,MB/pass. \quad (25)$$

PyroSat's science-data volume is expected to remain within this capability because raw GNSS-R I/Q samples will be processed onboard rather than downlinked. FMPL-2 uses a default GPS L1 DDm configuration of 100 delay bins, corresponding to 500 correlation-power cells per DDM [6]. Assuming PyroSat stores each correlation-power cell as a 32-bit value, a comparable DDM would require approximately 2 kB before metadata combined with Boson+ 640 thermal frame which is approximately 655 kB. When combined with the approximate 2 kB DDM and additional position, velocity, time, attitude, GNSS transmitter information, and further metadata, PyroSat conservatively budgets approximately 700 kB per single-frame science record. Therefore, the theoretical number of single-frame science records transferable during a 7-minute contact (about 25 three-frame observations, the unit used in the Data Management section) is,

$$N_{obs} = \frac{52500\,kB}{700\,kB/observation} \simeq 75 \text{ observations per pass.} \quad (26)$$

The actual science return is likely to be lower after accounting for overhead, link acquisition, reduced link quality, etc.

During a scheduled ground-station pass, the RP2350 OBC will command the satellite into Downlink Mode. The NanoCom will retrieve selected observations from its eMMC, organize the science products into downlink packets, and transfer those packets to the SRS-4 through a

dedicated RS-422 connection. The NanoCom's approximately 3 Mbps RS-422 throughput provides sufficient internal bandwidth for the proposed 1 Mbps science downlink. [35]. The SRS-4 itself supports RS-422 rates up to 12.5 Mbps, so the SRS-4 does not establish the limiting internal data rate [37]

Once the packetized science data reach the SRS-4 through RS-422, the transceiver performs the required framing, channel coding, and modulation before converting the digital data stream into an S-band RF signal. The SRS-4 includes its transmitter power amplifier and RF filtering, allowing its RF output to interface with a passive S-band antenna [37].

The SRS-4 modulations (BPSK, QPSK and 8PSK) set how many bits each symbol carries and do not by themselves provide redundancy. Resilience to noise and bit errors comes from the forward error correction and framing the SRS-4 provides: run-time configurable rate-1/2 convolutional coding, a (255,223) Reed-Solomon code that appends 32 parity bytes to each frame, and an optional CRC32C that lets the receiver detect frames that could not be corrected [37]. The SRS-4 does not provide retransmission or file transfer itself [37], so reliable delivery is handled by PyroSat's data-handling software. Each observation is stored as a file with its own CRC and transmitted in blocks, and it is kept on the NanoCom's eMMC until the ground station acknowledges the complete file, so a dropped pass resumes from the last good block instead of re-sending the file (see Data Management). The modulation and coding configuration will be set from the link budget, using BPSK or QPSK where margin is limited.

PyroSat proposes the SkyFox Labs piPATCH-S-DL/FM as its dedicated S-band downlink antenna. The antenna operates across the 2200–2290 MHz S-band downlink range, matching the SRS-4 transmit band, and uses RHCP polarization with approximately 5.55 dBic. The SRS-4 RF output will be routed through a coaxial connection to the piPATCH-S-DL/FM, which will radiate the signal toward the receiving ground station.

The completed science data pipeline is therefore,

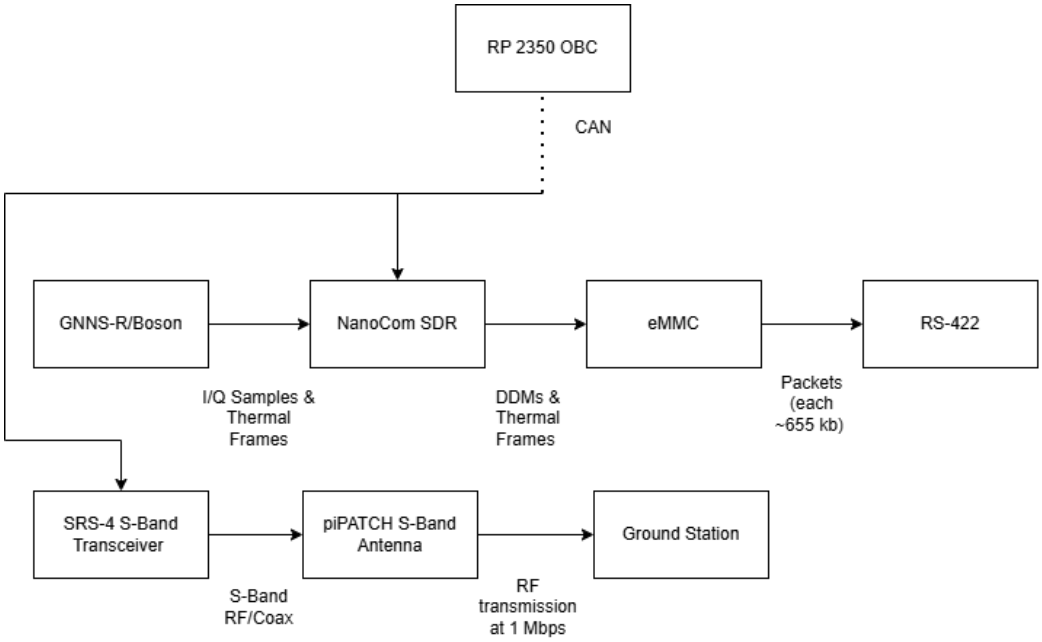


Figure 13. PyroSat science-data storage and S-band downlink architecture.

The final transmission frequency, modulation, coding configuration, operation data rate, and achievable science return will be verified through further testing and budgeting. These current values are to be treated as rough estimates, they will require further validation.

433-MHz TT&C

PyroSat will retain the SquidSat 433-MHz communication subsystem for low-rate telemetry, tracking, and command (TT&C). The SquidSat Deployable Antenna Board (DANT) incorporates an SX1278-based radio module and a deployable turnstile antenna tuned to 433 MHz. The radio includes transmit and receive amplification and provides approximately 1W RF transmit power without requiring an external power amplifier. SquidSat utilizes GMSK modulation at 19200 baud, for which the existing SquidSat link budget is reported to close with sufficient margin. The DANT contains an independent STM32F103 microcontroller running Zephyr and interfaces with the RP2350 OBC through a CAN Bus, which can also be kept. This allows the 433-MHz radio system to handle bus commands, beacon transmissions, housekeeping telemetry, and basic health/status information independently of the high-rate science-data path. Bulk GNSS-R and Boson+ science data will not be transmitted through the 433-MHz link and will instead use PyroSat's dedicated S-Band downlink [1].

Thermal Management

PyroSat preliminarily uses two Minco HAP6736 Space Flight Thermofoil heaters: one attached to the Boson thermal mass and one thermally coupled to the battery pack. Each heater

has a nominal resistance of approximately 454.49 Ω [70]. When operated from the spacecraft's 12 V regulated rail, each heater dissipates approximately 0.317 W. For power budgeting, each heater is allocated up to the approximately 36-minute maximum eclipse duration, corresponding to approximately 0.19 Wh per heater, or approximately 0.38 Wh combined per orbit. Actual operation will be thermostat-controlled, and heater sizing and duty cycles will be verified through transient thermal analysis and thermal-vacuum testing.

Two things drive the thermal design. The first is the Boson+ 640 radiometric imager, whose temperature readings are only as good as the stability of its own housing. The second is the S-band radio, which dissipates more heat than anything else on the satellite. The design is passive, with heaters as a backup, and it avoids a dedicated radiator because the heat load does not require one.

The total heat dissipated inside the satellite is about 5 W averaged over a day and about 7 W in the worst orbit (the load in the power table) [17], and the large short bursts come from the S-band radio, which draws about 10.8 W typical and up to 15 W peak only during a pass. As a first-order check, the power a surface can radiate to space is given by the Stefan-Boltzmann law [13],

$$q = \varepsilon\sigma T^4 \tag{27}$$

where q is the radiated power per unit area, ε is the emissivity of the surface, $\sigma = 5.67 \times 10^{-8}$ $W/(m^2 \cdot K^4)$ is the Stefan-Boltzmann constant, and T is the surface temperature in kelvin. For a surface at about 280 K with a high-emissivity finish ($\varepsilon \approx 0.85$), $q = 0.85(5.67 \times 10^{-8})(280)^4 \approx 296$ W/m^2, or roughly 300 W/m^2. Rejecting 10 W therefore needs an area of $A = P/q = 10/296 \approx 0.034$ m^2. One 3U side face is about 0.034 m^2 (100 by 340 mm), so one side face alone matches that area before the deployed wings are counted. This check ignores absorbed sunlight, albedo and Earth infrared, so it shows that a separate radiator is not necessarily required, not that the margin is large. A transient thermal model is planned to confirm it.

Radiometry depends on the temperature of the camera body, so the camera housing is isolated from the bus and tied to an aluminum thermal mass, as shown in Figure 2 (see LWIR Thermal Sensor). The stability target for the camera housing is about 1 K, matching the radiometric tolerance of the mission. The Boson+ datasheet gives an operating temperature range of -40 to +80 degrees C [3]. The 1 K target applies over a pass, meaning the capture window between shutter calibrations. The camera's own flat-field correction, which is available on shuttered variants, continues as normal and is not replaced by the thermal design [3]. PyroSat uses the shuttered variant, and a blackbody shutter resets the camera before any capture.

A 12.5 W transmission (between the 10.8 W typical and 15 W peak draw) lasting 10 minutes draws about 7.5 kJ, most of which becomes heat in the radio unless it is conducted away. If the radio sits on a 1 kg aluminum mass with no other path, that is an upper-bound rise of about 8 K (7.5 kJ divided by roughly 900 J per kg per K for aluminum), which is why the radio is mounted on its own thermal path to a bare side face, opposite the camera’s strap, and kept

physically away from the camera [14]. The radio is also not allowed to operate at the same time as the camera in the baseline plan, so the camera is never running during the transient, and the PEEK isolation limits how much of the residual heat reaches the camera housing afterward.

Once the wings are deployed, the side faces are the natural radiating surfaces. The exterior finish is low in solar absorptance and high in infrared emissivity. A strap links the thermal mass to a bare side face of the structure. The strap does not go to the wings or the zenith panel, because the solar cells there absorb sunlight and would heat the strap path instead of cooling it. Because anodized aluminum already has high emissivity, no separate radiator is added in the baseline. If the transient model shows too little margin, a small radiator patch on the side face is the first addition.

The camera opening must not face the Sun in the Sun-pointing attitude, because direct sunlight into the lens can upset the radiometric calibration of a thermal camera and may damage it. The camera is powered only after nadir pointing is reached, so the settling time before science resumes is the camera warm-up time. Heaters are the backup, and because a heater on a thermally connected mass can draw a lot of power, each one has a defined job and a limit. PyroSat has two. The heaters themselves are the Minco HAP6736 [70].

The battery survival heater (Minco HAP6736) protects the lithium-ion cells in the cold case (see the Power risk table). It runs from the bus under a thermostat and is the last load shed in a power emergency, because a cold battery is a worse failure than a missed image. The battery limit is the cold charging limit of the Jauch cells, 0 degrees C, which the cell datasheet allows to charge only between 0 and 45 degrees C.

The camera heater (Minco HAP6736) is mounted on the aluminum thermal mass and holds the housing above its cold limit during long eclipse periods. Its limit comes from the Boson+ datasheet operating range [3]. Unlike the battery heater, it is not a survival load, so it is shed together with the camera rather than last. Three measures keep it from burning power. First, the strap conductance is chosen so that the heater power needed in the cold case stays small, since at steady state that power equals the strap conductance times the difference between the setpoint and the panel temperature, and the transient thermal model sets this value. Second, the heater has a fixed power rating and a maximum on-time per orbit in flight software, and its energy is counted in the Thermal-control heaters row of the load budget in Table D. Third, the camera dissipates about 1.0 to 1.3 W when operating [3], which helps hold the housing, so the heater mostly runs when the camera is off, and then only to the survival limit rather than the tight imaging setpoint, unless a capture is coming. Temperature sensors on the camera housing, the thermal mass, the S-band radio and the battery feed the housekeeping telemetry. Verification is by thermal model and, if schedule allows, a thermal-vacuum test.

Attitude Determination & Control System

PyroSat will use an attitude determination and control system (ADCS) composed of triple axis reaction wheel system, magnetorquers, coarse sun sensors, inertial measurement unit (IMU), and a star tracker in order to achieve a pointing accuracy of 1 degree (2-sigma), a time to reorient by 90 degrees of under 20 seconds, and pointing knowledge of 0.04 degrees. Both payloads look straight down, so the satellite must hold nadir pointing during science.

The 0.04 degree attitude knowledge target comes from the size of one thermal camera pixel projected onto the ground from an altitude of 400 km. One Boson+ pixel covers about 270 m there, which is an angle of about 0.04 degrees (about 144 arcseconds). The looser requirement, for the GNSS-R payload, is 0.143 degrees, the angle that the 1 km geolocation target subtends at 400 km (see Geolocation of GNSS-R signal). Magnetometer and sun sensor setups cannot achieve better than about 1 degree of attitude knowledge when in eclipse, so the ADCS must include a star tracker. Commercial star trackers are specified in the 8 to 15 arcsecond class, or about 0.002 to 0.004 degrees [7], roughly ten times better than the 0.04 degree requirement, so a star tracker is suitable for PyroSat.

Commercial ADCS that meet these requirements can easily cost more than $20,000, and a star tracker alone costs $10,000 to $100,000 or more [9]. The $20,000 figure is an estimate drawn from the cited references. Integrated commercial units, such as the Blue Canyon XACT series [7] and the CubeSpace Gen 2 CubeADCS [8], are in this price range, so SDSL plans to develop its own system, which keeps the cost down and builds the club's capability. Two options are therefore presented: a custom in-house solution built by the SDSL ADCS team, and a solution assembled from commercial off-the-shelf (COTS) components. The second option exists because the in-house system is a new development. A late or underperforming ADCS would stop the science mission, since neither payload works without accurate pointing, so a COTS option is kept ready as the backup.

The custom ADCS needs its own microcontroller to read the sensors and drive the actuators, compute the attitude estimate and deliver it to the CAN bus. The microcontroller will likely be the RP2350 [4], because the club already has experience with it from the SquidSat bus and it is inexpensive. The sensors are an IMU, coarse sun sensors, the GNSS receiver already present for the GNSS-R payload and a star tracker developed by the SDSL ADCS division. The actuators are a three-axis reaction wheel assembly (Project Trident) and magnetorquers for detumbling and wheel desaturation.

To estimate the satellite’s attitude, a Bayes’ Filter method will be employed, such as the Multiplicative Extended Kalman Filter (MEKF). In a typical Kalman Filter, the state tracked by the filter would be the satellite’s attitude and the angular velocity and the control input would be the net torque applied by the magnetorquers and the reaction wheel system. In Kalman Filters, many variables have a corresponding covariance matrix which describes the independent uncertainty of each component of a variable and the correlations between them. The algorithm

for the Kalman Filter is roughly as follows: based on the values and covariances of the prior state estimate and control input, the current value and covariance of the state is predicted. Then, this estimate and its covariance is transformed into measurement space and compared against the actual measurements. This comparison is then used to update the state vector and covariance, effectively performing a weighted average. This process is then repeated every time new sensor data is received.

An MEKF is slightly modified from this procedure. For instance, the attitude error and gyro bias error are tracked by the filter, and the attitude and angular velocity are instead accumulated outside the filter. Based on the measurements, the attitude error and gyro bias error are updated. At the end of each iteration, the errors are accumulated into the outside attitude and gyro bias.

The COTS solution is built around the TM2Space AeroMagControl [18] as its core component. The AeroMagControl carries a 9-degree-of-freedom self-calibrating IMU on its driver board, the magnetorquer drive electronics with firmware for detumbling and telemetry. Three TM2Space SpaceWheel30 reaction wheels [19] provide three-axis control, each with 30 mNms of nominal momentum storage, 75 mNm of maximum torque and fine control to within 1 RPM above 100 RPM. Three TM2Space HyperMagCoil magnetorquers [20], available in 86 mm and 96 mm lengths with a 400 mA current rating, provide detumbling and wheel desaturation. The KSST-01 Star Tracker [52], mounted opposite the UHF antenna, will provide absolute attitude data with a design accuracy of 60 arcseconds (the conservative end of the vendor's 0.1 to 60 arcsecond range; the listing also claims better than 5 arcseconds at 3σ) to the AeroMagControl via RS-485.

A first-order check of the wheels against the reorientation time: treat the satellite as a uniform 100 by 100 by 340 mm box at 6 kg (conservative, since the NRCSD limit is 4.8 kg), which gives a largest moment of inertia of about 0.063 kg·m². Turning 180 degrees in 20 seconds (accelerating for 10 seconds, then decelerating for 10 seconds) needs an angular acceleration of about 0.031 rad/s², a torque of about 2.0 mNm, and a peak angular momentum of about 20 mNms, which is two thirds of one wheel's 30 mNms [19] and well inside the 75 mNm torque limit. This ignores disturbance torques and the momentum used by other maneuvers.

The satellite also points its solar panels at the Sun for a selected number of orbits each day (see Power Systems), which adds a second attitude mode and two slews at each change between modes. To switch between aiming the payload instruments at the earth and aiming the solar panels towards the sun requires at most a 180 degree slew, since the deployed solar panels face zenith and the payload instruments face nadir. Using a symmetric acceleration/deceleration with maximum torque of 2mNm in order to reduce power draw, it is possible to achieve a 180 degree slew in about 20 seconds. As such two slews can take at most 1 minute of the roughly 93-minute orbit before settling time is counted, and the Sun-pointing period must allow for that. The pointing and knowledge requirements above apply in Observation mode only. Sun-pointing accuracy can be much looser, because a 5-degree error costs less than 0.4% of panel power (the

cosine of 5 degrees is 0.996). More frequent slews also use more wheel momentum, so desaturation by the magnetorquers needs a check. Settling time applies to the nadir and payload modes only, not to Sun pointing. In Sun pointing, the zenith axis, which is perpendicular to the deployed solar panels, points at the Sun.

A third attitude mode, S-band downlink, applies during ground-station passes. The S-band patch antenna is on the nadir face in the CAD, so nadir pointing already aims it at the ground for passes near the sub-satellite track. For lower-elevation passes the attitude control system tilts the nadir axis toward the ground station, within the limit set by the S-band antenna beamwidth, and returns to nadir when the pass ends. The tilt limit is assumed to be 30 degrees from nadir, which is inside the roughly 40 degree half-angle that a typical single S-band patch covers, and will be set from the antenna datasheet once the patch is chosen.

The in-house design is used only if it passes the following tests before a yet-to-be-determined fixed date. The star tracker must achieve an accuracy better than 0.04 degrees in a hardware-in-the-loop test wherein star fields are simulated according to predicted satellite conditions. A physical prototype of the reaction wheel system must achieve a pointing accuracy of 1 degree (2-sigma) or better and reorient by 90 degrees in under 20 seconds. If the gate is missed, the COTS components are ordered. Lead times for ADCS hardware are long (reaction wheels about 12 weeks, star trackers 16 weeks to 4 months), and TM2Space lists its prices in Indian rupees, so the gate date must leave room for shipping, customs, and currency conversion [15], [18], [19].

A star tracker alone costs $10,000 to $100,000 or more [9], and Phoenix CubeSat paid $42,000 for its MAI-400 in 2015 [16], so attitude control is budgeted far above a hobby-level line. The COTS option is priced from the vendor's public listings in rupees [18, 19], and a written quote is not yet available. See the Mission Cost section. The ADCS broadcast attitude quaternions and rates over CAN to the on-board computer, payload microcontroller. The GNSS receiver time-synchronization signals over CAN so that the ADCS can stamp attitude with the same time base as the thermal frames.

Part/item	**Selected component/ model**	**Quantity**	**Dimensions (mm)**	**Mass/unit (g)**	**Remarks**
ADCS Core + IMU	AeroMagControl	1	95 x 95 x 30.6	410	*Mass not provided by datasheet*
Magnetorquers	HyperMagCoil	3	71.5 x 16 x 16	80	
Star Tracker	KSST-01	1	56 x 60 x 92.8	175	
Sun Sensors	BPW34	6	5 x 5 x 2	.078	Rough

					Dimensions
Reaction Wheels	SpaceWheel 30	3	55 x 55 x 40	280	
Total				1665	

Part/item	**Selected component /model**	**Nominal Total Power (W)**	**Max Total Power (W)**	**Remarks**
ADCS Core + IMU	AeroMagControl	1	1.5	Not provided by the datasheet. This is a rough estimate
Magnetorquers	HyperMag Coil	0.13 Earth / Sun facing 0.26 GS-pointing 0.4 Transition	4.5	Non-continuous. Probably at most 2 MT running concurrently.
Star Tracker	KSST-01	0.5	0.8 [52]	Vendor lists 0.3 W in regular mode and 0.8 W with cooler and calibration functions [52].
Sun Sensors	BPW34	0.002	0.002	
Reaction Wheels	SpaceWheel30	1.5 Earth / Sun facing 1.95 GS-pointing 3.6 Transition	36 at max velo	Max angular velocity will rarely be necessary, and not possible due to the power draw
Total		3.13 Earth / Sun facing 3.71 GS-pointing 5.50 Transition	42.8	

The tables above give the dimensions, mass and power of the ADCS component set used for the mass and power budgets; the priced plans are in Mission Cost.

Verification & Test Plan

The FMPL-2 development process demonstrates the value of verifying hardware and software interfaces before final integration. Its preintegration testing included platform-interface emulation, evaluation with an engineering model of the onboard computer, and GNSS reception tests and relevant functional tests were repeated after the instrument was mechanically integrated. PyroSat will apply a similar staged approach to identify incorrect supply voltage

levels, interface incompatibilities, excessive current consumption, and other integration faults before they affect the development schedule.

Verification will also assess the electrical and thermal behaviour of the integrated system under representative operating loads: voltage stability, current demand, regulator temperature, and possible electromagnetic interference with GNSS-R reception. These results will inform later design iterations and the environmental qualification and system level verification plan.

Test	Purpose
Staged interface tests	Catch supply voltage, interface, and current faults before integration
Solar panel current-voltage test	Measure real panel output
Battery capacity and cycle test	Confirm capacity and depth of discharge
Rail characterization	Rail voltage and current, converter efficiency, startup behaviour, load transient response, and regulator temperature
EPS functional and protection test	Overcurrent trip, latch-up trip, undervoltage lockout and cold start
Day-in-life power test	Run the full mode sequence , including simultaneous payload operation, with battery and solar simulation
Conducted noise at L1	Verify converters do not degrade the GNSS-R receiver
Patch antenna characterization	Measure the final frequency, bandwidth, gain and radiation pattern of the L1 patch antennas
Balloon flight	Measure real power and thermal behaviour in flight
Thermal vacuum	Qualify EPS, battery and thermal margins
Random Vibration	Launch survival(FMPL-2 precedent: 12.4g rms)
Deployment tests	Release of panels and antenna, including cold

Development Schedule

The following timeline shows the development schedule and goals that the PyroSat team plans to achieve over the project cycle.

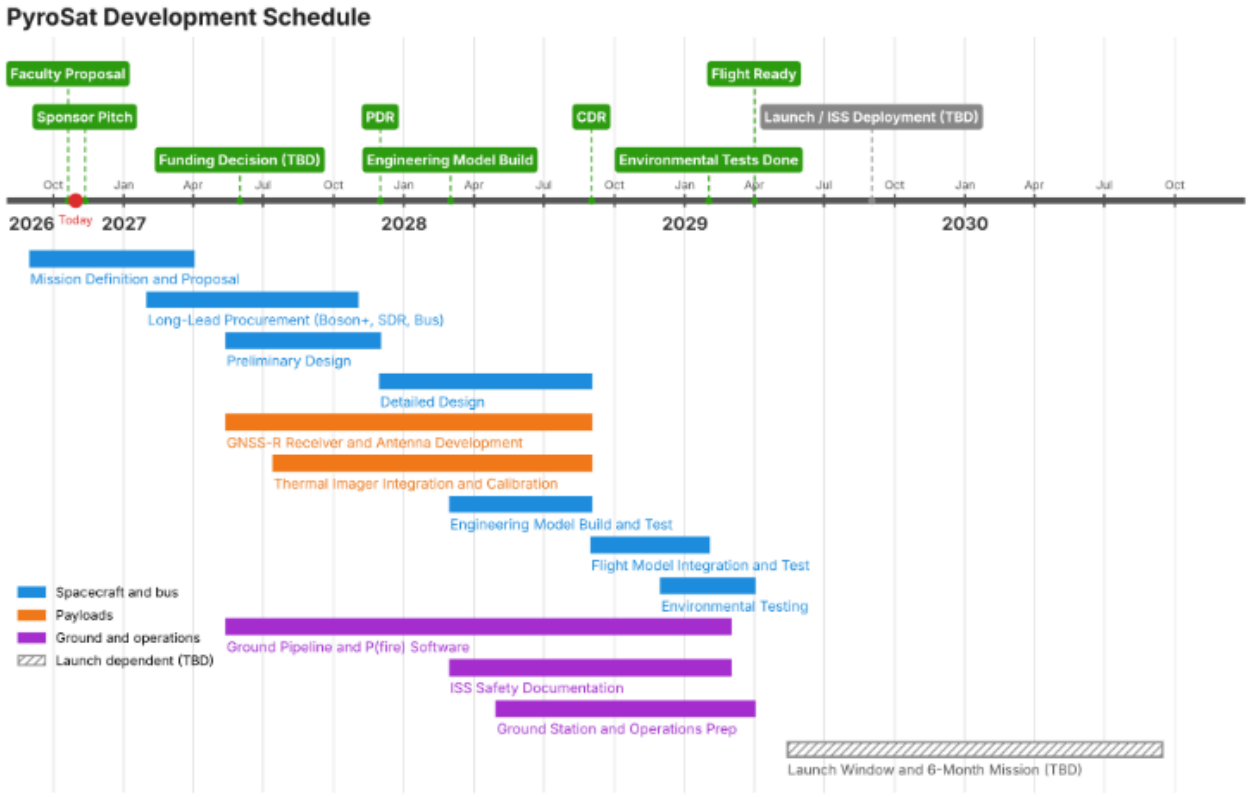


Figure 14. PyroSat development schedule from mission definition to flight readiness. The funding decision and the launch date are to be determined.

FlatSat development will begin in May 2027 and will include designing and manufacturing printed circuit boards for the subsystems, integrating the GNSS-R receiver and thermal imager with the onboard computer, creating flight software, and setting up a FlatSat with supporting ground support equipment. The team plans to hold a balloon test in which an early model of the satellite is flown on a weather balloon to test the radio and both payloads, including the collection of synchronized GNSS-R and thermal data. The preliminary design review is planned for December 2027 and the critical design review for September 2028, with the engineering model fully integrated and tested after that. The flight model will then be assembled and tested, followed by final environmental testing and the safety documentation required for deployment from the International Space Station. At the end of this timeline, PyroSat will be ready for integration with the launch provider in April 2029. The NanoCom SDR MK3 payload computer has a planning lead time of about 12 months (from the earlier $40,000 estimate) and the team is in talks with GomSpace for a quote, so the order date will set when GNSS-R

integration on the FlatSat can begin. An order placed in February 2027, when long-lead procurement starts, would arrive about February 2028, just before the engineering model build begins in March 2028.

Mission Cost

Thermal Payload:

Part/item	Selected component/ model	Qty.	Total Cost ($USD)	Cost Basis / Source	Remarks
Teledyne FLIR Boson+ 640 Radiometric with 18mm lens (SKU: 22640A018-6 IARX)	Industrial Grade <20mk model	1	$7,059	Manufacturer catalog price [66]	Formal vendor quotation pending.
Multi-layer insulation - polyimide film	Dunmore Aerospace STARcrest Satkit MLI	1 set	$1,500	Estimation	Based on the $500 Phoenix CubeSat cost from 2019 [48]; increased to a $1,500 upper-bound allowance to account for cost increases and design differences.
Copper thermal strap	TAI CuTS P7-501 OFHC Copper Thermal Strap	1 set	$1,000	Manufacturer catalog price [68]	The exact quantity is not determined yet.
Polyimide thermofoil heater	HAP6736 Space Flight Thermofoil Heater	1 set	$122/unit	Manufacturer catalog price [70]	The exact quantity is not determined yet.
Onboard thermal calibration reference	Custom high-emissivity on-board blackbody	1	$1000	Estimation, upper allowance	In-house fabrication allowance. Requirements: - temperature probe - metal blackbody - high-emissivity coating - heaters & coolers

Ground thermal calibration source	ASU calibrated blackbody equipment	1	$0	Existing university equipment	
Sub-Total			**$10,681**		

GNSS-R Antennas:

Part/item	**Selected component/ model**	**Qty.**	**Total Cost ($USD)**	**Cost Basis / Source**	**Remarks**
Zenith-facing RHCP GPS L1 patch antenna	microPATCH -L1/FM	1	Included price under GNSS navigation.	Manufacturer catalog price [71]	This separate patch antenna may not be needed, as the antenna provided in the GomSpace GPS kit may work for both GNSS-R and GPS navigation.
Nadir-facing LHCP GPS L1 patch antenna	Option 1: Custom 3-patch LHCP L1 array antenna substrate + feed network In-house built	2 array of 3 patch es each	$2,000	Estimation	Includes six patches, substrates and feed networks. The COTS option does not exist.
Sub-Total			**$2,000**		

GNSS-R RF Front End:

Part/item	**Selected component/ model**	**Qty.**	**Total Cost ($USD)**	**Cost Basis / Source**	**Remarks**
Custom RF front-end PCB	Custom PyroSat GNSS-R RF PCB	1	$500	Estimation	Includes PCB fabrication and assembly allowance.
RF shielding/encl	Custom shielding	1 set	$100	Estimation	Preliminary fabrication

osure	(aluminium)				allowance.
RF connectors/cabling	50 Ω SMA/U.FL + space-compatible coax	1 set	$150	Estimation	Includes payload RF interconnects.
Sub-Total			**$750**		

GNSS Navigation:

Part/item	**Selected component/ model**	**Qty.**	**Total Cost ($USD)**	**Cost Basis / Source**	**Remarks**
LEO-compatible GNSS receiver (OEM719, may require additional shielding)	Option 1: GomSpace NanoSense GPS Kit	1	$13,500 allowance	Formal vendor quotation pending. Priced based on Estimation	Additional shielding may be required. Antenna is included.
	Option 2: Pumpkin GPSRM 1 Kit	1	$11,485	Manufacturer catalog price [72]	Antenna is not included.
GNSS antenna	Option 2: SkyFox microPATCH -L1/FM	1	$2,545	Manufacturer catalog price [71]	Required only with the Pumpkin GPSRM 1 configuration.
Sub-Total			**$13,500-14,030**		Lower: $13,500 Upper: $14,030

Payload Computing:

Part/item	**Selected component/ model**	**Qty.**	**Total Cost ($USD)**	**Cost Basis / Source**	**Remarks**
Processing SoC	Option 1: GomSpace NanoCom SDR MK3	1	$40,000	Informal quotation, pending official quotation	Quote time valid for 12 months.

CAN-side microcontroller	RP2350A [4]	1	$2	Marketplace Catalog Price [73]	Component selection and integration are detailed in the SquidSat Interface section.
Sub-Total			**$40,002**		

S-band Communication:

Part/item	**Selected component/ model**	**Qty.**	**Total Cost ($USD)**	**Cost Basis / Source**	**Remarks**
S-band transceiver	Option 1: Satlab SRS-4	1	$23,697.26 each (min. Order 2)	Marketplace catalog price [69]	The quote tracker lists $13,200 for the transceiver and $14,800 for a transmitter [15], shown for comparison. The minimum order of 2 appears only in the vendor quote, not on Satlab's public SRS-4 page [37], so the quantity should be confirmed with Satlab. Higher-cost COTS option
	Option 2: GomSpace NanoCom Link S	1	N/A	Awaiting formal quotation	
	Option 3: EnduroSat S-BAND TRANSCEIVER	1	$17,500	Manufacturer catalog price [54]	A cheaper COTS alternative as GomSpace AX2150 may not be sufficient.
	Option 4: GomSpace AX2150	1	$7,300	Vendor informal quotation	Data-rate suitability requires confirmation.
S-band antenna / patch antenna	piPATCH-S-DL/FM		$2,778	Manufacturer catalog price [38]	The final antenna depends on radio selection.

Communications PCB/interface hardware	Custom interface PCB	1	$500	Estimation	Upper allowance
Sub-Total			**$10,578 -$50,672**		Lower: $10,578 Upper: 50,672

UHF / LoRa Radio:

Part/item	**Selected component/model**	**Qty.**	**Total Cost ($USD)**	**Cost Basis / Source**	**Remarks**
Plan A: Primary LoRa radio	Plan A: HopeRF RFM98PW	1	$12	Manufacturer catalog price [55]	Used in SquidSat-derived UHF architecture.
Plan A: Redundant LoRa radio	NiceRF LoRa1278F30	1	$32	Manufacturer catalog price [75]	Used in SquidSat-derived UHF architecture.
Plan A: Custom deployable antenna RF communications	(433 MHz) UHF/LoRa monopole, 173mm In-house built	1	$500	Estimation, upper allowance	Includes materials and prototyping.
Plan B: commercial UHF radio and antenna	Option 1: GomSpace	1 set			
	Option 2: EnduroSat UHF radio + antenna	1 set	$5,900 radio; $4,700 to $7,800 antenna	Catalog price [15]	COTS (upper) benchmark
Sub-Total			**$544 - $13,700**		Lower: $544 Upper: $13,700

Structure:

Part/item	Selected component/ model	Qty.	Total Cost ($USD)	Cost Basis / Source	Remarks
Aluminium structure/frame (6061 aluminium)	Option 1: Custom PyroSat 3U 6061-T6 aluminum structure	1 set	$750	Estimation	ASU/SDSL fabrication allowance.
	Option 2: EnduroSat 3U CubeSat Structure II	1 set	$3,500	Marketplace catalog price [76]	
Type-III hard anodizing process	Hard anodize on dispenser-contact surfaces	1	$250	Estimation	The CubeSat Design Specification calls for hard anodizing on these surfaces [2, 2.2.13]; Type III is the planned process. Vendor processing allowance.
PCB mounting hardware	Aluminum/stainless standoffs, fasteners, spacers & inserts	1 set	$150	Estimation	Upper-bound hardware allowance.
Thermal camera mounting / ruggedized housing (including cooling and heating)	Custom Boson+ 640 camera and lens aluminium housing	1	$1250	Estimation	Includes mechanical and thermal integration.
Antenna mounting / deployment hardware	Custom brackets, hinges, restraints & release hardware	1 set	$200	Estimation	Excludes antenna and PCB cost.

Sub-Total			**$2,600-$5,100**		Lower: $2,600 Upper: $5,100

Attitude determination & control (ADCS)

PyroSat is expected to use a mission-specific ADCS developed in-house by the SDSL ADCS team. The cost of this system has not yet been determined. ADCS with COTS option at $26,500-$33,100 cash cost, breakdown is below.

Part/item	**Selected component/ model**	**Qty.**	**Estimated Cost (USD)**	**Cost Basis**	**Remarks**
Control and Sensor board	TM2Space AeroMagControl	1	$3,950	Manufacturer catalog price [18]	Includes the IMU and six I2C sun sensors.
Reaction Wheels	TM2Space SpaceWheel30	3	$12,011.	Manufacturer catalog price [19]	
Magnetorquers	TM2Space HyperMagCoil [20]	3	$536	Manufacturer catalog price [20]	AeroMagControl already provides X and Y magnetorquers and a Z-axis air torquer, so these coils are optional and are needed only if its torque proves too small
Star Tracker	Option 1: TM2Space StarSense Lite star tracker	1	$16,594	Manufacturer catalog price [77]	
	Option 2: KSST-01 star tracker	1	No quotation yet. $10,000	Preliminary allowance	The star tracker in the ADCS text, mass, power and CAD tables (56 x 60 x 92.8 mm, 175 g, 60 arcseconds).
Sub-total:			**$26,500-$33,100**		Lower: $26,500 Upper: $33,100

Electrical Power System (EPS):

Part/item	Selected component/model	Quantity	Total cost ($USD)	Cost basis / source	Remarks
Power management and Distribution board	Option 1: GomSpace NanoPower P60	1	Awaiting quote	Quote requested [15]	This option might exceed PyroSat's needs
	Option 2: TM2Space PowerDrive - EPS for CubeSats	1	$8,268	Manufacture catalog price [78]	Benchmark for COTS price
	Option 3: Custom-built PyroSat EPS	1	$1,500	Estimation	EPS based of SquidSat structure
Battery	Option 1: GomSpace NanoPower BPX	1	Awaiting quote	Quote requested [15]	Works with the P60
	Option 2: Jauch 3350mAh	6	$96	Marketplace catalog price [79]	Battery-pack compatibility to be verified during integration.
Battery Heater	Minco HAP6736 Space Flight Thermofoil Heater	1	$122	Manufacturer catalog price [70]	Final heater sizing and duty cycle will be verified through thermal analysis.
1U solar panels	CubeSat Solar Panel DHV-CS-10	9	$14,661	Catalog price [33]	Baseline. The price is the catalog price per two-cell panel. DHV has to quote the custom 7-cell wings and 4-cell body panel before it is firm

Deployable solar-panel mechanisms (accounted in Structure section)			Included		The custom 80.5 mm wings carry the DHV cells priced in the line above. The substrate and hinge cost sits in the Structure section
Sub-total			**$16,379-$23, 147**		Lower: $16,379 Upper: $23,147

The SquidSat bus includes an electrical power system and a 72.2 Wh battery pack built from six Jauch 3350 mAh lithium-ion cells in a 2S3P configuration [1]. PyroSat's baseline reuses this design. SDSL will fabricate the EPS in house from SquidSat's published design, modified for the new solar array as described in PyroSat EPS Architecture, and will assemble the battery pack from the same Jauch cells. The baseline is therefore priced as the custom-built PyroSat EPS (Option 3) and the Jauch cells (Option 2), which sets the lower bound of the sub-total. The commercial options, the GomSpace NanoPower P60 and BPX and the TM2Space PowerDrive, serve as price benchmarks and as a contingency if the in-house EPS is not ready in time or does not pass qualification. The upper bound of the sub-total reflects the TM2Space PowerDrive, because the GomSpace quotes are still pending.

Testing:

Part/item	**Provider**	**Cost ($ USD)**	**Notes / Source**
Vibration testing	ASU IPL	0	
Thermal-vacuum testing	ASU IPL	0	
Basic RF testing	ASU IPL	0	
EMI/EMC/interference testing	Other labs at ASU - Dr. Trichopoulos	$5000 - Upper allowance estimate	Not confirmed on exact labs.
Payload calibration/testing	ASU	0	Depends on what PyroSat needs
Balloon test	ASU SDSL	$2000	Upper allowance, Estimation
Battery qualification/testing	ASU IPL	$2500	Upper allowance, Estimation

Deployment testing	ASU IPL	$500	Consumables and replacement hardware
Sub-Total		**$10,000**	

Launch: (Assuming ISS orbit)

Selected provider	**Cost ($USD)**	**Cost basis / source**	**Remarks**
NASA CSLI	$0	NASA CSLI	Parallel path; $0 launch cost applies only if PyroSat successfully applies for and is selected through CSLI.
Commercial rideshare	$120,000	$40,000 per U of volume as per industry standard	Contingency if no funded CSLI opportunity is available
Sub-Total	**$120,000**		The baseline is ISS deployment; $120,000 applies only if a commercial rideshare is needed.

Regulatory:

Part/item	**Agency**	**Cost ($USD)**	**Cost Basis / Source**	**Remarks**
Small Satellite FCC Application	FCC	$2,555	Current planning value; verify against fee schedule at filing.	Streamlined small-satellite license application under Part 25.
Annual small-satellite regulatory fee	FCC	$14,520	FCC FY2026 regulatory fee order	FY2026 fee per small-satellite license/call sign.
Sub-Total		**$17,075**		

Total Cost of Satellite:

Sub-section:	**Lower Estimate:**	**Higher Estimate:**

Thermal Payload	$10,681	$10,681
GNSS-R Antennas	$2,000	$2,000
GNSS-R RF Front End	$750	$750
GNSS Navigation	$13,500	$14,030
Payload Computing	$40,002	$40,002
S-band Communication	$10,578	$50,672
UHF/LoRa Radio	$544	$13,700
Structure	$2,600	$5,100
ADCS	$26,500	$33,100
EPS	$16,379	$23,147
Testing	$10,000	$10,000
Regulatory	$17,075	$17,075
Sub-total:	**$150,609**	**$220,257**

Ground Station Upgrade:

Part/item	**Selected component/model**	**Quantity**	**Total cost ($USD)**	**Cost basis / source**
SDR Receiver	Ettus USRP B206mini-i	1	$1,820	Manufacturer catalog price [57]
RX Low-Noise Amplifier	Mini-Circuits ZQL-2G22G70 W3+	1	$456	Manufacturer catalog price [58]
RX Bandpass Filter	Mini-Circuits ZVBP-2250-S+	1	$523	Manufacturer catalog price [59]

S-Band Receive Feed	KrakenRF Discovery S-Band Feed	1	$113	Manufacturer catalog price [60]
S-band parabolic reflector	Skyware Global 2.4 m Antenna Reflector	1	$2,495	Manufacturer catalog price [61]
Antenna Rotator	Existing ASU rotator	1	$0	Existing SDSL hardware
Low-Loss RF Coax	Times Microwave LMR-600	1	$200	Manufacturer catalog price [62]
Lightning / Surge Protection	PolyPhaser S-band coaxial protector	1	$160	Manufacturer catalog price [63]
LNA Bias / Power Hardware	Mini-Circuits ZFBT-4R2GW+	1	$112	Manufacturer catalog price [64]
Outdoor RF Enclosure	Polycase HP-060504	1	$31	Manufacturer catalog price [65]
Enclosure / Mounting Hardware	COTS glands, brackets, fasteners	1	$75	Estimation
RF Test / Integration Hardware	Loads, attenuators, jumpers, adapters	1	$300	Estimation
Ground Computer	Existing ASU workstation	1	$0	Existing SDSL hardware
Sub-Total			**$6,285**	

Launch is the largest cost factor for the mission. NASA's CubeSat Launch Initiative (CSLI) would cost the team nothing, and the schedule to an April 2029 readiness date makes it a realistic option. No call is currently open, but a launch follows selection by about two years, so a selection from a call opening in 2027 would still fit an April 2029 readiness date. Because CSLI selection is competitive and NASA sets the launch date, PyroSat treats it as a parallel path to pursue rather than the baseline, and it is the path that removes the launch cost if funding falls

short. A rideshare broker launch deployment costs $120,000 for PyroSat’s 3U size and gives an orbit near 400 km at 51.6 degrees that suits the GNSS-R geometry. It is the recommended baseline path because its date can be planned around PyroSat’s readiness. The orbital lifetime is six months to two years in one source and 5 to 12 months in the quote tracker.

PyroSat's current bottom-up satellite cost estimate ranges from $150,609 to $220,257. In addition to the spacecraft itself, the mission requires an estimated $6,285 ground-station upgrade to support the proposed S-band communications architecture while reusing existing SDSL infrastructure where possible. Including this ground-station upgrade cost increases the preliminary mission cost, prior to launch and integration, to approximately $156,894–$226,542. These values form the baseline used in the following launch-scenario comparison.

Total Price Estimate Summary:

Scenario	**Lower Estimate**	**Upper Estimate**	**Notes**
NASA CSLI	$156,894	$226,542	NASA sponsored launch covers all related launch fees.
Broker rideshare to ISS orbit	$276,894	$346,542	Includes preliminary $120,000 launch/integration allowance.

Based on the current preliminary estimates, PyroSat's total mission cost ranges from approximately $156,894 to $226,542 under the most favorable sponsored-launch scenario, which assumes that launch and deployment costs are externally supported. Under the currently preferred ISS deployment scenario, the estimated mission cost increases to approximately $276,894 to $346,542, depending on the final cost of launch and integration. These values include the current spacecraft estimate and the approximately $6,285 ground-station upgrade required to support the proposed communications S-Band architecture.

PyroSat's cost estimate is intended as a preliminary planning budget and will continue to evolve as the spacecraft design matures. Where possible, subsystem costs are based on identified commercial components and published vendor pricing. Components for which a final design or supplier has not yet been selected are represented using preliminary Estimations or planning allowances. These values will be replaced with material pricing and vendor quotations as procurement decisions are made.

Launch, integration, and deployment remain among the largest uncertainties in the overall mission cost. Because a launch opportunity and deployment provider have not yet been selected, PyroSat does not assume a single fixed ISS deployment price. Instead, the current budget evaluates a range of potential launch and integration costs, which will be replaced by

provider-specific pricing once a viable launch opportunity is identified. Additional areas of cost uncertainty include the electrical power system, environmental and qualification testing, GNSS-R antenna development, and portions of the S-band ground-segment upgrade. As these subsystems progress through prototyping, and component selection, the preliminary estimates will be updated to reflect the resulting hardware and testing requirements.

Funding Strategy

SDSL currently operates with a limited annual funding meaning PyroSat requires external funding from corporate or public benefactors to implement and eventually fly the PyroSat mission. Potential funding pathways under consideration include lab sponsors internal ASU, Arizona Space Grant opportunities, industry sponsorship, and other research or education grants. NASA's CubeSat Launch Initiative will also be investigated as a potential path for launch integration and deployment if PyroSat satisfies the program's eligibility, schedule, and selection requirements.

Cost reduction will be pursued through component reuse from previously CubeSat missions developed at SDSL, in-house development where technically feasible, reuse of existing SDSL ground-station equipment for LoRa's UHF transmission and receive, and continued subsystem research to further reduce to the cost as the design itself is matured. Any reduction that affects a science or mission requirement can be investigated; however, it will be evaluated under its effect on mission objectives before being adopted. For example, the removal of a payload capability, or communication system like S-Band or UHF.

Risk Analysis

Likelihood (L) and Consequence (C) are scored 1 to 5, multiplied to find the Total (T). The risks are ranked by their total risk to allow focusing on the most important risks. Mitigations are assigned to each risk to lower their likelihood or consequence, and encompass both planned implementations and possible solutions.

Thermal Imaging

Risk Description	L	C	T	Mitigation
Failure of on-board calibration of the Boson+ thermal imager	3	3	9	See calibration procedures and processing procedures, ground calibration,
Thermal control system cannot maintain a constant operating temperature for the Boson+	4	2	8	See thermal isolation and management procedures, vacuum testing

Outgassing occurs and contaminates the imager lens	2	3	6	See the bakeout procedure in the LWIR Thermal Sensor section [40], materials screening
The FOV of the imager is obstructed by a part of the satellite	1	4	4	CAD modeling, deployment testing, FOV analysis

GNSS-Receiver

Risk Description	**L**	**C**	**T**	**Mitigation**
Modified antenna performs worse/differently than expected	3	5	15	Prototype fabrication, RF characterization, simulations (HFSS/CST)
Geolocation algorithm produces large position errors	2	3	6	Validation against other methods (e.g. CYGNSS), DEM refinement
GNSS reflected signals are weaker than expected	2	3	6	Antenna optimization, front-end gain and filtering adjustments
DDM quality is insufficient for soil-moisture data	2	2	4	Balloon flight validation, calibration, algorithm refinement
Soil moisture model fails across varying vegetation or terrain types	1	2	2	Cross-calibration with other sources (e.g. SMAP) to confirm accuracy, possible incorporation of vegetation corrections from other models

ADCS

Risk Description	**L**	**C**	**T**	**Mitigation**
Reaction wheel failure or saturation	2	5	10	Wheel momentum management and testing, magnetorquer desaturation capability testing
In-house star tracker inaccuracies	3	2	6	Comparison against other models, back-up commercial variants
Post-slew settling time reduces science collection	2	2	4	Hardware testing, control law tuning

Data Management

Risk Description	L	C	T	Mitigation
Latch-up or upset in the Zynq, AD9361, or memory	3	4	12	Current-limited switches with automatic power cycling, error-correcting storage, scrubbing, shielding where mass allows
Storage corruption (eMMC)	3	3	9	Error-correcting code and redundant metadata
FPGA processing fails with GNSS-R workload or payload data	2	4	8	Benchmarking, prototype implementation, weather balloon test, optimization
NanoCom SDR MK3 lead time (about 12 months) delays FlatSat integration of the GNSS-R chain	3	4	12	Place the order as soon as the GomSpace quote closes, develop the FPGA chain on an evaluation Zynq board until delivery, keep the SDR on the critical path of the schedule

Power

Risk Description	L	C	T	Mitigation
Negative orbital energy balance	3	4	12	Generation model across the full beta cycle, 25% margin, added solar area and a larger Sun pointing share, duty-cycled science, low-power safe mode
EPS board or regulator failure	2	5	10	Conservative derating, independent supervisor and watchdog, full FlatSat and thermal vacuum testing
Battery cannot charge when cold	3	3	9	Charge inhibit logic, thermostat controlled heater, thermal vacuum test
Underestimated payload power	1	5	5	FlatSat power measurements, margin allocation, prototype calculations

Communications

Risk Description	L	C	T	Mitigation

S-Band transmitter interferes with the GNSS-R receiver	3	3	9	Mutually exclusive modes, filtering, interference test
Science data exceeds available downlink speed	2	3	6	Prioritized data selection, compression, adaptive collection strategies
S-band link margin insufficient	2	2	4	Link budget analysis, antenna optimization, lower data-rate fallback modes

Mechanical

Risk Description	**L**	**C**	**T**	**Mitigation**
Launch vibration damages payload	4	3	12	Vibration testing, structural analysis, fastener retention
Solar panels or antenna array fail to deploy	2	5	10	Redundant burn-wire releases, deployment tests, body-mounted cells as a fallback
Science payloads or onboard electronics do not remain in an operable thermal threshold	3	3	9	Thermal management systems, temperature monitoring, low-power modes
Center-of-gravity requirement is incorrect	1	3	3	Mass tracking throughout development, configuration audits, computer analysis

Science

Risk Description	**L**	**C**	**T**	**Mitigation**
Insufficient overlap of GNSS-R observations and thermal imagery	2	3	6	Weather balloon testing, ground-based co-registration, predefined calibration sites
Wildfire risk model shows little improvement to current analysis	2	2	4	Altering of risk model, use of outside data to increase accuracy, create databases for comparison
Cloud contamination reduces thermal data usability	4	2	8	See thermal processing procedure
Geolocation uncertainty exceeds GNSS-R footprint	1	4	4	Star tracker-based attitude implementation, DEM-refined

				processing

Proposal Appendix

References

1. A. Bournias, J. Dukowitz, M. Jaber, T. Field, M. Nguyen, K. Joshi, N. Campos, D. Riordan, R. La Ra, J. Luse, J. Sink, T. Nielsen, R. Ontivaros, M. Goryll and J. Dubois, "SquidSat: A Modular, Open-Source 3U CubeSat Bus for Low-Cost CubeSat Missions," paper SSC26-P1-30, Small Satellite Conference, 2026.
2. Cal Poly CubeSat Program, "CubeSat Design Specification, Rev. 14.1," 2022. https://www.nasa.gov/wp-content/uploads/2018/01/cubesatdesignspecificationrev14_12022-02-09.pdf
3. Teledyne FLIR, "Boson+ Product Datasheet, Rev. 114." https://www.acalbfi.com/nl/file/16133/Boson-Plus-Product-Datasheet-Rev-114.pdf
4. Raspberry Pi, "RP2350" product page, https://raspberrypi.com/products/rp2350, and "RP2350 Datasheet," https://pip-assets.raspberrypi.com/categories/1214-rp2350/documents/RP-008373-DS-3-rp2350-datasheet.pdf
5. S. Gleason, A. O'Brien, A. Russel, M. M. Al-Khaldi and J. T. Johnson, "Geolocation, Calibration and Surface Resolution of CYGNSS GNSS-R Land Observations," Remote Sensing 12(8), 1317, 2020. https://www.mdpi.com/2072-4292/12/8/1317
6. Muñoz-Martín et al., "The Flexible Microwave Payload-2: A SDR-Based GNSS-Reflectometer and L-Band Radiometer for CubeSats," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2020. https://upcommons.upc.edu/entities/publication/ecd71cd5-4ab3-439a-89d8-e63becb1c417
7. Blue Canyon Technologies, ADCS product data sheet, 2026. https://bluecanyontech.com/wp-content/uploads/ADCS-2026.pdf
8. CubeSpace, Gen2 CubeADCS. https://satcatalog.com/component/gen2-cubeadcs
9. TY-Space, "CubeSat Star Tracker Price and Features Explained." https://www.ty-space.net/cubesat-star-tracker-price-and-features-explained/
10. New Space Economy, "The Small Satellite Mission: A Guide to Development Costs and Timelines," 2025. https://newspaceeconomy.ca/2025/08/25/the-small-satellite-mission-a-guide-to-development-costs-and-timelines/
11. SpaceNexus, cost to launch a CubeSat. https://spacenexus.us/guide/cost-to-launch/cubesat
12. Federal Communications Commission, FY2025 regulatory fees, Federal Register, 7 July 2025. https://www.govinfo.gov/content/pkg/FR-2025-07-07/pdf/2025-12579.pdf.
13. D. G. Gilmore, ed., Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies, 2nd ed., The Aerospace Press, 2002.

14. F. P. Incropera, D. P. DeWitt, T. L. Bergman and A. S. Lavine, Fundamentals of Heat and Mass Transfer, Wiley.
15. PyroSat Quote Tracker (internal team file), prices found 8 October 2026, including EnduroSat listings and NASA CubeSat Launch Initiative status.
16. PyroSat Current Outline (internal team file), which cites Phoenix CubeSat budget figures and the passive magnetic stabilization estimate.
17. PyroSat team power analysis (internal team file).
18. TM2Space, AeroMagControl ADCS for CubeSats, product page. https://www.tm2.space/product-page/aeromagcontrol-adcs-for-cubesats?currency=GBP
19. TM2Space, SpaceWheel30 (30 mNms reaction wheel), product page. https://www.tm2.space/product-page/spacewheel30-30mnms-reaction-wheel
20. TM2Space, HyperMagCoil magnetorquers for CubeSats, product page. https://www.tm2.space/product-page/hypermagcoil
21. Nanoracks, CubeSat Deployer (NRCSD) Interface Definition Document. https://s3vi.ndc.nasa.gov/ssri-kb/static/resources/Nanoracks-CubeSat-Deployer-NRCSD-IDD.pdf
22. E. Rakhmatulina et al., "Soil moisture influences on Sierra Nevada dead fuel moisture content and fire risks," Forest Ecology and Management, 2021. https://www.sciencedirect.com/science/article/pii/S0378112721004679
23. S. O et al., "Observational evidence of wildfire-promoting soil moisture anomalies," Scientific Reports, 2020. https://pubmed.ncbi.nlm.nih.gov/32620812/
24. Nelson et al., "Integration of GNSS-R Derived Soil Moisture into the USGS Wildland Fire Potential Index," IGARSS, 2025. https://sreejanag.com/Documents/Nelson_IGARSS2025.pdf
25. W. H. Maes and K. Steppe, "Estimating evapotranspiration and drought stress with ground-based thermal remote sensing in agriculture: a review," Journal of Experimental Botany, vol. 63, no. 13, pp. 4671-4712, 2012. https://doi.org/10.1093/jxb/ers165
26. NASA Marshall Space Flight Center, "Ruggedized Infrared Camera," Technology Opportunity Sheet MFS-TOPS-1082, 2024. https://science.nasa.gov/wp-content/uploads/2024/03/mfs-tops-1082.pdf
27. Teledyne FLIR, "Boson+ Optical Specifications," support center knowledge base. https://www.oem.flir.com/en-150/support/support-center/knowledge-base/boson-plus-optical-specs/
28. J. E. Pryor et al., Ruggedized Miniaturized Infrared Camera System for Aerospace Environments, U.S. Patent 11,622,062 B1, 4 Apr. 2023.
29. R. Louw, PyroSat wildfire risk-index analysis (Zenodo record 23299028; Wildfire Risk Scoring), 2026. https://zenodo.org/records/23299028
30. CIRiS (Compact Infrared Radiometer in Space), eoPortal. https://www.eoportal.org/satellite-missions/ciris

31. C. C. Chew and E. E. Small, “Soil Moisture Sensing Using Spaceborne GNSS Reflections: Comparison of CYGNSS Reflectivity to SMAP Soil Moisture,” Geophysical Research Letters 45(9), 4049 to 4057, 2018. https://doi.org/10.1029/2018GL077905
32. SkyFox Labs, piPATCH-L1E1 Product Datasheet, Rev. B, 2024. http://www.skyfoxlabs.com/pdf/piPATCH-L1E1_Datasheet_rev_B_2024.pdf
33. DHV Technology, “CubeSat solar panel DHV-CS-10: 1U space-qualified solar array,” CubeSatShop, 2026. https://www.cubesatshop.com/product/cubesat-solar-panels-dhv/
34. J. R. Wertz, D. F. Everett and W. J. Larson, eds., Space Mission Engineering: The New SMAD, Microcosm Press, 2011.
35. GomSpace A/S, NanoCom SDR MK3 Datasheet, Rev. 3.0, Document No. 1026808, 2 Apr. 2025. https://gomspace.com/wp-content/uploads/2025/09/Data-Sheet-NanoCom-SDR-MK3-1.pdf
36. NanoAvionics, CubeSat S-band transceiver, product page. https://nanoavionics.com/cubesat-components/cubesat-s-band-transceiver/
37. Satlab A/S, SRS-4 Full-Duplex High-Speed S-Band Transceiver Datasheet, Rev. 1.2, 16 Aug. 2024. https://satlab.com/resources/SLDS-SRS4-1.2.pdf
38. SkyFox Labs s.r.o., “piPATCH-S-DL/FM (Flight Model),” 2026. https://www.skyfoxlabs.com/product/37-pipatch-s-dl?category=1
39. Jauch Quartz GmbH, Protected Li-Ion Rechargeable Battery 18650, LR1865HB, Jauch No. 250669, Version 1.2, Oct. 2022. https://www.farnell.com/datasheets/4128413.pdf
40. ASTM International, ASTM E2900-24, Standard Practice for Spacecraft Hardware Thermal Vacuum Bakeout.
41. BeaverCube CubeSat Mission, eoPortal. https://eoportal.org/satellite-missions/beavercube
42. S. Selvadurai, A. Chandran, D. Valentini, B. Lamprecht, “Passive Thermal Control Design Methods, Analysis, Comparison, and Evaluation for Micro and Nanosatellites Carrying Infrared Imager,” Applied Sciences, vol. 12, no. 6, 2858, 2022. https://doi.org/10.3390/app12062858
43. Victrex, VICTREX PEEK POLYMER 90G Datasheet. https://victrex.com/downloads/datasheets/victrex-peek-90g
44. NASA, “7.0 Thermal Control,” State-of-the-Art of Small Spacecraft Technology, Small Spacecraft Systems Virtual Institute, 7 May 2026. https://www.nasa.gov/smallsat-institute/sst-soa/thermal-control/
45. Thermal Space, Copper Thermal Straps. https://thermal-space.com/copper-thermal-straps/
46. Mouser Electronics, Thermistors. https://www.mouser.com/en/c/circuit-protection/thermistors/

47. Minco, Thermal Solutions Design Guide. https://www.minco.com/wp-content/uploads/Minco-Thermal-Solutions-Design-Guide.pdf
48. Arizona State University, “Phoenix: Thermal Imaging to Explore the Impact of Urban Heat Islands on the Environment,” NASA Space Grant proposal, 2015. https://phxcubesat.asu.edu/sites/default/files/general/phoenix_proposal_fall2015.pdf
49. “ASTERIA,” Gunter’s Space Page. https://space.skyrocket.de/doc_sdat/asteria.htm
50. NASA Marshall Space Flight Center, MSFC-SPEC-1238 Rev. A, Thermal Vacuum Bakeout Specification for Contamination Sensitive Hardware, 29 June 2007. https://s3vi.ndc.nasa.gov/ssri-kb/static/resources/MSFC-SPEC-1238-A.pdf
51. G. C. Hulley, S. J. Hook, E. Abbott, N. Malakar, T. Islam, M. Abrams, “The ASTER Global Emissivity Dataset (ASTER GED): Mapping Earth’s emissivity at 100 meter spatial scale,” Geophysical Research Letters, vol. 42, no. 19, pp. 7966-7976, 2015. https://doi.org/10.1002/2015GL065564
52. KAIROSPACE Co., Ltd, KSST-01 5 arcsec Star Tracker for CubeSat Missions, Satnow product listing. https://www.satnow.com/products/star-trackers/kairospace-co-ltd/37-1189-ksst-01
53. C. Maffei, R. Lindenbergh, M. Menenti, “Combining multi-spectral and thermal remote sensing to predict forest fire characteristics,” ISPRS Journal of Photogrammetry and Remote Sensing, vol. 181, pp. 400-412, 2021. https://doi.org/10.1016/j.isprsjprs.2021.09.016
54. EnduroSat. “S-Band Transceiver.” *EnduroSat*, https://www.endurosat.com/products/s-band-transceiver/. Accessed 25 Oct. 2026.
55. DWMZone. “RFM98PW/RFM95PW +33dBm Enhanced Power LoRa Long Range Transceiver Module.” *DWMZone*, https://dwmzone.com/en/lora-wireless-modules/353-rfm98pw-rfm95pw-33dbm-enhanced-power-lora-long-range-transceiver-module.html. Accessed 25 Oct. 2026.
56. NiceRF. “2PCS LoRa1278F30 1W 8km Long Distance RF Module.” *Tindie*, https://www.tindie.com/products/nicerf/2pcs-lora1278f30-1w-8km-long-distance-rf-module/. Accessed 25 Oct. 2026.
57. Ettus Research. “USRP B206mini-i.” *Ettus Research*, https://www.ettus.com/all-products/USRP-B206mini-i/. Accessed 26 Oct. 2026.
58. Mini-Circuits. “ZQL-2G22G70W3+ Low Noise Amplifier.” *Mini-Circuits*, https://www.minicircuits.com/WebStore/dashboard.html?model=ZQL-2G22G70W3%2B. Accessed 26 Oct. 2026.
59. Mini-Circuits. “ZVBP-2250-S+ Bandpass Filter.” *Mini-Circuits*, https://www.minicircuits.com/WebStore/dashboard.html?model=ZVBP-2250-S%2B. Accessed 26 Oct. 2026.
60. KrakenRF. “DISCOVERY-S-BAND-FEED-01 S-Band Discovery Feed.” *Mouser Electronics*,

https://www.mouser.com/en/ProductDetail/KrakenRF/DISCOVERY-S-BAND-FEED-01. Accessed 26 Oct. 2026.
61. Satcom Solutions. “Skyware Global 2.4m C-Band Antenna Reflector.” *Satcom Solutions*, https://satcomsolutions.org/product/skyware-global-2-4m-c-band-antenna-reflector/. Accessed 26 Oct. 2026.
62. Pasternack. “RF Adapters.” *Pasternack*, https://www.pasternack.com/rf-adapters-category.aspx. Accessed 26 Oct. 2026.
63. PolyPhaser. “TSX-NFF-P Connectorized Surge Protector.” *Mouser Electronics*, https://www.mouser.com/en/ProductDetail/PolyPhaser/TSX-NFF-P. Accessed 26 Oct. 2026.
64. Mini-Circuits. “ZFBT-4R2GW+ Bias Tee.” *Mini-Circuits*, https://www.minicircuits.com/WebStore/dashboard.html?model=ZFBT-4R2GW%2B. Accessed 26 Oct. 2026.
65. Polycase. “HP-060504 NEMA Polycarbonate Enclosure.” *Polycase*, https://www.polycase.com/hp-060504. Accessed 26 Oct. 2026.
66. FLIR Boson+ 640.” FLIR Cameras, https://flircameras.com/products/22640as24-sl-r . Accessed 27 Oct. 2026
67. Use of AI tools for research.
68. Thermal Applications, Inc. CuTS® Catalog. 2026 ed., 2026. Internal manufacturer catalog provided to Sun Devil Satellite Laboratory.
69. “CubeSat and SmallSat S-Band Transceivers: SRS-4.” *Kongsberg NanoAvionics*, https://nanoavionics.com/cubesat-components/cubesat-s-band-transceiver/. Accessed 27 Oct. 2026.
70. “HAP6736 Space Flight Thermofoil Heater.” *Minco*, https://www.minco.com/catalog/?catalogpage=product&cid=3_6-all-polyimide-thermofoil-heaters&id=HAP6736. Accessed 27 Oct. 2026.
71. “microPATCH-L1.” *SkyFox Labs*, https://skyfoxlabs.com/product/23-micropatch-l1. Accessed 27 Oct. 2026.
72. “GNSS Receiver Module (GPSRM 1) Kit.” *Pumpkin Space Systems*, https://www.pumpkinspace.com/store/p58/GNSS_Receiver_Module_%28GPSRM_1%29_Kit.html. Accessed 27 Oct. 2026.
73. “SC1509-13 RP2350A Microcontroller.” *DigiKey Electronics*, https://www.digikey.com/en/products/detail/raspberry-pi/SC1509-13/26241107. Accessed 27 Oct. 2026.
74. “piPATCH-S-DL.” *SkyFox Labs*, https://skyfoxlabs.com/product/37-pipatch-s-dl. Accessed 27 Oct. 2026.
75. “2PCS LoRa1278F30 1W 8KM Long Distance RF Module.” *Tindie*, https://www.tindie.com/products/nicerf/2pcs-lora1278f30-1w-8km-long-distance-rf-module/. Accessed 27 Oct. 2026.

76. “3U CubeSat Structure II.” *SatCatalog*, https://www.satcatalog.com/component/3u-cubesat-structure-ii/. Accessed 27 Oct. 2026.
77. “Star Tracker.” *TM2Space*, https://www.tm2.space/product-page/startracker. Accessed 27 Oct. 2026.
78. “PowerDrive EPS for CubeSats.” *TM2Space*, https://www.tm2.space/product-page/powerdrive-eps-for-cubesats. Accessed 27 Oct. 2026.
79. “LI18650JLS-HB-3.35AH Protected Lithium-Ion Battery.” *DigiKey Electronics*, https://www.digikey.com/en/products/detail/jauch-quartz/LI18650JLS-HB-3-35AH-PROTECTED/16837989. Accessed 27 Oct. 2026.
80. “Received Power and DDM Track for the Selected Track at the North of Russia.” *ResearchGate*, figure from *In-Orbit Validation of the FMPL-2 Instrument—The GNSS-R and L-Band Microwave Radiometer Payload of the FSSCat Mission*, www.researchgate.net/figure/Received-power-and-DDM-track-for-the-selected-track-at-the-North-of-Russia-collected-on_fig18_348144205. Accessed 29 Oct. 2026.
81. USGS, “Landsat Burned Area (Collection 2 Level-3),” Landsat Missions. https://www.usgs.gov/landsat-missions/landsat-burned-area
82. Muon Space, “Muon Space Deploys First Three Operational FireSat Satellites for Earth Fire Alliance,” 2026. https://www.muonspace.com/muon-space-deploys-first-three-operational-firesat-satellites-for-earth-fire-alliance/
83. European Space Agency, “Hellenic Fire System satellites launched for Greece,” 2026. https://www.esa.int/Applications/Observing_the_Earth/Hellenic_Fire_System_satellites_launched_for_Greece
84. NASA Jet Propulsion Laboratory, “ECOSTRESS.” https://ecostress.jpl.nasa.gov/
85. European Space Agency, “Copernicus Sentinels work together to monitor wildfires,” Sentinel Online. https://sentinel.esa.int/web/success-stories/-/copernicus-sentinels-work-together-to-monitor-wildfires
86. Arizona State University Ira A. Fulton Schools of Engineering, “CubeSat project propels students forward,” ASU Engineering News, July 28, 2026. https://news.engineering.asu.edu/2026/07/cubesat-project-propels-students-forward/
87. R. Ontiveros, Q. H. Dinh, T. Nielsen, J. Sink, N. Campos, M. Goryll, J. Dubois, “Coconut: Student-Led Development and On-Orbit Operation of a 1U LoRa Amateur Radio CubeSat,” Small Satellite Conference, 2026. https://digitalcommons.usu.edu/smallsat/2026/all2026/99
88. Pumpkin, Inc. “SUPERNOVA™ Firmware Reference Manual, Rev 4.4.” *Pumpkin Space Systems*, 2019, https://www.pumpkininc.com/space/SupMCU/SFR_Manual_4.4.pdf
89. K. Bonilla-Ormachea, H. Cuizaga, E. Salcedo, S. Castro, S. Fernandez-Testa and M. Mamani, “ForestProtector: An IoT Architecture Integrating Machine Vision and Deep

Reinforcement Learning for Efficient Wildfire Monitoring," arXiv:2501.09926, 2025. https://arxiv.org/abs/2501.09926

90. NovAtel Inc., OEM719 Installation and Operation User Manual, Version 7, 2026. https://docs.novatel.com/OEM7/Content/PDFs/PreviousVersions/OEM7_Installation_Operation_Manual_v7.pdf
91. HopeRF, RFM98PW LoRa module. https://www.hoperf.com/modules/lora/RFM98PW.html

Appendix A: Resumes

Remy Louw

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Aerospace Engineering student in Barrett, The Honors College seeking a software or embedded systems internship. CubeSat mission Principal Investigator and published preprint author, now building programming skills in MATLAB.

EDUCATION

Arizona State University, Ira A. Fulton Schools of Engineering *Expected Graduation: May 2030*

B.S. in Aerospace Engineering (Astronautics) | Barrett, The Honors College

- **Current Coursework:** Intro to Programming in MATLAB (MAE 215), Linear Algebra (MAT 242), Physics I: Mechanics and Lab (PHY 121/122), Intro to Engineering (FSE 100), EPICS Gold I (FSE 104), The Human Event I (HON 171)
- **Planned Spring 2027:** Circuits I (EEE 202), Physics II: Electricity and Magnetism and Lab (PHY 131/132), Calculus III (MAT 267), Differential Equations (MAT 275), Particles and Rigid Bodies I (MAE 201), EPICS in Action (FSE 404), The Human Event II (HON 272)
- **Prior Coursework:** Completed 15 AP and 19 dual enrollment courses at Arizona College Prep High School

PROJECTS & LEADERSHIP

PyroSat CubeSat | Principal Investigator *Oct 2026 – Present*

Sun Devil Satellite Lab (SDSL), Arizona State University

- Lead a student team defining a CubeSat mission that estimates wildfire risk from GNSS-R soil moisture and thermal imagery.
- Coordinate Risk, Cost, Feasibility, Citation, and Overview subteams to deliver the mission proposal to SEMTE leadership and faculty.
- Preparing a funding proposal for industry sponsors (Nov 2026) for a satellite estimated at $150k to $350k.
- Hand-sketched the 3U layout, then built an interactive 3D CAD model with AI-assisted code: dragonfly-pyrosat-bus.netlify.app

Wildfire Risk Scoring Methods Preprint | Sole Author *Oct 2026*

Published on Zenodo: zenodo.org/records/23299028

- Developed P(fire), a probabilistic wildfire risk framework fusing soil moisture and thermal data; basis of the PyroSat proposal.

EPICS Wind Tunnel Rebuild | Design Lead *Fall 2026 – Present*

Arizona State University, EPICS Program

- Own the design, budget, and build plan for a rebuilt subsonic educational wind tunnel for an EPICS community partner.
- Authored a 24-page construction proposal with a three-tier budget ($3,705 recommended), risk register, and 15-week build schedule.
- Estimated fan power and test-section speed from loss-coefficient analysis, and set six acceptance criteria (including ±3% flow uniformity) accepted by the partner.
- Diagnosed the root cause of recurring airflow problems by reviewing five semesters of team records and rig measurements.

SDSL Payload Pitch Competition | Payload Proposal Author *Sep 2026*

- Co-led an 8-person team on Ori-Tech, a motor-free origami deployable antenna; expected on future SDSL satellites.
- Co-wrote Eddy's Social Distancing Protocol, a contactless debris-manipulation demo using camera-guided electromagnets.

WORK EXPERIENCE

Cutco | Independent Sales Representative *July 2026 – Oct 2026*

- Managed a business-to-consumer sales pipeline, scheduling appointments through outbound calls and referrals.
- Delivered consultative in-person and virtual product demonstrations to prospective customers.

Walmart | Fashion Team Associate *Aug 2025 – May 2026*

- Assisted customers with product selection and maintained merchandising standards in the fashion department.
- Restocked inventory, organized the sales floor, and processed customer transactions.

VOLUNTEER EXPERIENCE & ACTIVITIES

St. Vincent de Paul (SVDP) Food Bank | Volunteer *Fall 2023 – Present*

- Assist with food distribution and pantry operations three mornings a week during school breaks.

Church Choir | Member *Fall 2023 – Present*

ASU Kendo Club | Member *Fall 2026 – Present*

Varsity Track and Field | Arizona College Prep High School *2024 – 2026*

SKILLS

- **Programming:** MATLAB (beginner, MAE 215 coursework)
- **Technical:** CubeSat mission and payload design, wind tunnel loss analysis, proposal writing, budgeting and scheduling

Erick Alvarez

Tempe, AZ | enalvar1@asu.edu | (928) 510-0735 | Github | Portfolio

Education

Arizona State University **Tempe, AZ**

Computer Science, BS, Honors, GPA: 4.00/4.00 *Expected December 2028*

- **Relevant Coursework:** Object-Oriented Programming, Multivariable Calculus, Discrete Math, Principles of Programming, Introduction to Programming Languages, Digital Design Fundamentals, Applied Linear Algebra
- **Extracurriculars:** Software Developers Association (SoDA), Sun Devil Satellite Lab (SDSL)
- **Honors:** Dean's List (Fall 2025 - Present), Horatio Alger State Scholar, Hispanic Scholarship Fund Scholar

Technical Skills

Languages: C++, Java, English (Native), Spanish (Fluent)
Tools & APIs: OpenGL, GLSL, LWJGL, Raylib, Dear ImGui, GitHub, Visual Studio, IntelliJ
Certifications: Google Cybersecurity Professional, Fortinet Certified Fundamentals (FCF), Google IT Support

Projects

Procedural Terrain Generation with Tesselation in C++

[GitHub Repository]

- Engineered a real-time 3D terrain rendering engine in C++ using modern OpenGL and GLSL shaders.
- Implemented dynamic Level of Detail (LOD) via tessellation evaluation shaders to subdivide geometry based on camera distance, exposing real-time controls to balance visual quality and target framerates.
- Generated realistic heightmaps via Perlin noise layered with fractal Brownian motion (fBm), incorporating custom distance fog and height-based texture blending shaders for grass and mud texturing.

CPU Raytracer with SFML in C++

[GitHub Repository]

- Built a CPU-based raytracer in C++ utilizing SFML for frame rendering and image export.
- Implemented ray-sphere intersection math, surface normal calculations, dynamic material properties (Lambertian diffuse, metallic reflection, and dielectric refraction), and anti-aliasing via jittered sumpersampling.

Procedural Terrain Generation with LWJGL in Java | *CSE 205 Object-Oriented Programming*

[GitHub Repository]

- Built an interactive 3D terrain generator using Java and LWJGL (OpenGL bindings) with real-time parameter controls via Dear ImGui controls.
- Integrated a Phong illumination model with dynamic light positioning, smooth normal calculation, and WASD camera control matrices.
- Architected a modular rendering engine using Object-Oriented Programming (OOP) principles, ensuring clean abstraction between the LWJGL graphics pipeline, ImGui interface, and procedural generation algorithms.

Work Experience

America Reads **Phoenix, AZ**

Tutor *September 2025 - Current*

- Received **Tutor of the Month** (March 2026) out of 24 other peer tutors for exceptional reliability, student engagement, and lesson quality.
- Designed and executed custom weekly lesson plans for 4+ K-8 students for math and reading proficiency.

Andrew Curtis

480-930-1341 | Chandler, AZ | atcurtis08@gmail.com | github.com/meiszwflz | linkedin.com/in/meiszwflz/ | wflz.dev

Education

Arizona State University Aug 2026 — May 2030
Mechanical Engineering (Computational Mechanics) *Tempe, AZ*
- Barrett, The Honors College

Basha High School Jul 2022 — May 2026
High School Diploma *Chandler, AZ*
- Cumulative GPA: 4.0/4.0
- Relevant Coursework: AP Physics C: Mechanics, Multivariable Calculus, Differential Equations, Linear Algebra, AP CompSci, Security Fundamentals and Ethics in IT, Linux OS, Intro to LAN & Security

Extracurricular Activities

Vex V5 Robotics Competition (V5RC) May 2022 — Apr 2026
- Served as Lead Programmer on team 781X Dark Matter
- Used collaboration, brainstorming, and prototyping to solve technical problems and build solutions under tight time constraints
- Documented engineering design processes and software architecture, contributing to multiple Design and Think awards
- Researched and developed advanced navigation algorithms, improving efficiency and competition performance
- 90 hours of mentoring middle school robotics teams, translating complex technical concepts into clear instruction

Scouts BSA Mar 2019 — Present
- Earned Eagle Scout rank by directing 100+ volunteer hours and 10 volunteers to design and construct a mobile pegboard for a middle school robotics team
- 300 hours of volunteering at and leading service projects
- Served as Senior Patrol Leader, leading planning, delegation, and execution of large group activities
- Delivered formal presentations on servant leadership and communication skills to groups of 50+ participants
- Developed tools and documentation to streamline training and course material production

Institute of Cyber Operations and Networking Club at Basha High School Jul 2022 — May 2025
- Attended Future Business Leaders of America (FBLA) conferences, learning key leadership and business skills
- Crafted and delivered a professional network design proposal to a company executive in a FBLA role play competition

Projects

Community Lead, LemLib (lemlib.readthedocs.io) Nov 2023 — Present
- Developed open source tools to streamline build, deployment, and review processes, such as: lemlib/pros-depot and pr-comment.
- Guided a team in developing their own libraries and tools, such as: lemlib/pros-build and lemlib/gamepad.
- Collaborated with contributors to gather requirements, prioritize features, and resolve technical issues.
- Provided technical support and documentation to help users adopt and effectively use the library.

Work Experience

Cybersecurity Intern Jan 2026 — May 2026
AaronLokoteJones LLC. *AZ*
- Enrolled in courses focused on incident management and response, cybersecurity best practices, and technical documentation
- Collaborated with a team of interns to produce cybersecurity educational materials for law enforcement

Skills & Certificates

- **Programming Languages**: Python, C/C++, JavaScript/TypeScript, Java, Bash, Rust, Nushell, Makefile
- **Technologies**: Onshape, Git, Linux, Docker, GitHub Actions, SQL, GDB, VSCode Extension Development, Typst
- **IT Fundamentals (ITF+)**, CompTIA Feb 2024

MEHRAB RAHMAN

Tempe, AZ | 304-840-6577 | mehrab.wv@gmail.com
linkedin.com/in/mehrab-rahman-383713400 | github.com/Mehrabwv | valleystemtutoring.com

ENGINEERING PROFILE

First-year Electrical Engineering student with practical experience building custom electronics and microcontrollers. Skilled in hardware prototyping, leading STEM workshops and tutoring experience.

EDUCATION

Arizona State University - Ira A. Fulton Schools of Engineering **Expected 2030**
BSE, Electrical Engineering | Tempe, AZ
Relevant Coursework: Calculus I, C++ Programming, Introduction to Engineering, General Chemistry
Activities: IEEE Student Branch, Combat Robotics, Sun Devil Satellite Laboratory (SDSL)

Catholic Junior College **2024 - 2025**
Singapore | Singapore-Cambridge GCE A-Level curriculum

TECHNICAL SKILLS

Programming: C++, Python | **Hardware/Platforms:** Arduino, ESP32, Raspberry Pi | **Prototyping:** 3D printing, laser cutting, introductory PCB design

ENGINEERING PROJECTS

Custom PCB-Based MP3 Player **In Progress**
- Designing a custom portable audio player around a purpose-built PCB and 3D housing; currently developing the circuit architecture and learning component selection, PCB layout and routing.
- Researching audio-path components, including capacitor selection, with the goal of improving signal quality while building practical PCB design experience.

ESP32-S3 Plant Watering System **Personal Project**
- Built an ESP32-S3-based plant watering system with internet connectivity for remote monitoring and control, combining microcontroller programming with practical sensor/actuator integration.

EXPERIENCE & LEADERSHIP

Founder & STEM Tutor - Valley STEM Tutoring **Jun 2025 - Aug 2026**
- Created and independently ran a one-on-one tutoring service, teaching 5 students ages 10-16 in mathematics, sciences and hands-on STEM topics; two students continued for approximately one year.
- Developed individualized lesson plans and managed scheduling, parent communication and the full student experience without an agency.

Secretary / Executive Committee Member - STEM Club, Catholic Junior College **Jan 2024 - Jun 2025**
- Helped lead an approximately 80-member STEM club and co-led a 3-person committee that planned and delivered two outreach workshops for about 50 primary-school students.
- Served as main instructor and trained 13 student volunteers for workshops on Arduino line-tracing robots and laser cutting; club-built robots used laser-cut acrylic and 3D-printed parts.

AWARDS & ADDITIONAL

Awards: National STEM Championship Quarterfinalist (2023); Australian Mathematics Competition Credit Award (2024, top 33% in region); Outstanding Academic Performance, Catholic Junior College (2024); Young Engineer Award - Applied Learning (2021)
Languages: English (native/bilingual), Bengali (native/bilingual)

Abhay Padiyar

Bachelor of Science in Engineering (Robotics)

abhaypadiyar8@gmail.com
510-750-2642

SUMMARY & SKILLS

- Focused on developing hardware through embedded systems design and multi-sensor integration for autonomous, space-constrained robots.
- Proficient in hands-on circuit debugging, custom PCB layout in KiCad, low-level firmware development across PSoC and ESP32 microcontrollers using oscilloscopes, logic analyzers, and LTspice, as well as FPGA logic design, hardware description, and RTL verification using VHDL and Verilog.
- Demonstrated technical leadership by directing cross-functional engineering teams, managing project budgets, and driving multi-disciplinary systems integration across complex hardware and robotics builds.

PROJECT EXPERIENCE

Aerial Search & Rescue Drone | *In Progress (Expected 2027)*
Tools: ESP32, LiDAR, Thermal / Night Vision, Electromagnetic Sensors, Multi-Sensor Fusion, Sensor Telemetry

- Engineering an autonomous ESP32-powered drone using LiDAR, thermal imaging, and RF detection to locate targets in low-visibility or GPS-denied environments.
- Developing embedded firmware for real-time multi-sensor data fusion, collision avoidance, and ground-station telemetry.

Subterranean Pipe Inspection Mobile Robot | *In Progress (Expected Dec 2026)* | *Sundt Construction*
Tools: ESP32, Custom Surface-Mount PCBs (SMT), Camera Module, Motor Control, Multi-Sensor Fusion, Power Management

- Architecting a space-constrained mobile robot by designing custom surface-mount (SMT) PCBs to integrate an ESP32, power management, motor drivers, and sensor buses.
- Developing embedded firmware to aggregate onboard camera feeds and multi-sensor telemetry for real-time defect localization and logging over a custom interface.

Articulated Robotic Arm System | *In Progress (Expected Dec 2026)*
Tools: PSoC, C/C++, Python, Serial Communication, Kinematics, Motor Control

- Architecting a multi-tier control pipeline pairing Python high-level forward/inverse kinematics with low-level PSoC firmware for multi-axis trajectory execution.
- Developing C/C++ embedded firmware to manage real-time PWM motor driving, encoder position feedback loops, and high-speed serial communications.

Autonomous Firewatch Robotic System | *Sundt Construction*
Tools: PSoC, Custom Multi-Layer PCBs, Motor Control Circuitry, Sensor Bus Networks, Hardware Isolation

- Co-engineered multi-layer custom PCBs for a Sundt Construction embedded robotic platform to interface with an external AI vision system for automated fire mitigation.
- Designed and routed localized high-speed motor control circuitry and multi-sensor bus networks to ensure reliable physical movement and fluid delivery.
- Implemented hardware isolation strategies between high-power actuator switching circuits and sensitive low-voltage digital control paths to prevent telemetry disruption.

EDUCATION

Arizona State University – ***Expected Grad: December 2027***
Major GPA: 3.17
Vertical Flight Sun-Devils - *President and Co-founder*
January 2026 - Present

- Co-founded a Vertical Flight Society university chapter, leading a 5-student executive team and successfully defending operational budgets to secure founding resources.
- Developing hands-on instructional workshops on VTOL and eVTOL principles to train the student body in rotorcraft fundamentals for upcoming competitions.
- Establishing technical learning tracks covering flight systems and aerodynamics to build a competitive team pipeline from the ground up.

Relevant Coursework: Embedded Systems Design Project 1 and 2, Robotic Systems 1 and 2, Electrical Fundamentals, Design of Electrical Systems, Computational Modeling

Joshua Hayes

Chandler, AZ 85226 | 480.490.8778 | jdhaye15@asu.edu | linkedin.com/in/joshua-hayes623

SUMMARY

Materials Science and Engineering student in Barrett, the Honors College. Currently involved and developing skills in laboratory research, data analysis, and technical projects.

EDUCATION

Arizona State University, Tempe, AZ — Expected Graduation - May 2030
B.S.E. Materials Science and Engineering | Ira A. Fulton Schools of Engineering — Current GPA - 4.00
Barrett, the Honors College

Corona del Sol High School, Phoenix, AZ — May 2026
High School Diploma — 4.00 GPA

- Top 1% of graduating class
- Arizona Seal of Fine Arts Proficiency, Finance, and Biliteracy (French)

EXPERIENCE

Zero Mass Labs, Tempe, AZ: Lab Aide — June 2026 – present

- Superabsorbent polymer synthesization and analysis of sorption kinetics
- General laboratory organization and data analysis

EXTRACURRICULAR ACTIVITIES

Sun Devil Satellite Lab (SDSL), Arizona State University — Fall 2026 – present

- Researched and wrote the initial proposal for PyroSat for the SDSL Payload Pitch Competition
- Aided in the creation of a funding proposal for PyroSat, a CubeSat dedicated to the prediction of wildfire inflammation.

FRC Team 6479 – AZTECH Robotics, Corona del Sol High School, Tempe, AZ — Fall 2023 – Spring 2026

- Served as the Director of Secretary—managed Gantt charts and timelines, led PDR and CDR meetings, managed membership fees, registration, and onboarding (Notion)
- Aided in the design and fabrication of a competitive robot (Onshape)
- Strategic analysis of a competitive game including data collection and analysis

We the People, Corona del Sol High School, Tempe, AZ — Fall 2025 – Spring 2026

- Competed in the We the People National Competition at Washington D.C. as the Arizona Wildcard team
- Member of Unit 5, examining what rights are protected under the Bill of Rights

PROJECTS

Superabsorbent Polymer Kinetics, Zero Mass Labs — June 2026 – present

- Examined the fitting of the Avrami model of sorption kinetics to various superabsorbent polymers (MATLAB, Origin)
- Continuing to analyze the effects of crosslinker length and concentration in sorption kinetics (DVSA, Origin, Excel)

SKILLS

Technical and Organization: Onshape, Excel, Notion
Analysis: Origin, MATLAB, Granta

Dhyanam Shah

Tempe, Arizona | dshah207@asu.edu | linkedin.com/in/dhyanamshah

PROJECTS

Robo-Weeder

- Developed an autonomous vision-based agricultural robot capable of detecting and removing weeds in real-time.
- Built a computer vision pipeline to distinguish weeds from crops using visual features under varying field conditions.
- Designed a servo-actuated gripper system for precise weed extraction with minimal crop disturbance.
- Reduced dependence on herbicides through targeted mechanical weed removal.

8-bit Central Processing Unit

- Designed and simulated an 8/16-bit CPU architecture in Logisim, implementing core digital logic and processor components.
- Implemented an ALU, register system, control unit, and instruction for executing processor operations.
- Designed and integrated combinational and sequential logic circuits to implement CPU functionality.
- Validated processor behavior through systematic simulation and testing of instruction execution.

Passion Projects

- Hosting a server with an old CPU, training LLMs locally, and tinkering with development boards.

EXPERIENCE

Team D.MechatronicX — Mumbai, IND
Competitive Robotics Engineer — Dec 2024 – Feb 2026

- Collaborated in team-based environments to rapidly prototype, test, and refine robotic systems under time-constrained conditions.
- Engineered robotic manipulation systems (gripper mechanisms) for object handling, optimized for stability and precision under variable terrain conditions.
- Developed sensor-based navigation systems for line-following using real-time signal processing and feedback loops.
- Integrated embedded systems (Arduino/ESP32) with motor drivers and sensor modules for real-time control and responsiveness.

EDUCATION

Arizona State University, Tempe — Tempe, AZ
BSE, Electrical and Electronics Engineering — May 2030

SKILLS

Programming: C++, Python, C
Hardware: Arduino, ESP32, Microcontrollers, Digital Logic, Embedded Systems
Design: PCB Design, CAD, Computer Vision
Tools: Logisim, KiCad, Git
Additional Skills: Public Speaking, Writing, Research, Leadership, Teamwork, Financial Analysis

Lisbeth Arroyo

CONTACT

480-742-9427
Larroyop@asu.edu
Phoenix, AZ

SKILLS

- Problem-solving and analytical reasoning in engineering contexts
- Laboratory experience in physics and chemistry experimentation
- Technical writing and structured communication
- Team-based academic project experience

LANGUAGES

English

Spanish

OBJECTIVE

Earth and Space Exploration (Exploration Systems Design) student seeking hands-on research experience in space systems, robotics, and exploration technologies. Interested in developing technical and problem-solving skills applicable to future aerospace and STEM outreach work.

EDUCATION

High School Diploma — Sierra Linda High School May 2018
Phoenix, AZ | GPA: 4.0 | Salutatorian

- AP Scholar
- National Honor Society

B.S. Earth and Space Exploration (Exploration Systems Design) — Arizona State University Expected Spring 2028
Tempe, AZ | GPA: 3.04

- Chicanos por la Causa Scholar

LEADERSHIP & INVOLVEMENT

Sun Devil Satellite Laboratory November 2026 – Present
Research Lead, Payload Member

- Student organization where we are trained in proposal writing and work in teams to develop CubeSat satellites for launch
- My current team is working on PyroSat, a 3U CubeSat satellite meant to collect thermal data and water/soil vegetation data for wildfire prevention
- My research is focused on satellite structure, thermal imagery, GNSS-R payload, attitude control, power, and cost

PERSONAL PROJECT

Engineering Proposal October 2026
Proposed to the Sun Devil Satellite Laboratory

- The Medusa Project aims to improve the architecture of deployables for satellites. Through the use of biomimicry engineering, the design would be based on the flexibility of a jellyfish's medusa membrane, which allows for contraction and deformation. The architecture could later be adapted as the structural support for a deployable antenna, reflectarray, solar/sensor array, or other large-area small-spacecraft system.
- This project is still in development and was proposed to SDSL as follows: https://silent-glade-746.linkyhost.page

RELEVANT COURSEWORK

- Calculus I-II
- Physics I (Mechanics)
- Chemistry for Engineers
- In progress: Physics II (Electromagnetism), Calculus III

SUNNY THIEN CHI TRAN

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EDUCATION

MET Professional Academy (MET) – Technology Strand, Peoria, AZ **May 2026**
Relevant Coursework: MET Cybersecurity Honors – Cisco, IT Ethics, Linux, Ethical Hacker, MS Windows

Sunrise Mountain High School (SMHS), Peoria, AZ **May 2026**
Relevant Coursework: Physics (GCC Dual Enrollment), AP Calculus I, AP Calculus II **GPA: 3.9 / 4.0 (Unweighted)**

CERTIFICATES & HONORS

Certifications: TestOut Client Pro – Windows Administration **Dec. 2025**
TestOut Ethical Hacker Pro – Organizational Cybersecurity **May 2025**

Certificates: Red Hat System Administration I (RH124) – Linux Foundation **Apr. 2025**
CCNA: Intro to Networks – Networking Fundamentals **Dec. 2024**

WORK EXPERIENCE

Tech Intern – Highway 85 Creative, Peoria, AZ **Sep. 2025 – Aug. 2025**
Part-time

- Optimized IT readiness, configuring Windows user accounts, updating and testing laptops, and preparing dozens of TVs and mini-PCs for shipment
- Supported multiple concurrent, time-sensitive projects, responding to short-notice requests to meet tight deadlines without compromising quality
- Installed and maintained LED display systems, troubleshooting hardware faults, and repairing fine-pitch (<2.5 mm) LED pixels to restore full system functionality and visual consistency

IMT Intern – Peoria Unified School District Cholla Annex IMT, Glendale, AZ **May 2025 – Jul. 2025**
Full-time

- Contributed to reimaging 35,000+ student laptops in two months as part of a district-wide refresh, handling operating system installation, software setup, and hardware preparation
- Designed a time-efficient 12-device batch imaging workflow by analyzing network switch bandwidth at each school, notably improving imaging speed and turnaround time
- Accurately inventoried and organized large volumes of new laptops, ensuring proper tracking, labeling, and smooth deployment
- Diagnosed and replaced various faulty laptop keyboards, glass screens, and LCD displays

PROJECT EXPERIENCE

Liberty Wildlife VR Training Simulation **Jul. 2025 – Apr. 2026**
Project Manager

- Oversaw a team of 6 students to develop an educational VR training simulation, learning Unity and Blender from scratch to create an immersive experience for potential volunteers
- Managed project workflow, including task delegation, mentor communication, and detailed technical documentation, ensuring steady progress and high-quality deliverables
- Partnered with stakeholders, including Liberty Wildlife and Microsoft industry mentor, to align project goals with organizational needs and improve volunteer training efficiency

ROMAN CALURE FIRST YEAR

443-355-7212 • rcalure@asu.edu• linkedin profile

SUMMARY

Aerospace Engineering Student (Astronautics) with experience in a mentoring position (LTF Cafe) and community service building and social projects (ASP Service Project and Camden Service Trip).

EDUCATION

B.S.E., Aerospace Engineering (Astronautics) May 2030
Arizona State University N/A GPA
Our Lady of Good Counsel High School Olney, MD May 2026
Achieved 2 AP/Dual Enrollment credits 3.88 unweighted GPA

TECHNICAL SKILLS

Design and Modeling Tools: Microsoft Office, Floorplanner, AutoCAD.
Low voltage wiring: Home outdoor lighting implementation (2024).
Professional writing: Experience in writing proposals for Sun Devil Satellite Laboratory.

PROJECTS

ASU SDSL Pyrosat Mechanical Lead Aug 2026 – present

- Founding member of a new CubeSat mission to detect wildfires
  - Workday Learning courses: General lab safety and Fire Safety
  - Eventual learning of systems such as deployers, structural frame subsystem, Attitude Determination and Control Systems.
  - Technical skills in professional proposal writing from helping with three proposals (Inkjet, PyroSat, and Solaris).

2024 Appalachia Service Project

- Part of a service group who built a back porch for an underprivileged senior
  - Gained basic knowledge on power tools and building (Electric Drill, jigsaw, mitersaw, measuring tape)

2023 & 2025 Camden Service Trip

- Participated in a service trip where I volunteered for a week with underprivileged youth and elderly. I gained valuable social and leadership skills.

WORK EXPERIENCE

Part-time LifeTime Cafe' Employee (Fall 2024-Summer 2026)≈20hrs/week

- Trained 3 employees on smoothie line and housekeeping developing my leadership and communication skills.

JASON LUO

Chandler, AZ • 480.930.8136 • Jiaxuluo@asu.edu

SUMMARY

Electrical engineering student with hands-on circuit design and testing experience through EPICS and independent projects. Interested in circuit design, PCB, sensors, and related fields.

EDUCATION

B.S.E., Electrical Engineering — Barrett Honors College — Expected May 2029
Arizona State University — GPA: 4.00

Arizona College Prep High School, Chandler, AZ — May 2026
National Merit Semifinalist • 16 AP Classes — GPA: 4.77

Relevant Coursework: Calculus 3 (A), Differential Equations (A), Applied Linear Algebra (A), AP Physics C: E&M (5), AP Physics C: Mechanics (5), Intro to C++ (A)

TECHNICAL SKILLS

Design & Modeling Tools: KiCad, LTspice
Programming: C++, MATLAB
Lab Equipment: Oscilloscopes, function generators, multimeters

PROJECTS

Elephant Detection System — EPICS (Engineering Projects in Community Service), ASU — Fall 2026 – Present
Electrical subteam member (4-person) on a 12-person team developing a low-cost geophone-based early-warning system to help the Maasai Mara community reduce human-elephant conflict

- Designing and testing signal-conditioning circuitry to filter and amplify a 4–10mV geophone signal for input to an ESP32 microcontroller
- Ran controlled tests on the instrumentation amplifier stage using an oscilloscope and function generator (learning both tools hands-on), identifying a critical design flaw before hardware integration

Technical Proposal Author — SDSL (Sun Devil Satellite Club), ASU — Fall 2026 – Present
Member of SDSL NewSat team (~40 people) developing official proposals for a new satellite to be reviewed by industry partners for funding. Expected to be built, launched, and operating by 2030.

- Developed the SOLARIS proposal, an in-flight technology-validation payload testing the feasibility of perovskite-based solar cells. Placed 4rd of 25 proposals in the SDSL Internal Design Competition.
- Researched the power feasibility for Pyrosat, a proposed 3U satellite building a wildfire prediction model based on GNSS-R and thermal Imaging data.

Analog Guitar Pedal (Personal Project) — Summer 2026 – Present
Designing and building a custom analog distortion pedal from scratch to deepen understanding of AC circuit analysis and op-amp design

- Designed a non-inverting op-amp gain stage and a diode hard-clipping stage in LTspice to shape the pedal's distortion characteristics
- Currently building and testing the physical prototype circuit

WORK EXPERIENCE

High School Math Tutor (Freelance) — Fall 2025 – Spring 2026
Mentored high school students ages 13–16, building strong algebraic foundations through individualized instruction

- Created individualized lesson plans tailored to each student's pace and learning style

Ethan Bialis

Tucson, AZ | 520-404-8183 | ebialis@asu.edu

Highly motivated student with a passion for learning across disciplines, including a strong interest in space science. Collaborative leader who enjoys building inclusive teams and working with others to solve problems and create positive impact.

EDUCATION

ASU | IRA. A. Fulton Schools of Engineering | Barret the Honors College
B.S. Aerospace Engineering (Astronautics)
Expected graduation May 2030

High School Diploma, The Gregory School, Class of 2026 • Tucson, Arizona GPA: 4.55
Course Highlights: AP Chemistry, AP Biology, AP Physics, AP Calculus, Engineering, Design and Build, Stagecraft and Technical Design

ACADEMIC AWARDS, PUBLICATIONS & HONORS

2025/2026 School Year (Senior)

- Flinn Scholar
- School's Ingersol Award Winner
- School's Math Department Award Winner
- High Honor Roll
- Stage Manager School Play
- Stage Manager School Musical
- Technical Advisor to StuCo
- Odyssey of the Mind Regional & States First Place Team Captain
- Robotics Team Captain Bid to State
- Robotics First Place Design Award State Winner
- Robotics Inspire Award
- Robotics International Connect Award 2nd Place
- Writing Center Tutor
- Entrepreneurship People's Choice Winner at 10 West
- Odyssey of the Mind Scholarship
- Creative Competitions Scholarship
- Founder of O2Art

2024/2025 School Year (Junior)

- High Honor Roll
- Dean's list First Tech Challenge Semifinalist
- Stage Manager School Musical
- Technical Advisor to StuCo
- Odyssey of the Mind Regional & States First Place Team Captain
- Robotics Team Captain Bid to State
- Writing Center Tutor
- Entrepreneurship Winner

2023/2024 School Year (Sophomore)

- High Honor Roll
- Theater Light Manager
- Robotics Team Vice-Captain
- Odyssey of the Mind Regional Second Place Team Captain
- Commander's Cup Space Camp Awardee
- Film Festival Technical Coordinator

2022/2023 School Year (Freshman)

- High Honor Roll
- Ran lights in musical and play
- Robotics Team member bid to State
- Film Festival Technical Coordinator

Other Accolades

- Mentor of the Year FLL Season 2022
- Space Camp Mentor Award
- Started 501 C 3 charity and head of board

ADDITIONAL COURSEWORK

AP Physics 1 — University of Alabama, Huntsville
UC SCOUT
Intro to Space Science

EXTRACURRICULAR ACTIVITIES

- **Avid Astrophotographer:** Member of TAAA (2023)
- **Odyssey of the Mind,** 2026, 2025, 2024 School Years
- **First Tech Challenge Robotics,** 2025, 2024, 2023 School Years
- **Soccer Team Member,** 2025, 2024, 2023 School Years
- **Varsity Tennis Team,** 2024, 2023 School Years
- **Musical Theater,** Light Designer, Stage Manager
- **Clubs:** Hawks Robotics, Film Club, Medical Club, STEM Club
- **National Honors Society Member**
- **National Spanish Honor Society**

WORK EXPERIENCE

- **Starizona** - Summer Internship 2025
  - Worked designing parts for telescope accessories using CAD, 3D printing
  - Assembled the Hyperstar
  - Assisted customers at Starviewing night events
- **Nanomaterials International Corporation** - Summer Internship 2024
  - Worked milling metal parts, designing them and milling
  - Advanced 3D scanning methods
  - Skills: CAD, CAM, 3D Printing, Laser cutting, Waterjet cutter, Metal Mill
- **Writing Center Tutor** - School Year 2024-2026
  - Worked tutoring all the students in the school twice a week
  - Additional course work in how to tutor and how to write

COMMUNITY SERVICE & LEADERSHIP

- **Boo's Whisker Wishes** - Founded and run a non-profit that provides funding to people for their vet bills so that they can keep their animals.
- **Technical Advisor to StuCo.** - Create decorations, trophies, lighting effects, and editing of videos and technical resources for the Student Council.
- **Minds Alive** Volunteered helping with the broken light board.
- **First Lego League** - Ran a team, volunteered, and helped at the tournaments.
- **Gregory Admission Days** - Ran tours of the Fab Lab and worked with younger students to learn to use the equipment.
- **Volunteer at TGS Events:** Homecoming, concerts, after school events
- **Mentor High School Robotics team:** Teaching CAD and helping them to prep for their competition

REFERENCES

- Mr. Dennis Conner, The Gregory School, dconner@gregoryschool.org
- Ms. Lorie Heald, The Gregory School, lheald@gregoryschool.org

Appendix B: Full Mass, Dimensions, and Power Tables

Mass and Dimensions:
https://docs.google.com/spreadsheets/d/1WHJpG7GDutD4zj7hWMic2UL9TpPHD-5hI-2AKUWcneI/edit?usp=sharing
(too large to paste)

Component	Operating Power	Operation	Assumed Time / 93-min Orbit	Energy / Orbit
NanoCom SDR MK3	5.46 W [35]	Science	10 min	0.91 Wh
Boson+ 640	1.0–1.3 W [3]	Science	10 min	0.17–0.22 Wh
Nadir RF-FE A	~0.25 W*	Science	10 min	~0.042 Wh
Nadir RF-FE B	~0.25 W*	Science	10 min	~0.042 Wh
OEM719	~0.90 W [90]	Continuous	93 min	~1.40 Wh
piPATCH-L1E1/FM	0.066 W [32]	Continuous	93 min	~0.10 Wh
RP2350 OBC	~0.10 W [4]	Continuous	93 min	~0.16 Wh
ADCS	3.13 W [17]	Earth / Sun facing	80 min	4.17 Wh
ADCS	3.71 W [17]	GS-pointing	10 min/pass	0.618 Wh
ADCS	5.5 W [17]	Transition	3 min	0.275 Wh
SRS-4 S-band Transceiver	10.8 W [37]	Downlink at 33 dBm	10 min/pass	1.80 Wh
SRS-4 RS-422 interface	0.06 W [37]	Downlink	10 min/pass	~0.01 Wh
SX1278 LoRa Radio	3.25–4.5 W [91]	Transmit	10 min/pass	~0.54–0.75 Wh
SX1278 LoRa Radio	0.065 W [91]	Receive	83 min	~0.090 Wh
SkyFox piPATCH-S-DL/FM	0 W [38]	Downlink	10 min/pass	0 Wh

Boson thermal-mass heater (HAP6736)	~0.32 W* [70]	Cold-case thermal control	36 min	0.19 Wh
Battery survival heater (HAP6736)	~0.32 W* [70]	Thermostat-controlled battery heating	36 min	0.19 Wh
EPS conversion losses	80%*	All powered modes	Variable	Applied to mode totals